STA 2002, Summer 2024
Probability and Statistics II

Midterm Review

Midterm

- Sunday June 30, 2:30pm – 4:30pm, Beijing Time
- Location: TD206
- One A4 Cheat-sheet allowed, double-sided, handwritten without stickers
- No make-up exam
- Please bring your student ID card
- Calculator is NEEDED
- Zero tolerance on cheating and plagiarism

Outline

- **Probabilities and Distributions (STA 2001 review)**
- **Point Estimation**
- **Confidence Interval**
- **Hypothesis Testing**

Discrete Distribution

Continuous Distribution

Moment generating function

Central Limit Theorem

PROBABILITY (STA2001)

Bernoulli Distribution

$X \sim Bernoulli(p) \quad x = 0,1$ *p*: probability of success

pmf, mean and variance:

$$f(x) = p^x(1-p)^{1-x}, \qquad x = 0, 1$$

$$\mu = E(X) = \sum_{x=0}^{1} x p^x (1-p)^{1-x} = (0)(1-p) + (1)(p) = p$$

$$\sigma^2 = \text{Var}(X) = \sum_{x=0}^{1} (x-p)^2 p^x (1-p)^{1-x}$$

$$= (0-p)^2(1-p) + (1-p)^2 p = p(1-p)$$

Binomial Distribution

$X \sim Bin(n, p)$ $x = 0,1,\ldots,n$ p: probability of success

pmf, mean and variance:

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, 2, \ldots, n$$

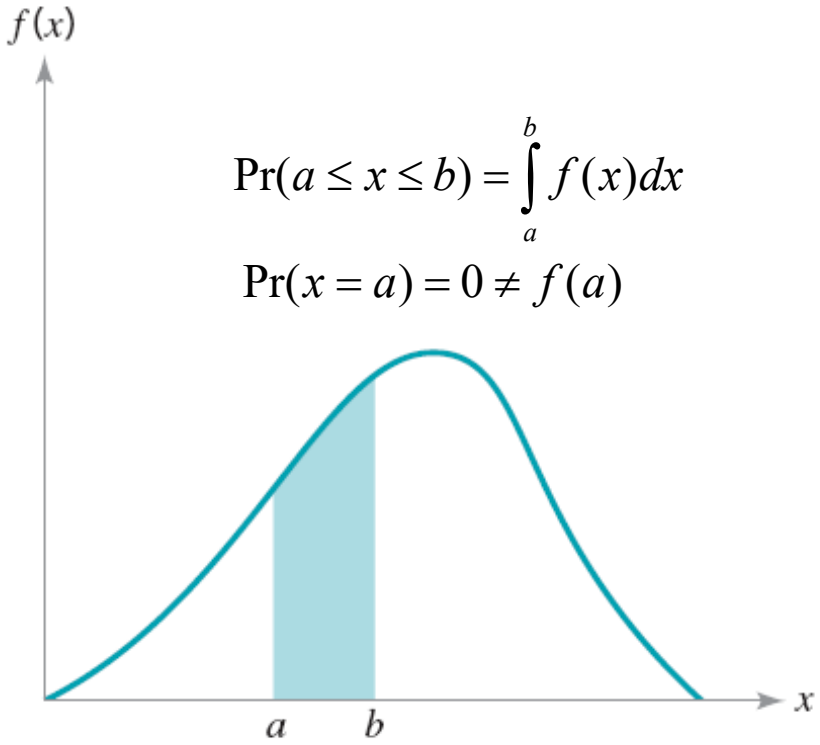
$$\mu = E(X) = np \qquad \sigma^2 = E(X^2) - [E(X)]^2 = np(1-p)$$

Fact: sum of n independent Bernoulli variables (p) is Binomial (n,p)

Continuous Distributions

- When the outcome takes continuous values

Probability density function (pdf): $f(x)$ is pdf $\leftrightarrow \begin{cases} f(x) \geq 0 \\ \int_{-\infty}^{+\infty} f(x)dx = 1 \end{cases}$



$$\Pr(a \leq x \leq b) = \int_a^b f(x)dx$$

$$\Pr(x = a) = 0 \neq f(a)$$

CDF: $F(x) = \int_{-\infty}^{x} f(u)du$

Mean: $E(X) = \int xf(x)dx = \mu$

Variance: $V(X) = E(X^2) - [E(X)]^2$

Moment Generating Function (MGF)

- The moment generating function of $X \sim f(x)$, if exists, is defined as

$$M(t) = \int_{-\infty}^{\infty} e^{tx} f(x)\, dx, \qquad -h < t < h.$$

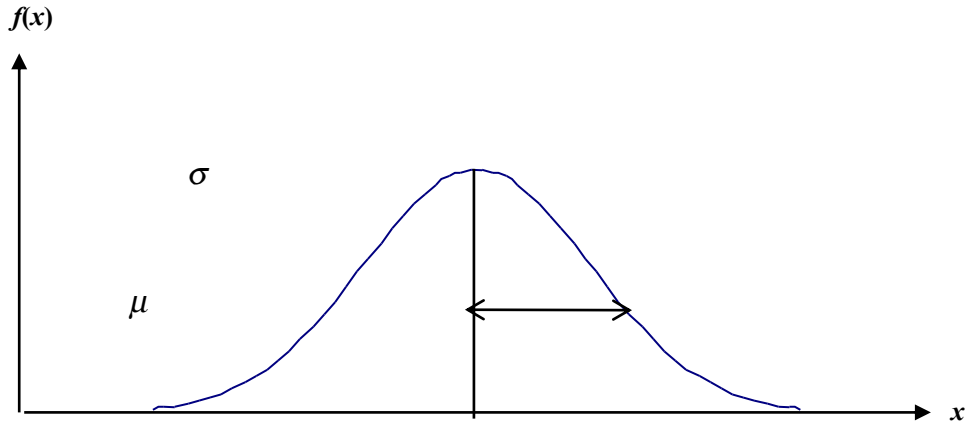
$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x)\, dx = M'(0)$$

$$\sigma^2 = \mathrm{Var}(X) = E[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx = M''(0) - [M'(0)]^2$$

Normal (Gaussian) Distribution

$$X \sim N(\mu,\sigma^2); \qquad -\infty < x < +\infty$$

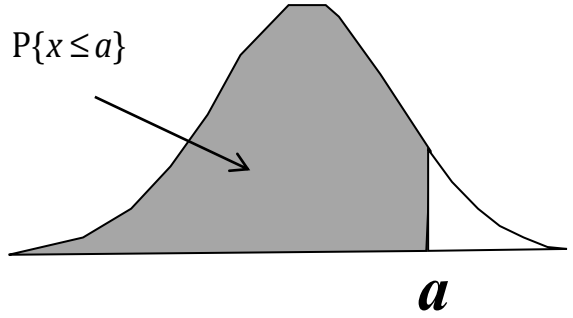
$$f(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$



MGF: $$M(t) = \exp\left(\frac{2\mu\sigma^2 t + \sigma^4 t^2}{2\sigma^2}\right) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right).$$

$$E(X) = M'(0) = \mu,$$

$$\mathrm{Var}(X) = M''(0) - [M'(0)]^2 = \mu^2 + \sigma^2 - \mu^2 = \sigma^2$$



$$\mathrm{P}\{x \le a\} = \int_{-\infty}^{a} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\} dx$$

Chi-square distribution

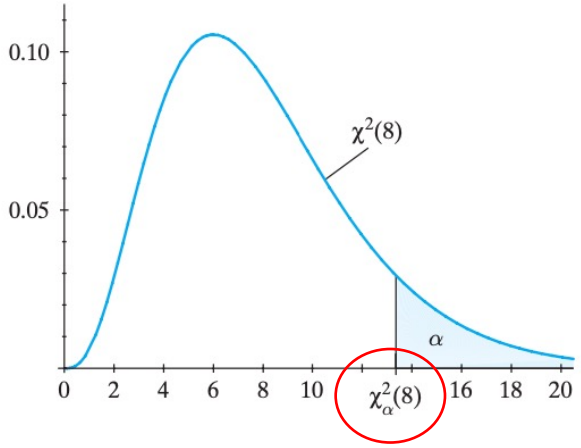
- Chi-square distribution with r degrees of freedom:

$$X \sim \chi^2(r) \qquad f(x) = \frac{1}{\Gamma(r/2)2^{r/2}} x^{r/2-1} e^{-x/2}, \qquad 0 < x < \infty$$

$$Gamma\left(\frac{r}{2}, 2\right)$$

Recall Gamma function: $\Gamma(t) = \int_0^\infty y^{t-1} e^{-y}\, dy, \qquad 0 < t$

$$M(t) = (1-2t)^{-r/2}, \qquad t < \frac{1}{2} \qquad \mu = \alpha\theta = \left(\frac{r}{2}\right)2 = r \qquad \sigma^2 = \alpha\theta^2 = \left(\frac{r}{2}\right)2^2 = 2r$$



	$P(X \le x)$							
	0.010	0.025	0.050	0.100	0.900	0.950	0.975	0.990
r	$\chi^2_{0.99}(r)$	$\chi^2_{0.975}(r)$	$\chi^2_{0.95}(r)$	$\chi^2_{0.90}(r)$	$\chi^2_{0.10}(r)$	$\chi^2_{0.05}(r)$	$\chi^2_{0.025}(r)$	$\chi^2_{0.01}(r)$
1	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.34
4	0.297	0.484	0.711	1.064	7.779	9.488	11.14	13.28
5	0.554	0.831	1.145	1.610	9.236	11.07	12.83	15.09
6	0.872	1.237	1.635	2.204	10.64	12.59	14.45	16.81
7	1.239	1.690	2.167	2.833	12.02	14.07	16.01	18.48
8	1.646	2.180	2.733	3.490	13.36	15.51	17.54	20.09
9	2.088	2.700	3.325	4.168	14.68	16.92	19.02	21.67
10	2.558	3.247	3.940	4.865	15.99	18.31	20.48	23.21
11	3.053	3.816	4.575	5.578	17.28	19.68	21.92	24.72
12	3.571	4.404	5.226	6.304	18.55	21.03	23.34	26.22
13	4.107	5.009	5.892	7.042	19.81	22.36	24.74	27.69
14	4.660	5.629	6.571	7.790	21.06	23.68	26.12	29.14
15	5.229	6.262	7.261	8.547	22.31	25.00	27.49	30.58

Useful Results for Normal and Chi-square

- If $x_1 \sim N(\mu_1, \sigma_1^2), x_2 \sim N(\mu_2, \sigma_2^2)$ are independently normally distributed variables, then

$$y = x_1 + x_2$$

also follows the normal distribution:

$$y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

Theorem 3.3-1 If X is $N(\mu, \sigma^2)$, then $Z = (X - \mu)/\sigma$ is $N(0, 1)$.

$$X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2) \rightarrow \bar{X} \sim N(\mu, \frac{\sigma^2}{n}) \rightarrow \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0,1)$$

Sampling distribution of sample mean is normal, when samples are normal

Useful Results for Normal and Chi-square

Theorem 3.3-2 If the random variable X is $N(\mu, \sigma^2)$, $\sigma^2 > 0$, then the random variable $V = (X-\mu)^2/\sigma^2 = Z^2$ is $\chi^2(1)$.

Facts:

- (connection to normal) $X_1, X_2, \ldots, X_r \sim N(0,1)$, and they are independent, then

$$X_1^2 + X_2^2 + \cdots + X_r^2 \sim \chi^2(r)$$

- (sum of chi-squares) $X \sim \chi^2(r_1), Y \sim \chi^2(r_2)$, and they are independent, then

$$X + Y \sim \chi^2(r_1 + r_2)$$

F distribution

- Let $U \sim \chi^2(r_1), V \sim \chi^2(r_2)$, and they are independent

$$W = \frac{U/r_1}{V/r_2} \sim F(r_1, r_2) : F \text{ distribution with } r_1 \text{ and } r_2 \text{ degrees of freedom}$$

Facts:

- $W \sim F(r_1, r_2) \xrightarrow{\text{yields}} \frac{1}{W} \sim F(r_2, r_1)$
- Denote right-tail probability $P[W \geq F_\alpha(r_1, r_2)] = \alpha$

$$\alpha = P[W \leq F_{1-\alpha}(r_1, r_2)] = P[1/W \geq 1/F_{1-\alpha}(r_1, r_2)]$$

$$\Longrightarrow \boxed{\frac{1}{F_{1-\alpha}(r_1, r_2)} = F_\alpha(r_2, r_1)} \text{ or } \boxed{F_{1-\alpha}(r_1, r_2) = \frac{1}{F_\alpha(r_2, r_1)}}$$

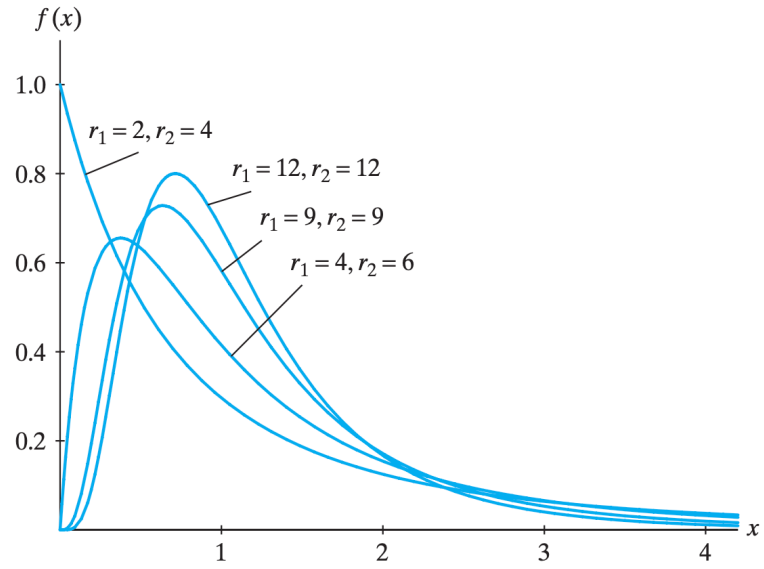


Figure 5.2-5 Graphs of F pdfs

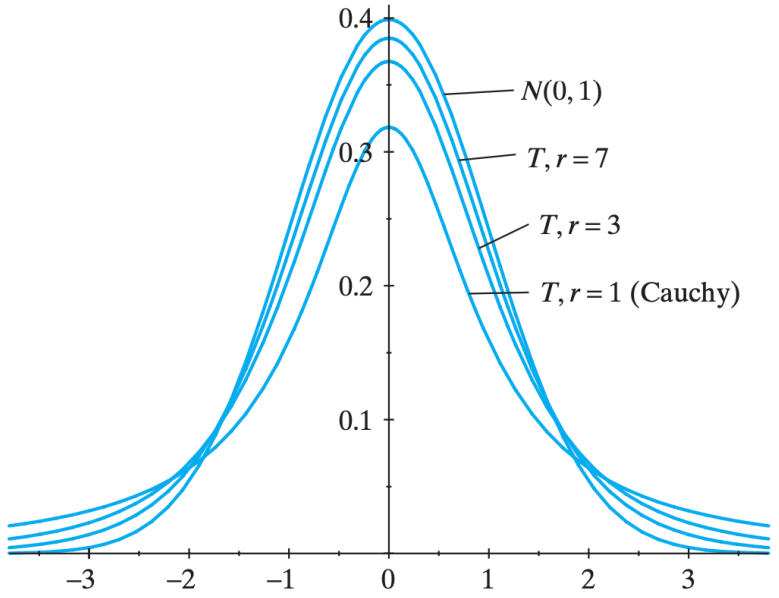
Student's t distribution

- Let $Z \sim N(0,1), U \sim \chi^2(r)$, and they are independent

$$T = \frac{Z}{\sqrt{U/r}} \sim t(r)$$: t distribution with r degrees of freedom

Facts:

The larger the degrees of freedom r, the more the t distribution looks like the normal distribution



Sample mean and variance

- Suppose we have a sequence of i.i.d. (independent and identically distributed) random variables $X_1, X_2, \ldots, X_n$. Let μ denotes their common mean, and σ^2 the common variance

- The **sample mean** is defined as

$$\overline{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

$$\mathbb{E}(\overline{X}) = \mu, \quad \mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}$$

- The **sample variance** is defined

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

Sample mean and variance

Corollary 5.5-1 If $X_1, X_2, \ldots, X_n$ are observations of a random sample of size n from the normal distribution $N(\mu, \sigma^2)$, then the distribution of the sample mean $\overline{X} = (1/n)\sum_{i=1}^{n} X_i$ is $N(\mu, \sigma^2/n)$.

Theorem 5.5-2 Let $X_1, X_2, \ldots, X_n$ be observations of a random sample of size n from the normal distribution $N(\mu, \sigma^2)$. Then the sample mean,

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i,$$

and the sample variance,

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n} (X_i - \overline{X})^2,$$

are independent and

$$\frac{(n-1)S^2}{\sigma^2} = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{\sigma^2} \text{ is } \chi^2(n-1).$$

Sample mean and variance

- As a consequence, we have

$$U = \sum_{i=1}^{n} \frac{(X_i - \mu)^2}{\sigma^2} \sim \chi^2(n) \qquad W = \sum_{i=1}^{n} \frac{(X_i - \overline{X})^2}{\sigma^2} \sim \chi^2(n-1)$$

$$T = \frac{\overbrace{\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}}^{N(0,1)}}{\sqrt{\underbrace{\frac{(n-1)S^2}{\sigma^2}}_{\chi^2(n-1)} \Big/ (n-1)}} = \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

Central Limit Theorem (CLT)

Theorem 5.6-1 **(Central Limit Theorem)** If $\overline{X}$ is the mean of a random sample $X_1, X_2, \ldots, X_n$ of size n from a distribution with a finite mean μ and a finite positive variance σ^2, then the distribution of

$$W = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sqrt{n}\,\sigma}$$

is $N(0,1)$ in the limit as $n \to \infty$.

Sampling distribution of sample mean is approximately normal, even when samples are NOT normal

Rule of thumb: when n > 30 this works pretty well.

		One population	Two populations
Sample Mean	Known Variance	$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$	$\frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}} \sim N(0,1)$
	Unknown Variance	$\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1)$	$\frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t(n_1+n_2-2)$
Sample Variance		$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$	$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1-1,n_2-1)$
Sample proportion		$\hat{p} \sim N(p, p(1-p)/n)$	$\frac{(\hat{p}_1-\hat{p}_2)-(p_1-p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1}+\frac{p_2(1-p_2)}{n_2}}} \sim N(0,1)$

Definition

Bias, standard error, and MSE

Maximum Likelihood

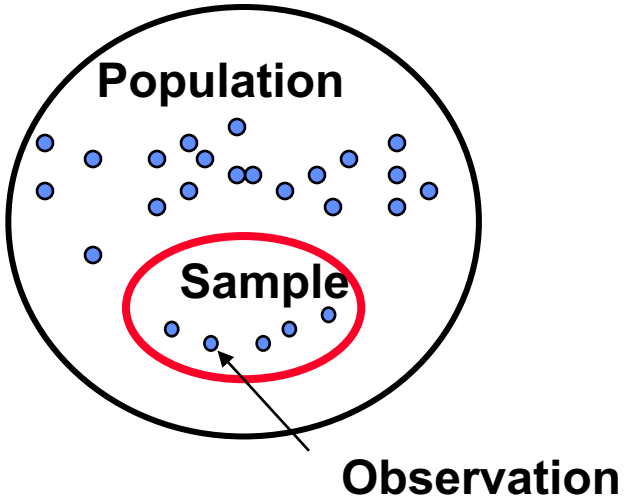
Method of Moments

POINT ESTIMATOR

Population Vs. Sample

- **Population:** a finite well-defined group of ALL objects which, although possibly large, can be enumerated in theory (e.g. investigating ALL the bearings manufactured today).

- **Sample:** A sample is a SUBSET of a population



Estimator

Suppose X is a random variable with $f(x;\theta)$ as the pdf. If $X_1, X_2, \ldots X_n$ is a random sample of size n from X, the statistic

$$\hat{\Theta} = h(X_1, X_2, \ldots, X_n)$$

Is called a **point estimator** of θ.

After the sample has been selected, $\hat{\Theta}$ takes on a particular numerical value called the **point estimate** of θ.

Note that $\hat{\Theta}$ is a random variable because it is a statistic (function of random variables)

Unbiased Estimator

The point estimator $\hat{\Theta}$ is an **unbiased estimator** for the parameter θ if

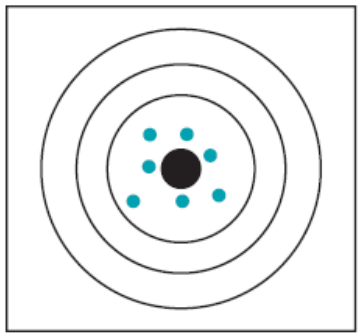
$$E(\hat{\Theta}) = \theta$$

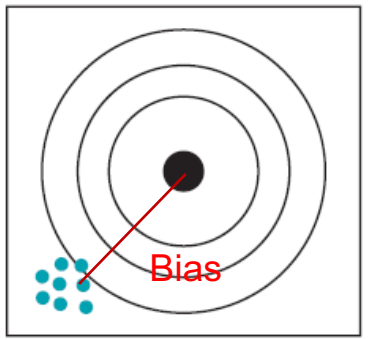
ties to sampling distribution

If the estimator is not unbiased, then the difference

$$E(\hat{\Theta}) - \theta$$

is called the **bias** of the estimator $\hat{\Theta}$.





Variance of a Point Estimator

If two estimators are unbiased, the one with smaller variance is preferred.

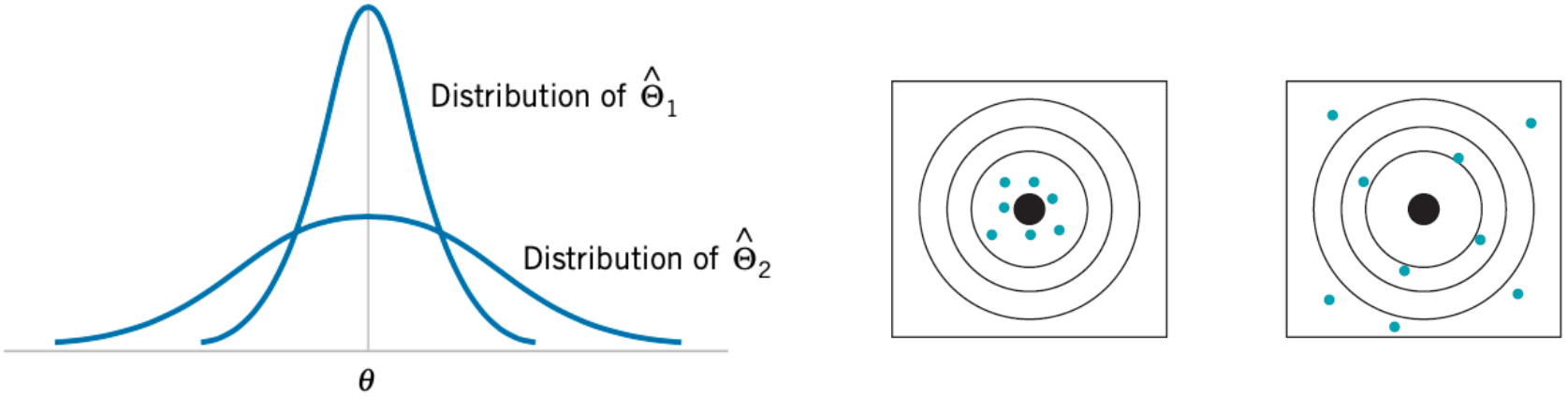


Figure The sampling distributions of two unbiased estimators

$$\text{var}(\hat{\Theta}_1) < \text{var}(\hat{\Theta}_2)$$

ties to sampling distribution

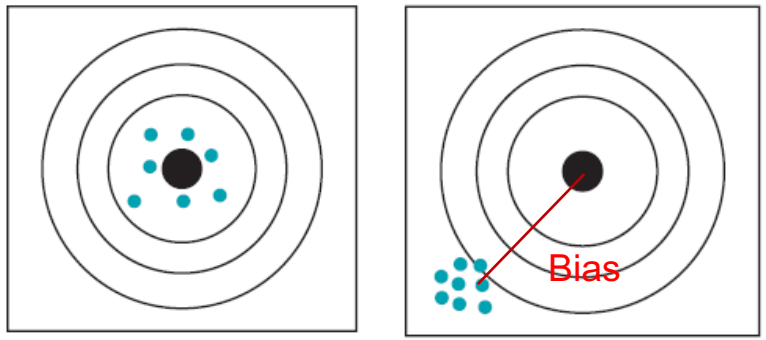
Mean Square Error (MSE)

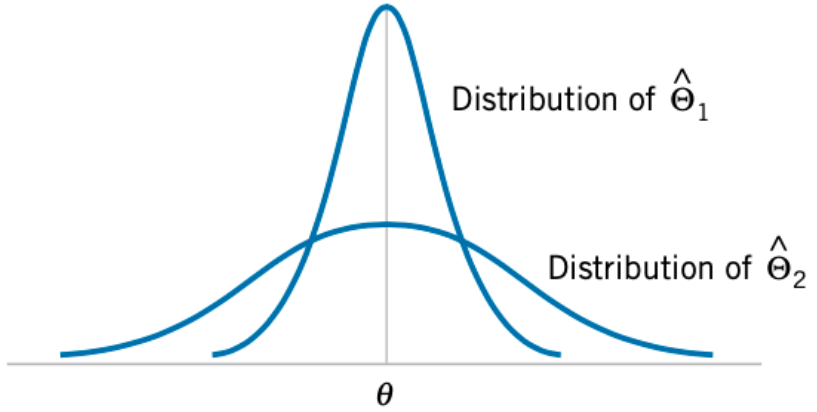
The **mean square error** of an estimator $\hat{\Theta}$ of the parameter θ is defined as

$$\text{MSE}(\hat{\Theta}) = E(\hat{\Theta} - \theta)^2$$

$$MSE(\hat{\Theta}) = E(\hat{\Theta} - \Theta)^2 = \left[E(\hat{\Theta} - \Theta)\right]^2 + \text{var}(\hat{\Theta} - \Theta)$$

$$MSE(\hat{\Theta}) = \left[\text{Bias}(\hat{\Theta})\right]^2 + \text{var}(\hat{\Theta})$$





Methods for Finding Estimators

- Assume a distribution for the samples
- Estimate the parameter of the distribution
- Several methods
  - Maximum likelihood
  - Method of moment

Method of Maximum Likelihood

Suppose that X is a random variable with probability distribution $f(x; \theta)$, where θ is a single unknown parameter. Let $x_1, x_2, \ldots, x_n$ be the observed values in a random sample of size n. Then the **likelihood function** of the sample is

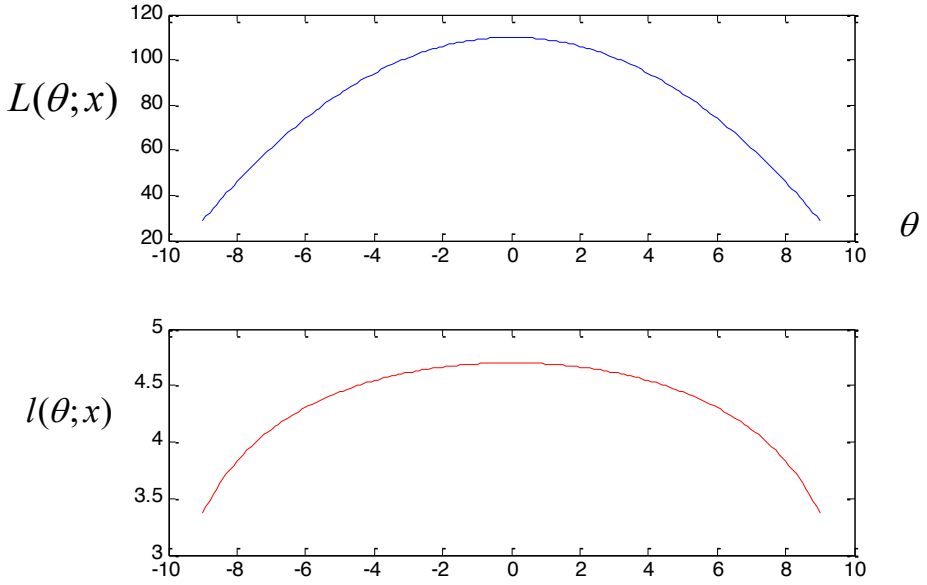
$$L(\theta) = f(x_1; \theta) \cdot f(x_2; \theta) \cdot \cdots \cdot f(x_n; \theta) \qquad (7\text{-}5)$$

Note that the likelihood function is now a function of only the unknown parameter θ. The **maximum likelihood estimator** of θ is the value of θ that maximizes the likelihood function $L(\theta)$.

$$L(\theta; x) = \prod_{i=1}^{n} f(x_i; \theta) = f(x_1; \theta) \ldots f(x_n; \theta)$$

$$l(\theta; x) = \sum_{i=1}^{n} \log\left[f(x_i; \theta)\right]$$

$$\hat{\Theta}(x) = \arg\max_{\theta} L(\theta; x) = \arg\max_{\theta} l(\theta; x)$$



Example: Normal

Let X be normally distributed with unknown μ and known variance σ^2. The likelihood function of a random sample of size n, say $X_1, X_2, \ldots, X_n$, is

$$L(\mu) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} e^{-(x_i-\mu)^2/(2\sigma^2)} = \frac{1}{(2\pi\sigma^2)^{n/2}} e^{-(1/2\sigma^2)\sum_{i=1}^{n}(x_i-\mu)^2}$$

Now

$$\ln L(\mu) = -(n/2)\ln(2\pi\sigma^2) - (2\sigma^2)^{-1}\sum_{i=1}^{n}(x_i-\mu)^2$$

and

$$\frac{d \ln L(\mu)}{d\mu} = (\sigma^2)^{-1}\sum_{i=1}^{n}(x_i-\mu)$$

$\longrightarrow$ What is the MLE for μ?

Example (Continued, unknown variance)

$$\ln L(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

$$\frac{\partial \ln L(\mu, \sigma^2)}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu) = 0$$

$$\frac{\partial \ln L(\mu, \sigma^2)}{\partial (\sigma^2)} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i - \mu)^2 = 0$$

The solutions to the above equation yield the maximum likelihood estimators

$$\hat{\mu} = \bar{X} \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

Example

6.4-14. Let $X_1, X_2, \ldots, X_n$ be a random sample of size n from a normal distribution.

(a) Show that an unbiased estimator of σ is cS, where

$$c = \frac{\sqrt{n-1}\,\Gamma\left(\frac{n-1}{2}\right)}{\sqrt{2}\,\Gamma\left(\frac{n}{2}\right)}.$$

HINT: Recall that the distribution of $(n-1)S^2/\sigma^2$ is $\chi^2(n-1)$.

$\chi^2(r)$ $\quad f(x) = \dfrac{1}{\Gamma(r/2)2^{r/2}}\, x^{r/2-1}e^{-x/2}, \qquad 0 < x < \infty$

$X \sim Gamma(\alpha, \theta)$ $\quad f(x) = \dfrac{1}{\Gamma(\alpha)\theta^{\alpha}}\, x^{\alpha-1}e^{-x/\theta}, \qquad 0 \le x < \infty$

Example

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$$

Recall $X \sim \chi^2(n-1)$

$$f(x) = \frac{1}{\Gamma(\frac{n-1}{2})2^{\frac{n-1}{2}}} x^{\frac{n-1}{2}-1} e^{-\frac{x}{2}}$$

$$\mathbb{E}\left[\frac{\sqrt{n-1}\,S}{\sigma}\right] = \mathbb{E}[\sqrt{X}] = \int_0^{+\infty} \sqrt{x} \cdot \frac{1}{\Gamma(\frac{n-1}{2})2^{\frac{n-1}{2}}} x^{\frac{n-1}{2}-1} e^{-\frac{x}{2}}\, dx$$

$$= \frac{1}{\Gamma(\frac{n-1}{2})2^{\frac{n-1}{2}}} \underbrace{\int_0^{+\infty} x^{\frac{n}{2}-1} e^{-\frac{x}{2}}\, dx}_{= \Gamma(\frac{n}{2})2^{\frac{n}{2}}}$$

Gamma$(\frac{n}{2}, 2)$

$$= \frac{\Gamma(\frac{n}{2})\sqrt{2}}{\Gamma(\frac{n-1}{2})}$$

$$\Rightarrow \frac{\sqrt{n-1}\cdot \mathbb{E}[S]}{\sigma} = \frac{\Gamma(\frac{n}{2})\sqrt{2}}{\Gamma(\frac{n-1}{2})}$$

$$\Rightarrow \frac{\sqrt{n-1}\cdot\Gamma(\frac{n-1}{2})}{\Gamma(\frac{n}{2})\sqrt{2}} E[S] = \sigma$$

Example

8. Find the maximum likelihood estimator of the unknown parameter θ where $X_1, X_2, \ldots, X_n$ is a sample from the distribution whose density function is

$$f_X(x) = \begin{cases} e^{-(x-\theta)} & \text{if } x \geq \theta \\ 0 & \text{otherwise.} \end{cases}$$

Is the MLE an unbiased estimator of θ?

Likelihood

$$L(\theta) = \prod_{i=1}^{n} e^{-(x_i-\theta)} \cdot 1\{x_i \geq \theta\}$$

$$= e^{-\sum_{i=1}^{n} x_i + n\theta} \; 1\{x_{(1)} \geq \theta\}$$

$$x_{(1)} = \min\{x_1, x_2, \cdots x_n\}$$

Obviously $L(\theta)$ is an increasing function of θ for $\theta \leq x_{(1)}$ and the maximum is obtained at $\hat{\theta} = x_{(1)}$

$\Rightarrow$ MLE is $\hat{\theta} = x_{(1)}$

Example

Expectation of $X_{(1)}$:

Recall pdf of $X_{(1)}$: $f_{X_{(1)}}(x) = n(1-F(x))^{n-1} f(x)$

$$F(x) = \int_\theta^x e^{-(w-\theta)} dw = -e^{-(w-\theta)} \Big|_\theta^x = 1 - e^{-(x-\theta)}$$

$$\Rightarrow 1 - F(x) = e^{-(x-\theta)}$$

$$\Rightarrow f_{X_{(1)}}(x) = n \cdot e^{-(n-1)(x-\theta)} e^{-(x-\theta)} = n e^{-n(x-\theta)}$$

$$\mathbb{E}[X_{(1)}] = \int_\theta^{+\infty} n x e^{-n(x-\theta)} dx = \int_\theta^{+\infty} x \, d(-e^{-n(x-\theta)})$$

$$= -x e^{-n(x-\theta)} \Big|_\theta^{+\infty} - \int_\theta^{+\infty} -e^{-n(x-\theta)} dx$$

$$= \theta + \left[-\frac{1}{n} e^{-n(x-\theta)} \Big|_\theta^{+\infty} \right] = \theta + \frac{1}{n}$$ biased.

Methods of Moments

Let $X_1, X_2, \ldots, X_n$ be a random sample from the probability distribution $f(x)$, where $f(x)$ can be a discrete probability mass function or a continuous probability density function. The kth **population moment** (or **distribution moment**) is $E(X^k)$, $k = 1, 2, \ldots$. The corresponding kth **sample moment** is $(1/n)\sum_{i=1}^{n} X_i^k$, $k = 1, 2, \ldots$.

Population moments

$$\mu_k' = \begin{cases} \int_x x^k f(x)dx & \text{If } x \text{ is continuous} \\ \sum_x x^k f(x) & \text{If } x \text{ is discrete} \end{cases}$$

Sample moments

$$m_k' = \frac{\sum_{i=1}^{n} X_i^k}{n}$$

Method of Moments

- **Equating empirical moments to theoretical moments**

Let $X_1, X_2, \ldots, X_n$ be a random sample from either a probability mass function or probability density function with m unknown parameters $\theta_1, \theta_2, \ldots, \theta_m$. The **moment estimators** $\hat{\Theta}_1, \hat{\Theta}_2, \ldots, \hat{\Theta}_m$ are found by equating the first m population moments to the first m sample moments and solving the resulting equations for the unknown parameters.

***m* equations for *m* parameters**

$$\begin{cases} m_1' = \mu_1' \\ m_2' = \mu_2' \\ \vdots \\ m_m' = \mu_m' \end{cases}$$

Example

MoM estimator for exponential parameter?

MoM estimator for normal distribution?

Definition

Two-sided vs. one-sided

CI for mean

CI for mean differences

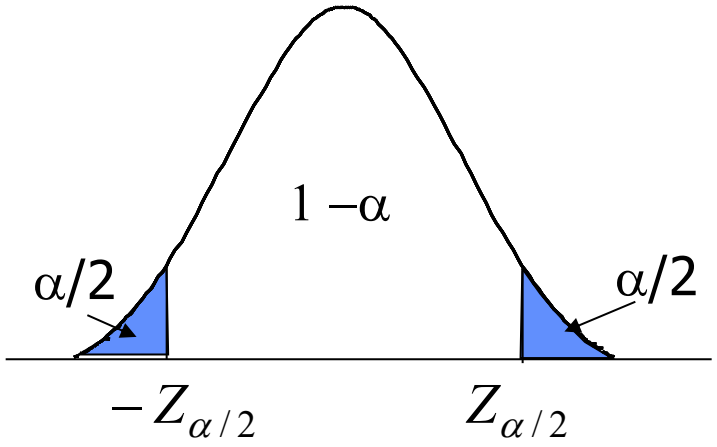
CI for proportion

CONFIDENCE INTERVAL

100(1-α)% CI for Mean of a Normal Distribution (two-sided, known variance)

- **For mean μ from a normal population with known σ,**

$$X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2) \rightarrow \bar{X} \sim N(\mu, \frac{\sigma^2}{n}) \quad \Rightarrow \quad \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$



$$P\left\{-z_{\alpha/2} \le \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \le z_{\alpha/2}\right\} = 1 - \alpha$$

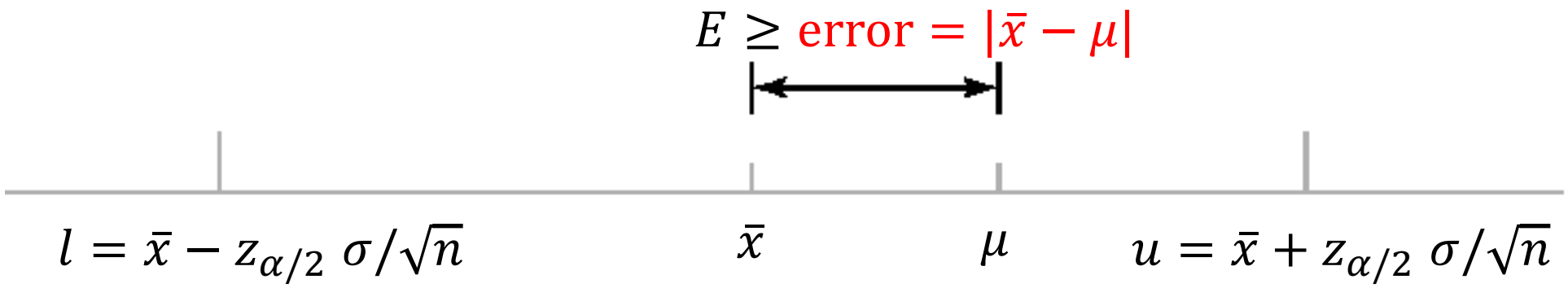
Confidence level

CI Lower CI Upper

$$\bar{x} - Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} \le \mu \le \bar{x} + Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

This CI can be used for mean of non-normal distributions (CLT)

Choice of Sample Size



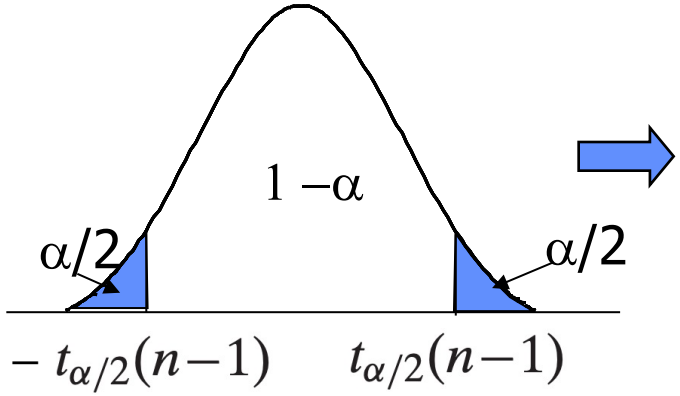
If $\bar{x}$ is used as an estimate of μ, we can be $100(1-\alpha)\%$ confident that the error $|\bar{x} - \mu|$ will not exceed a specified amount E when the sample size is

$$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2$$

width / length of the CI $= \mathbf{2}\, z_{\alpha/2}\, \sigma/\sqrt{n}$

100(1-α)% CI for Mean of a Normal Distribution (two-sided, unknown variance)

$$X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2) \rightarrow \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$



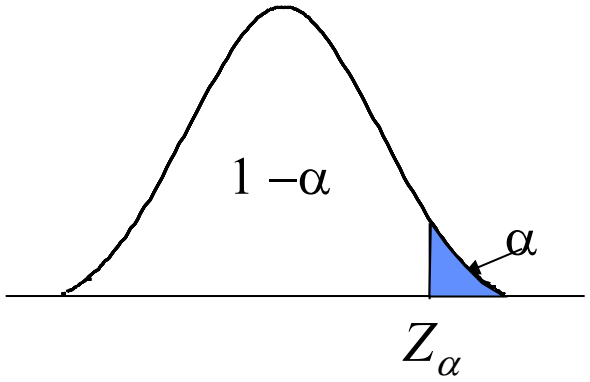
$$P\left(-t_{\alpha/2}(n-1) \leq \frac{\overline{X} - \mu}{S/\sqrt{n}} \leq t_{\alpha/2}(n-1)\right) = 1 - \alpha$$

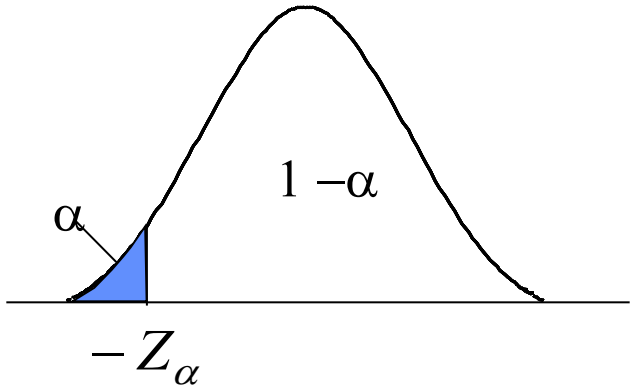
$$\overline{X} - t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right) \leq \mu \leq \overline{X} + t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right)$$

Note that $t_{\alpha/2}(n-1) \approx z_{\alpha/2}$ for *n* >= 30

This CI can be used as approximate CI for mean of non-normal distributions with unknown variance

100(1-α)% CI for Mean of a Normal Distribution (one-sided, known variance)





$$\Pr\left\{\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \le Z_\alpha\right\} = 1-\alpha$$

$$\Pr\left\{-Z_\alpha \le \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}\right\} = 1-\alpha$$

$$\mu \ge \bar{X} - Z_\alpha \times \frac{\sigma}{\sqrt{n}}$$

Lower bound

$$\mu \le \bar{X} + Z_\alpha \times \frac{\sigma}{\sqrt{n}}$$

Upper bound

This CI can be used for mean of non-normal distributions when n>30

100(1-α)% CI for Mean of a Normal Distribution (one-sided, unknown variance)

$$X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2) \rightarrow \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

- **Upper confidence bound for μ:** $\mu \leq \bar{x} + t_\alpha(n-1)(s/\sqrt{n})$

- **Lower confidence bound for μ:** $\mu \geq \bar{x} - t_\alpha(n-1)(s/\sqrt{n})$

100(1-α)% CI for Variance of a Normal Distribution

- Using chi-square, we can estimate a CI for the variance, σ^2, of a normal population.

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1} \quad \Rightarrow \quad P\left\{\chi^2_{1-\alpha/2}(n-1) \le \frac{(n-1)s^2}{\sigma^2} \le \chi^2_{\alpha/2}(n-1)\right\} = 1-\alpha$$

Two-sided CI

$$\Rightarrow \quad \frac{(n-1)s^2}{\chi^2_{\alpha/2}(n-1)} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}(n-1)}$$

One-sided Upper bound

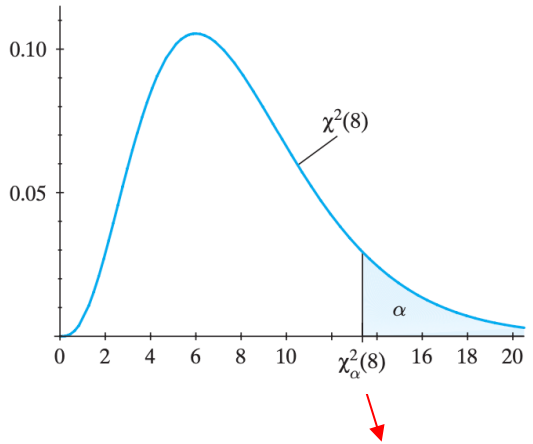
$$\sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha}(n-1)}$$

$$\left[0, \frac{(n-1)s^2}{\chi^2_{1-\alpha}(n-1)}\right]$$

One-sided Lower bound

$$\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_{\alpha}(n-1)}$$

$$\left[\frac{(n-1)s^2}{\chi^2_{\alpha}(n-1)}, +\infty\right]$$

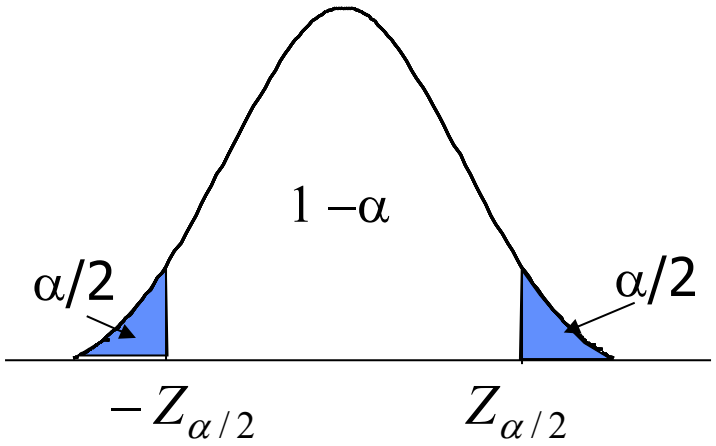


chi-squared critical Value

100(1-α)% CI for a Population Proportion with large sample size

$$X_1, X_2, \cdots, X_n \sim Bernoulli(p)$$

- Using CLT, we can estimate a CI for the population proportion.



$$\hat{p} = \frac{\sum_{i=1}^{n} X_i}{n} \sim N(p, \frac{p(1-p)}{n})$$

$$\hat{p} - Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p \le \hat{p} + Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

(Approximate) Two-sided CI

(Approximate) One-sided Lower bound

$$\hat{p} - Z_{\alpha}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p$$

$$\left[\hat{p} - Z_{\alpha}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, 1\right]$$

(Approximate) One-sided Upper bound

$$p \le \hat{p} + Z_{\alpha}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\left[0, \hat{p} + Z_{\alpha}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right]$$

Choice of Sample Size

The sample size for a specified value *E* is given by

$$n = \left(\frac{z_{\alpha/2}}{E}\right)^2 p(1-p)$$

An upper bound on *n* is given by (i.e., at least (1-α)100% confidence)

$$n = \left(\frac{z_{\alpha/2}}{E}\right)^2 0.25$$

before

If $\bar{x}$ is used as an estimate of μ, we can be $100(1-\alpha)\%$ confident that the error $|\bar{x}-\mu|$ will not exceed a specified amount E when the sample size is

$$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2$$

Confidence Intervals for Two-Samples

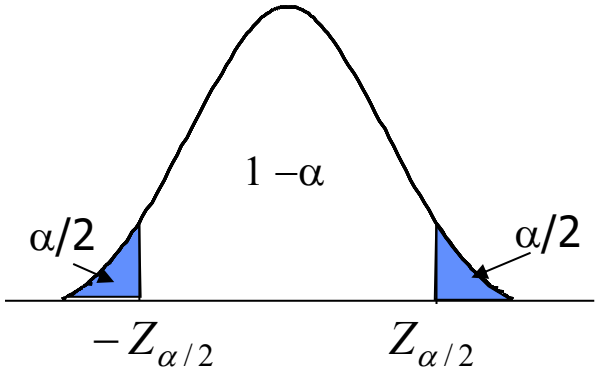
100(1-α)% CI for Differences in Means of Two Normal Distributions (two-sided, known variances)

Two independent random samples from two Normal distributions with the known variances

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$$

$$Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$$

$$\Rightarrow \bar{X} - \bar{Y} \sim N\left(\mu_X - \mu_Y, \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}\right) \Rightarrow \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}} \sim N(0,1)$$



$$P\left\{-z_{\alpha/2} \leq \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}} \leq z_{\alpha/2}\right\} = 1 - \alpha$$

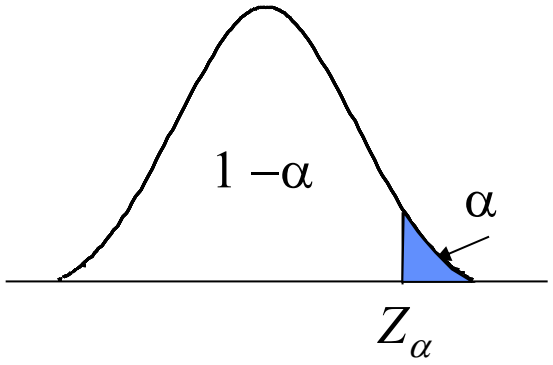
← **Confidence level**

CI Lower **CI Upper**

$$\bar{X} - \bar{Y} - z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}} \leq \mu_X - \mu_Y \leq \bar{X} - \bar{Y} + z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$$

This CI can be used for mean of non-normal distributions when *n,m*>30

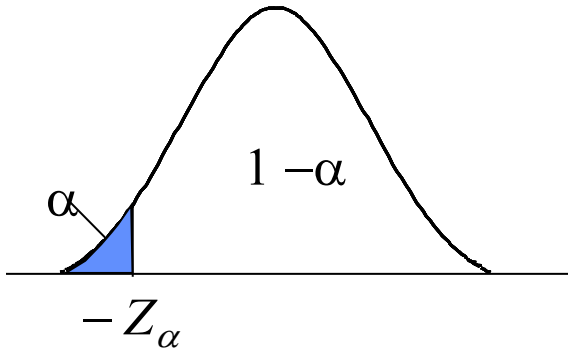
100(1-α)% CI for Differences in Means of Two Normal Distributions (one-sided, known variances)



$$P\left\{\frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}} \le z_\alpha\right\} = 1-\alpha$$

$$\bar{X}-\bar{Y}-z_\alpha\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}} \le \mu_X-\mu_Y$$

Lower bound



$$P\left\{\frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}} \ge -z_\alpha\right\} = 1-\alpha$$

$$\mu_X-\mu_Y \le \bar{X}-\bar{Y}+z_\alpha\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}$$

Upper bound

This CI can be used for mean of non-normal distributions when n,m>30

100(1-α)% CI for Difference in Means of Two Normal Distributions (two-sided, unknown but equal variances)

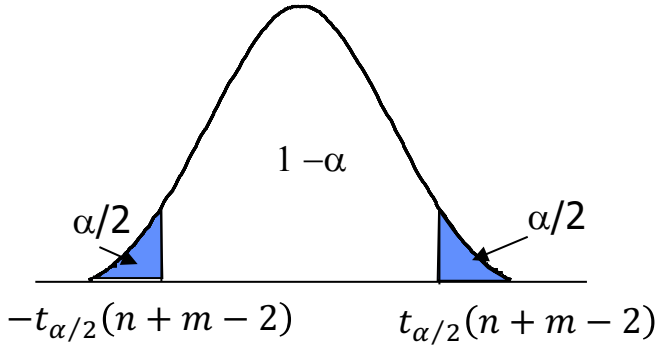
Two independent random samples from two Normal distributions with unknown (but equal) variances

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma^2)$$

$$Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma^2)$$

$$\Rightarrow \quad \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim t(n + m - 2)$$

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n + m - 2}$$



$$\Rightarrow \quad P\left\{-t_{\alpha/2}(n+m-2) \le \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \le t_{\alpha/2}(n+m-2)\right\} = 1 - \alpha$$

$$\bar{X} - \bar{Y} - t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}} \le \mu_X - \mu_Y \le \bar{X} - \bar{Y} + t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$$

Two-sided

100(1-α)% CI for Difference in Means of Two Normal Distributions (one-sided, unknown but equal variances)

Two independent random samples from two Normal distributions with unknown (but equal) variances

$$X_1, \dots, X_n \sim N(\mu_X, \sigma^2)$$

$$Y_1, \dots, Y_m \sim N(\mu_Y, \sigma^2)$$

$$\Rightarrow \quad \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim t(n + m - 2)$$

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n + m - 2}$$

$$\Rightarrow \quad \mu_X - \mu_Y \geq \bar{X} - \bar{Y} - t_\alpha(n + m - 2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$$

Lower bound

$$\mu_X - \mu_Y \leq \bar{X} - \bar{Y} + t_\alpha(n + m - 2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$$

Upper bound

100(1-α)% CI for Difference in Means of Two Normal Distributions (unknown and unequal variances)

Two independent random samples from two Normal distributions with unknown (and unequal) variances

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$$
$$Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$$

$$\Rightarrow \quad \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}} \begin{cases} \sim N(0,1) & \text{If } n \text{ and } m \text{ are large enough and the underlying distributions are close to normal} \\ \sim t(r) & \text{for smaller } n \text{ and } m \end{cases}$$

Two-sided **Welch's t-interval**

$$\bar{X} - \bar{Y} - t_{\alpha/2}(r)\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}} \leq \mu_X - \mu_Y \leq \bar{X} - \bar{Y} + t_{\alpha/2}(r)\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}$$

$$r = \left\lfloor \frac{\left(\frac{S_X^2}{n} + \frac{S_Y^2}{m}\right)^2}{\frac{1}{n-1}\left(\frac{S_X^2}{n}\right)^2 + \frac{1}{m-1}\left(\frac{S_Y^2}{m}\right)^2} \right\rfloor$$

$$\mu_X - \mu_Y \geq \bar{X} - \bar{Y} - t_{\alpha}(r)\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}$$

Lower bound

$$\mu_X - \mu_Y \leq \bar{X} - \bar{Y} + t_{\alpha}(r)\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}$$

Upper bound

$\lfloor x \rfloor$ *is the floor function of* x

$$\lfloor 3.4 \rfloor = \lfloor 3.9 \rfloor = \lfloor 3 \rfloor = 3$$

100(1-α)% CI for Difference in Means of Two Normal Distributions (paired t-interval)

n independent pairs of (X, Y), but X_i, Y_i for each given *i-th* pair are dependent.

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2) \qquad Y_1, \ldots, Y_n \sim N(\mu_Y, \sigma_Y^2)$$

$$(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n) \qquad n \text{ independent pairs}$$

$$D_i = X_i - Y_i \qquad i = 1, \ldots, n.$$

$$\Rightarrow \quad \frac{\overline{D} - (\mu_X - \mu_Y)}{S_D/\sqrt{n}} \sim t(n-1) \quad \Rightarrow \quad P\left\{\overline{D} - t_{\alpha/2}(n-1)\frac{S_D}{\sqrt{n}} \le \mu_X - \mu_Y \le \overline{D} + t_{\alpha/2}(n-1)\frac{S_D}{\sqrt{n}}\right\} = 1 - \alpha$$

Two-sided

$$\overline{D} - t_{\alpha/2}(n-1)\frac{S_D}{\sqrt{n}} \le \mu_X - \mu_Y \le \overline{D} + t_{\alpha/2}(n-1)\frac{S_D}{\sqrt{n}}$$

$$\mu_X - \mu_Y \ge \overline{D} - t_{\alpha}(n-1)\frac{S_D}{\sqrt{n}}$$

Lower bound

$$\mu_X - \mu_Y \le \overline{D} + t_{\alpha}(n-1)\frac{S_D}{\sqrt{n}}$$

Upper bound

100(1-α)% CI for Variances Ratio of Two Normal Distributions

Two independent random samples from two Normal distributions

$$X_1, \dots, X_n \sim N(\mu_X, \sigma_X^2)$$

$$Y_1, \dots, Y_m \sim N(\mu_Y, \sigma_Y^2)$$

$$\Rightarrow \quad \frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F(n-1, m-1)$$

$$F_{1-\alpha}(r_1, r_2) = \frac{1}{F_\alpha(r_2, r_1)}$$

$$\frac{S_X^2}{S_Y^2} F_{1-\alpha/2}(m-1, n-1) \le \frac{\sigma_X^2}{\sigma_Y^2} \le \frac{S_X^2}{S_Y^2} F_{\alpha/2}(m-1, n-1)$$

Two-sided

$$\frac{\sigma_X^2}{\sigma_Y^2} \ge \frac{S_X^2}{S_Y^2} F_{1-\alpha}(m-1, n-1)$$

Lower bound

$$\left[\frac{S_X^2}{S_Y^2} F_{1-\alpha}(m-1, n-1), +\infty\right]$$

One-sided

$$\frac{\sigma_X^2}{\sigma_Y^2} \le \frac{S_X^2}{S_Y^2} F_{\alpha}(m-1, n-1)$$

Upper bound

$$\left[0, \frac{S_X^2}{S_Y^2} F_{\alpha}(m-1, n-1)\right]$$

100(1-α)% CI for Difference in Population Proportions with large sample size

$$\left.\begin{array}{l} X_1, X_2, \cdots, X_{n_1} \sim Bernoulli(p_1) \\ Y_1, Y_2, \cdots, Y_{n_2} \sim Bernoulli(p_2) \end{array}\right\} \rightarrow \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}} \sim N(0,1)$$ For large sample sizes

Two-sided CI

$$(\hat{p}_1 - \hat{p}_2) - Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \le (p_1 - p_2) \le (\hat{p}_1 - \hat{p}_2) + Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

One-sided CI

Lower bound

$$(\hat{p}_1 - \hat{p}_2) - Z_{\alpha}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \le (p_1 - p_2)$$

$$\left[\hat{p}_1 - \hat{p}_2 - Z_{\alpha}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}, \mathbf{1}\right]$$

Upper bound

$$(p_1 - p_2) \le (\hat{p}_1 - \hat{p}_2) + Z_{\alpha}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$$\left[-\mathbf{1}, \hat{p}_1 - \hat{p}_2 + Z_{\alpha}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}\right]$$

Recap:Summary of Sampling Distributions

		One population	Two populations
Sample Mean	Known Variance	$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$	$\frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}} \sim N(0,1)$
	Unknown Variance	$\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1)$	$\frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{S_p\sqrt{\frac{1}{n}+\frac{1}{m}}} \sim t(n+m-2)$
Sample proportion		$\hat{p} \sim N(p, p(1-p)/n)$	$\frac{(\hat{p}_1-\hat{p}_2)-(p_1-p_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}} \sim N(0,1)$
Sample Variance		$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$	$\frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F(n-1, m-1)$

Summary

		One population	Two populations
Sample Mean	Known Variance	$\bar{x} - z_{\alpha/2}\sigma/\sqrt{n} \le \mu \le \bar{x} + z_{\alpha/2}\sigma/\sqrt{n}$ $\mu \ge \bar{X} - Z_\alpha \times \frac{\sigma}{\sqrt{n}}$ $\quad$ $\mu \le \bar{X} + Z_\alpha \times \frac{\sigma}{\sqrt{n}}$	$\bar{X} - \bar{Y} - z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}} \le \mu_X - \mu_Y \le \bar{X} - \bar{Y} + z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$ $\bar{X} - \bar{Y} - z_\alpha\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}} \le \mu_X - \mu_Y$ $\quad$ $\mu_X - \mu_Y \le \bar{X} - \bar{Y} + z_\alpha\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$
	Unknown Variance	$\bar{X} - t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right) \le \mu \le \bar{X} + t_{\alpha/2}(n-1)\left(\frac{S}{\sqrt{n}}\right)$ $\mu \le \bar{x} + t_\alpha(n-1)(s/\sqrt{n})$ $\mu \ge \bar{x} - t_\alpha(n-1)(s/\sqrt{n})$	$\bar{X} - \bar{Y} - t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}} \le \mu_X - \mu_Y \le \bar{X} - \bar{Y} + t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$ $\mu_X - \mu_Y \ge \bar{X} - \bar{Y} - t_\alpha(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$ $\mu_X - \mu_Y \le \bar{X} - \bar{Y} + t_\alpha(n+m-2)S_p\sqrt{\frac{1}{n} + \frac{1}{m}}$
Sample proportion		$\hat{p} - Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p \le \hat{p} + Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ $\hat{p} - Z_\alpha\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p$ $\quad$ $p \le \hat{p} + Z_\alpha\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$(\hat{p}_1 - \hat{p}_2) - Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \le (p_1 - p_2) \le (\hat{p}_1 - \hat{p}_2) + Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$ $(\hat{p}_1 - \hat{p}_2) - Z_\alpha\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \le (p_1 - p_2)$ $\quad$ $(p_1 - p_2) \le (\hat{p}_1 - \hat{p}_2) + Z_\alpha\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$
Sample Variance		$\frac{(n-1)s^2}{\chi^2_{\alpha/2}(n-1)} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}(n-1)}$ $\sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha}(n-1)}$ $\quad$ $\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_\alpha(n-1)}$	$\frac{S_X^2}{S_Y^2}F_{1-\alpha/2}(m-1,n-1) \le \frac{\sigma_X^2}{\sigma_Y^2} \le \frac{S_X^2}{S_Y^2}F_{\alpha/2}(m-1,n-1)$ $\frac{\sigma_X^2}{\sigma_Y^2} \ge \frac{S_X^2}{S_Y^2}F_{1-\alpha}(m-1,n-1)$ $\quad$ $\frac{\sigma_X^2}{\sigma_Y^2} \le \frac{S_X^2}{S_Y^2}F_\alpha(m-1,n-1)$

3 ways to perform hypothesis test

1) Rejection region
2) P-value
3) Use confidence interval

Hypothesis testing for normal mean, variance

Hypothesis testing for two samples, equality of two means, variances

Hypothesis testing for proportion

HYPOTHESIS TESTING

Hypothesis Testing Procedures

1. **Parameter of interest:** From the problem context, identify the parameter of interest
2. **Null hypothesis H_0:** State the null hypothesis – initial belief/claim/assumption
3. **Alternative hypothesis H_1:** Specify an appropriate alternative hypothesis
4. **Test statistic:** Determine an appropriate test statistic
5. **Reject H_0 if :** State the rejection criteria for the null hypothesis
6. **Computation:** Compute any necessary sample quantities, substitute these into the equation for the test statistic, and compute that value
7. **Draw conclusion:** Decide whether or not H_0 should be rejected and report that in the problem context
8. **Report p-values**

- **Three ways to perform hypothesis test**
  - **Rejection region**
  - **Compare p-value with α**
  - **Using confidence interval**

Errors in Hypothesis Test

Type I Error

Rejecting the null hypothesis H_0 when it is true is defined as a **type I error.**

Probability of Type I Error

$\alpha = P(\text{type I error}) = P(\text{reject } H_0 \text{ when } H_0 \text{ is true})$

Type II Error

Failing to reject the null hypothesis when it is false is defined as a **type II error.**

Probability of Type II Error

$\beta = P(\text{type II error}) = P(\text{fail to reject } H_0 \text{ when } H_0 \text{ is false})$

Decision	H_0 Is True	H_0 Is False
Fail to reject H_0	no error	type II error
Reject H_0	type I error	no error

Significance Level

- Generally, the statistician controls type-I error probability α when she selects the critical values
- Set type-I error probability at a desired level called significance level
- Deal with the corresponding type-II error β resulted from this

Typical value for significance level: 0.05

p-value

- The p-value is the probability that the test statistic will take on a value that is at least as extreme as the observed value of the statistic when the null hypothesis H_0 is true.
  - Given a significance level α
  - If the p-value is smaller than α $\Rightarrow$ Reject H_0
  - If the p-value is larger than α $\Rightarrow$

    there is not enough evidence to reject H_0

Hypothesis testing for normal mean

Inference on the Mean of a Normal Population – Known Variance

Null Hypothesis $H_0 : \mu = \mu_0$

Test Statistic $Z_0 = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$

Distribution under H_0 $Z_0 \sim N(0,1)$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1 : \mu \neq \mu_0$	$\lvert Z_0 \rvert > Z_{\alpha/2}$	$2\mathrm{P}(Z > \lvert Z_0 \rvert)$	$[\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}]$
$H_1 : \mu > \mu_0$	$Z_0 > Z_\alpha$	$\mathrm{P}(Z > Z_0)$	$[\bar{X} - z_\alpha\frac{\sigma}{\sqrt{n}}, +\infty)$
$H_1 : \mu < \mu_0$	$Z_0 < -Z_\alpha$	$\mathrm{P}(Z < Z_0)$	$(-\infty, \bar{X} + z_\alpha\frac{\sigma}{\sqrt{n}}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include μ_0

Inference on the Mean of a Normal Population – Unknown Variance

Null Hypothesis $H_0 : \mu = \mu_0$

Test Statistic $t_0 = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}}$

Distribution under H_0 $t_0 \sim t(n-1)$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1 : \mu \neq \mu_0$	$\lvert t_0 \rvert > t_{\alpha/2}(n-1)$	$2\mathrm{P}(T > \lvert t_0 \rvert)$	$[\bar{X} - t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}}]$
$H_1 : \mu > \mu_0$	$t_0 > t_{\alpha}(n-1)$	$\mathrm{P}(T > t_0)$	$[\bar{X} - t_{\alpha}(n-1)\frac{S}{\sqrt{n}}, +\infty)$
$H_1 : \mu < \mu_0$	$t_0 < -t_{\alpha}(n-1)$	$\mathrm{P}(T < t_0)$	$(-\infty, \bar{X} + t_{\alpha}(n-1)\frac{S}{\sqrt{n}}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include μ_0

Hypothesis testing for normal variance

Inference on the Variance of a Normal Population

Null Hypothesis $H_0 : \sigma^2 = \sigma_0^2$

Test Statistic $\chi_0^2 = \dfrac{(n-1)S^2}{\sigma_0^2}$

Distribution under H0 $\chi_0^2 \sim \chi^2(n-1)$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1 : \sigma^2 \neq \sigma_0^2$	$\chi_0^2 > \chi^2_{\alpha/2}(n-1)$ *or* $\chi_0^2 < \chi^2_{1-\alpha/2}(n-1)$	Approximation $2\min\{P(\chi^2 > \chi_0^2), P(\chi^2 < \chi_0^2)\}$	$[\dfrac{(n-1)s^2}{\chi^2_{\alpha/2}(n-1)}, \dfrac{(n-1)s^2}{\chi^2_{1-\alpha/2}(n-1)}]$
$H_1 : \sigma^2 > \sigma_0^2$	$\chi_0^2 > \chi^2_{\alpha}(n-1)$	$P(\chi^2 > \chi_0^2)$	$[\dfrac{(n-1)s^2}{\chi^2_{\alpha}(n-1)}, +\infty)$
$H_1 : \sigma^2 < \sigma_0^2$	$\chi_0^2 < \chi^2_{1-\alpha}(n-1)$	$P(\chi^2 < \chi_0^2)$	$[0, \dfrac{(n-1)s^2}{\chi^2_{1-\alpha}(n-1)}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include σ_0^2

Hypothesis testing for two-samples

Inference for Differences in Means of Two Normal Distributions (known variances)

Two independent random samples from two Normal distributions

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$$
$$Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$$

$$\Rightarrow \bar{X} - \bar{Y} \sim N\left(\mu_X - \mu_Y, \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}\right) \Rightarrow \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}} \sim N(0,1)$$

Null Hypothesis

$$H_0: \mu_X - \mu_Y = \Delta_0$$

e.g., $\Delta_0 = 0$ $(\mathrm{H}_0: \mu_X = \mu_Y)$

Test Statistic

$$Z_0 = \frac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}$$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

Inference for Differences in Means of Two Normal Distributions (known variances)

Null Hypothesis

$$H_0: \mu_X - \mu_Y = \Delta_0$$

Test Statistic

$$Z_0 = \frac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}$$

Distribution under H0

$$Z_0 \sim N(0,1)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1: \mu_X - \mu_Y \neq \Delta_0$	$\lvert Z_0 \rvert > Z_{\alpha/2}$	$2\mathrm{P}(Z > \lvert Z_0 \rvert)$	$[\bar{X} - \bar{Y} - z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}, \bar{X} - \bar{Y} + z_{\alpha/2}\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}]$
$H_1: \mu_X - \mu_Y > \Delta_0$	$Z_0 > Z_\alpha$	$\mathrm{P}(Z > Z_0)$	$[\bar{X} - \bar{Y} - z_\alpha\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}, +\infty)$
$H_1: \mu_X - \mu_Y < \Delta_0$	$Z_0 < -Z_\alpha$	$\mathrm{P}(Z < Z_0)$	$(-\infty, \bar{X} - \bar{Y} + z_\alpha\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include Δ_0

Inference Difference in Means of Two Normal Distributions (unknown and equal variances)

pooled t-test

Two independent random samples from two Normal distributions

$$X_1, \dots, X_n \sim N(\mu_X, \sigma^2)$$

$$Y_1, \dots, Y_m \sim N(\mu_Y, \sigma^2)$$

$$\Rightarrow \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim t(n + m - 2)$$

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n + m - 2}$$

Null Hypothesis

$$H_0: \mu_X - \mu_Y = \Delta_0$$

e.g., $\Delta_0 = 0$ $(H_0: \mu_X = \mu_Y)$

Test Statistic

$$t_0 = \frac{\bar{X} - \bar{Y} - \Delta_0}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}}$$

Distribution under H_0

$$t_0 \sim t(n + m - 2)$$

Inference Difference in Means of Two Normal Distributions (unknown and equal variances)

pooled t-test

Null Hypothesis

$$H_0: \mu_X - \mu_Y = \Delta_0$$

Test Statistic

$$t_0 = \frac{\bar{X} - \bar{Y} - \Delta_0}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}}$$

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}$$

Distribution under H0

$$t_0 \sim t(n+m-2)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1: \mu_X - \mu_Y \neq \Delta_0$	$\lvert t_0 \rvert > t_{\alpha/2}(n+m-2)$	$2\mathrm{P}(T > \lvert t_0 \rvert)$	$[\bar{X} - \bar{Y} - t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n}+\frac{1}{m}}, \bar{X} - \bar{Y} + t_{\alpha/2}(n+m-2)S_p\sqrt{\frac{1}{n}+\frac{1}{m}}]$
$H_1: \mu_X - \mu_Y > \Delta_0$	$t_0 > t_\alpha(n+m-2)$	$\mathrm{P}(T > t_0)$	$[\bar{X} - \bar{Y} - t_\alpha(n+m-2)S_p\sqrt{\frac{1}{n}+\frac{1}{m}}, +\infty)$
$H_1: \mu_X - \mu_Y < \Delta_0$	$t_0 < -t_\alpha(n+m-2)$	$\mathrm{P}(T < t_0)$	$(-\infty, \bar{X} - \bar{Y} + t_\alpha(n+m-2)S_p\sqrt{\frac{1}{n}+\frac{1}{m}}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include Δ_0

Inference Difference in Means of Two Normal Distributions (unknown and unequal variances)

Welch's t-test

Two independent random samples from two Normal distributions

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$$

$$Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$$

$$\Rightarrow \quad \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}} \sim t(r)$$

$$r = \left\lfloor \frac{\left(\frac{S_X^2}{n} + \frac{S_Y^2}{m}\right)^2}{\frac{1}{n-1}\left(\frac{S_X^2}{n}\right)^2 + \frac{1}{m-1}\left(\frac{S_Y^2}{m}\right)^2} \right\rfloor$$

Null Hypothesis

$$H_0: \mu_X - \mu_Y = \Delta_0$$

e.g., $\Delta_0 = 0$ $(\mathrm{H}_0: \mu_X = \mu_Y)$

Test Statistic

$$t_0 = \frac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}}$$

Distribution under H_0

$$t_0 \sim t(r)$$

Inference Difference in Means of Two Normal Distributions (unknown and unequal variances)

Welch's t-test

Null Hypothesis

$$H_0: \mu_X - \mu_Y = \Delta_0$$

Test Statistic

$$t_0 = \frac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}}$$

$$r = \left\lfloor \frac{\left(\frac{S_X^2}{n} + \frac{S_Y^2}{m}\right)^2}{\frac{1}{n-1}\left(\frac{S_X^2}{n}\right)^2 + \frac{1}{m-1}\left(\frac{S_Y^2}{m}\right)^2} \right\rfloor$$

Distribution under H0

$$t_0 \sim t(r)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1: \mu_X - \mu_Y \neq \Delta_0$	$\lvert t_0 \rvert > t_{\alpha/2}(r)$	$2\mathrm{P}(T > \lvert t_0 \rvert)$	$[\bar{X} - \bar{Y} - t_{\alpha/2}(r)\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}, \bar{X} - \bar{Y} + t_{\alpha/2}(r)\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}]$
$H_1: \mu_X - \mu_Y > \Delta_0$	$t_0 > t_\alpha(r)$	$\mathrm{P}(T > t_0)$	$[\bar{X} - \bar{Y} - t_\alpha(r)\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}, +\infty)$
$H_1: \mu_X - \mu_Y < \Delta_0$	$t_0 < -t_\alpha(r)$	$\mathrm{P}(T < t_0)$	$(-\infty, \bar{X} - \bar{Y} + t_\alpha(r)\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include Δ_0

Inference Difference in Means of Two Normal Distributions (paired t-test)

n independent pairs of (X, Y), but X_i, Y_i for each given *i-th* pair are dependent.

$$X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2) \qquad Y_1, \ldots, Y_n \sim N(\mu_Y, \sigma_Y^2)$$

$$(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n) \quad n \text{ independent pairs}$$

$$D_i = X_i - Y_i \qquad i = 1, \ldots, n. \quad \Rightarrow \quad \frac{\overline{D} - (\mu_X - \mu_Y)}{S_D/\sqrt{n}} \sim t(n-1)$$

Null Hypothesis

$$H_0: \mu_X - \mu_Y = \Delta_0 \ (\mu_D = \Delta_0)$$

e.g., $\Delta_0 = 0$ $(\mathrm{H}_0: \mu_X = \mu_Y)$

Test Statistic

$$t_0 = \frac{\overline{D} - \Delta_0}{S_D/\sqrt{n}}$$

Distribution under H_0

$$t_0 \sim t(n-1)$$

Inference Difference in Means of Two Normal Distributions (paired t-test)

Null Hypothesis

$$H_0: \mu_X - \mu_Y = \Delta_0$$

Test Statistic

$$t_0 = \frac{\bar{D} - \Delta_0}{S_D/\sqrt{n}}$$

Distribution under H0

$$t_0 \sim t(n-1)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1: \mu_X - \mu_Y \neq \Delta_0$	$\lvert t_0 \rvert > t_{\alpha/2}(n-1)$	$2\mathrm{P}(T > \lvert t_0 \rvert)$	$[\bar{D} - t_{\alpha/2}(n-1)\frac{S_D}{\sqrt{n}}, \bar{D} + t_{\alpha/2}(n-1)\frac{S_D}{\sqrt{n}}]$
$H_1: \mu_X - \mu_Y > \Delta_0$	$t_0 > t_\alpha(n-1)$	$\mathrm{P}(T > t_0)$	$[\bar{D} - t_\alpha(n-1)\frac{S_D}{\sqrt{n}}, +\infty)$
$H_1: \mu_X - \mu_Y < \Delta_0$	$t_0 < -t_\alpha(n-1)$	$\mathrm{P}(T < t_0)$	$(-\infty, \bar{D} + t_\alpha(n-1)\frac{S_D}{\sqrt{n}}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include Δ_0

Inference on the Variance of Two Normal Populations

Two independent random samples from two Normal distributions

$$X_1, \dots, X_n \sim N(\mu_X, \sigma_X^2)$$

$$Y_1, \dots, Y_m \sim N(\mu_Y, \sigma_Y^2)$$

$$\Rightarrow \quad \frac{S_X^2 / \sigma_X^2}{S_Y^2 / \sigma_Y^2} \sim F(n-1, m-1)$$

$$F_{1-\alpha}(r_1, r_2) = \frac{1}{F_\alpha(r_2, r_1)}$$

Null Hypothesis

$$H_0: \sigma_X^2 = \sigma_Y^2$$

Test Statistic

$$F_0 = \frac{S_X^2}{S_Y^2}$$

Distribution under H_0

$$F_0 \sim F(n-1, m-1)$$

Inference on the Variance of Two Normal Populations

Null Hypothesis $H_0: \sigma_X^2 = \sigma_Y^2$

Test Statistic $F_0 = \frac{S_X^2}{S_Y^2}$

Distribution under H0 $F_0 \sim F(n-1, m-1)$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach (CI for $\frac{\sigma_X^2}{\sigma_Y^2}$)
$H_1: \sigma_X^2 \neq \sigma_Y^2$	$F_0 > F_{\alpha/2}(n-1, m-1) or$ $F_0 < F_{1-\alpha/2}(n-1, m-1)$	Approximation $2\min\{P(F > F_0), P(F < F_0)\}$	$[\frac{S_X^2}{S_Y^2} F_{1-\alpha/2}(m-1, n-1), \frac{S_X^2}{S_Y^2} F_{\alpha/2}(m-1, n-1)]$
$H_1: \sigma_X^2 > \sigma_Y^2$	$F_0 > F_{\alpha}(n-1, m-1)$	$P(F > F_0)$	$[\frac{S_X^2}{S_Y^2} F_{1-\alpha}(m-1, n-1), +\infty)$
$H_1: \sigma_X^2 < \sigma_Y^2$	$F_0 < F_{1-\alpha}(n-1, m-1)$	$P(F < F_0)$	$[0, \frac{S_X^2}{S_Y^2} F_{\alpha}(m-1, n-1)]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include 1

Inference on a Population Proportion

$$X_1, X_2, \cdots, X_n \sim Bernoulli(p)$$

- Using CLT, we can estimate the distribution of the sample proportion.

$$\hat{p} = \frac{\sum_{i=1}^{n} X_i}{n} \sim N(p, \frac{p(1-p)}{n})$$

Null Hypothesis

$$H_0 : p = p_0$$

Test Statistic

$$Z_0 = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Asymptotic Distribution under H_0

$$Z_0 \sim N(0,1)$$

Inference on a Population Proportion

Null Hypothesis

$$H_0 : p = p_0$$

Test Statistic

$$Z_0 = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Asymptotic Distribution under H0

$$Z_0 \sim N(0,1)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1 : p \neq p_0$	$\lvert Z_0 \rvert > Z_{\alpha/2}$	$2\mathrm{P}(Z > \lvert Z_0 \rvert)$	$[\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}]$
$H_1 : p > p_0$	$Z_0 > Z_\alpha$	$\mathrm{P}(Z > Z_0)$	$[\hat{p} - z_\alpha\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, 1]$
$H_1 : p < p_0$	$Z_0 < -Z_\alpha$	$\mathrm{P}(Z < Z_0)$	$[0, \hat{p} + z_\alpha\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include p_0

Hypothesis testing for equality of two proportions

Inference on equality of two Population Proportions

$$\left.\begin{array}{l} X_1, X_2, \cdots, X_{n_1} \sim Bernoulli(p_1) \\ Y_1, Y_2, \cdots, Y_{n_2} \sim Bernoulli(p_2) \end{array}\right\} \rightarrow \frac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}} \sim N(0,1)$$

For large sample sizes (CLT)

Null Hypothesis

$$H_0: p_1 = p_2$$

Test Statistic

$$Z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

$$\hat{p} = \frac{\sum_{i=1}^{n_1} X_i + \sum_{i=1}^{n_2} Y_i}{n_1 + n_2}$$

Estimate of $p = p_1 = p_2$

Inference on equality of two Population Proportions

Null Hypothesis

$$H_0: p_1 = p_2$$

Test Statistic

$$Z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} \qquad \hat{p} = \frac{\sum_{i=1}^{n_1} X_i + \sum_{i=1}^{n_2} Y_i}{n_1 + n_2}$$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1: p_1 \neq p_2$	$\lvert Z_0 \rvert > Z_{\alpha/2}$	$2\mathrm{P}(Z > \lvert Z_0 \rvert)$	$[\hat{p}_1 - \hat{p}_2 - z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}, \hat{p}_1 - \hat{p}_2 + z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}]$
$H_1: p_1 > p_2$	$Z_0 > Z_\alpha$	$\mathrm{P}(Z > Z_0)$	$[\hat{p}_1 - \hat{p}_2 - z_\alpha\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}, 1]$
$H_1: p_1 < p_2$	$Z_0 < -Z_\alpha$	$\mathrm{P}(Z < Z_0)$	$[-1, \hat{p}_1 - \hat{p}_2 + z_\alpha\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}]$

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include 0

Power for testing normal mean

Testing normal mean
(Known Variance)

Data samples

$$X_1, \ldots, X_n \sim N(\mu, \sigma^2)$$

Null Hypothesis

$$H_0 : \mu = \mu_0$$

Alternative Hypothesis

$$H_1 : \mu \neq \mu_0 \quad \text{two-sided}$$

$$H_1 : \mu > \mu_0$$

$$H_1 : \mu < \mu_0 \quad \text{one-sided}$$

Test Statistic

$$Z_0 = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

Critical region

$$\mathbf{C:} \quad |Z_0| > Z_{\alpha/2}$$

$$Z_0 > Z_\alpha$$

$$Z_0 < -Z_\alpha$$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

two-sided **Power function** $\boldsymbol{K(\mu_1) = 1 - \beta = P(Z_0 \in C; \mu_1) = P(|Z_0| > Z_{\alpha/2}; \mu_1)}$

Under $\mu = \mu_1 > \mu_0$ **Power function** $\boldsymbol{K(\mu_1) = 1 - \beta = P(Z_0 \in C; \mu_1) = P(Z_0 > Z_\alpha; \mu_1)}$

Under $\mu = \mu_1 < \mu_0$ **Power function** $\boldsymbol{K(\mu_1) = 1 - \beta = P(Z_0 \in C; \mu_1) = P(Z_0 < -Z_\alpha; \mu_1)}$

Power function, denoted by $K(\mu)$, is a function of the parameter μ.

Testing normal mean

Factors Affecting the Power

(1) If the significance level is made smaller (α decreases), z_α increases and hence the power decreases.

(2) If the alternative mean is shifted farther away from the null mean ($|\mu_0 - \mu_1|$ increases), then the power increases.

(3) If the standard deviation of the distribution of individual observations increases (σ increases), then the power decreases.

(4) If the sample size increases (n increases), then the power increases.

$$\Phi\left(-Z_{\alpha/2} + \frac{|\mu_1 - \mu_0|}{\sigma/\sqrt{n}}\right) + \Phi\left(-Z_{\alpha/2} - \frac{|\mu_1 - \mu_0|}{\sigma/\sqrt{n}}\right)$$

two-sided

$$\Phi\left(-Z_{\alpha} + \frac{|\mu_0 - \mu_1|}{\sigma/\sqrt{n}}\right)$$

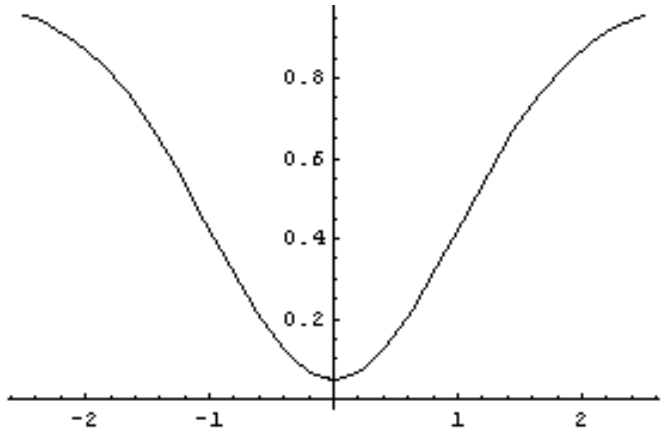
one-sided

Example:
power function for the test

$$H_0 : \mu = 0$$

$$H_1 : \mu \neq 0$$

P(reject H_0)



Sample Size: Testing normal mean
(Known Variance)

A test will be conducted at level α

The true alternative mean is assumed to be μ_1

$$H_0 : \mu = \mu_0$$

$$H_1 : \mu \neq \mu_0 \quad \text{two-sided}$$

$$H_1 : \mu > \mu_0 \quad \text{one-sided}$$

$$H_1 : \mu < \mu_0 \quad \text{one-sided}$$

What sample size is needed to be able to detect a significant difference (reject H_0) with probability 1-β ?

one-sided

$$K(\mu_1) = \Phi\left(-Z_\alpha + \frac{|\mu_0 - \mu_1|}{\sigma/\sqrt{n}}\right) = 1 - \beta$$

$$n = \frac{\left(Z_\beta + Z_\alpha\right)^2 \sigma^2}{(\mu_1 - \mu_0)^2}$$

two-sided

$$1 - \beta = K(\mu_1) \approx \Phi\left(-Z_{\alpha/2} + \frac{|\mu_1 - \mu_0|}{\sigma/\sqrt{n}}\right)$$

$$n \approx \frac{\left(Z_\beta + Z_{\alpha/2}\right)^2 \sigma^2}{(\mu_1 - \mu_0)^2}$$

Testing normal mean

Null Hypothesis	Test Statistic	Distribution under H_0
$H_0 : \mu = \mu_0$	$Z_0 = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$	$Z_0 \sim N(0,1)$

Alternative Hypothesis	Critical Region (H_0 is rejected)	Power under $\mu = \mu_1$	Sample size needed for $K(\mu_1) = 1 - \beta; K(\mu_0) = \alpha$
$H_1 : \mu \neq \mu_0$	$\lvert Z_0 \rvert > Z_{\alpha/2}$	$\Phi\left(-Z_{\alpha/2} + \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right) + \Phi\left(-Z_{\alpha/2} - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right)$	$n = \dfrac{(Z_\beta + Z_{\alpha/2})^2 \sigma^2}{(\mu_1 - \mu_0)^2}$
$H_1 : \mu > \mu_0$	$Z_0 > Z_\alpha$	$\Phi\left(-Z_\alpha + \dfrac{\lvert\mu_0 - \mu_1\rvert}{\sigma/\sqrt{n}}\right)$	$n = \dfrac{(Z_\beta + Z_\alpha)^2 \sigma^2}{(\mu_1 - \mu_0)^2}$
$H_1 : \mu < \mu_0$	$Z_0 < -Z_\alpha$		

Sample size

Factors Affecting the Sample Size

(1) The sample size increases as σ^2 increases.

(2) The sample size increases as the significance level is made smaller (α decreases).

(3) The sample size increases as the required power increases ($1 - \beta$ increases).

(4) The sample size decreases as the absolute value of the distance between the null and alternative means ($|\mu_0 - \mu_1|$) increases.

$$n = \frac{(Z_\beta + Z_{\alpha/2})^2 \sigma^2}{(\mu_1 - \mu_0)^2}$$

two-sided

$$n = \frac{(Z_\beta + Z_\alpha)^2 \sigma^2}{(\mu_1 - \mu_0)^2}$$

one-sided

Power for two-samples

Testing Equality of Two Normal Means (known variances)

Data samples

$$X_1, \dots, X_n \sim N(\mu_X, \sigma_X^2)$$

$$Y_1, \dots, Y_m \sim N(\mu_Y, \sigma_Y^2)$$

Null Hypothesis

$$H_0: \mu_X = \mu_Y$$

Alternative Hypothesis

$$H_1: \mu_X \neq \mu_Y$$
$$H_1: \mu_X > \mu_Y$$
$$H_1: \mu_X < \mu_Y$$

Test Statistic

$$Z_0 = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}$$

Critical region

C: $|Z_0| > Z_{\alpha/2}$

$Z_0 > Z_\alpha$

$Z_0 < -Z_\alpha$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

Power function $K(\Delta) = 1 - \beta = P(Z_0 \in C; \Delta)$

$\mu_X - \mu_Y = \Delta$

Under $\mu_X - \mu_Y = \Delta$

$$Z_0 = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}} \sim N\left(\frac{\Delta}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}, 1\right)$$

Testing Equality of Two Normal Means (known variances)

power function:

$$K(\Delta)= \mathrm{P}(|Z_0| > Z_{\alpha/2};\Delta) = \Phi\left(-Z_{\alpha/2}+\frac{\Delta}{\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}}\right)+\Phi\left(-Z_{\alpha/2}-\frac{\Delta}{\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}}\right)$$

$$\xrightarrow{n=m,\sigma_X=\sigma_Y=\sigma} \Phi\left(-Z_{\alpha/2}+\frac{\Delta}{\sigma}\sqrt{\frac{n}{2}}\right)+\Phi\left(-Z_{\alpha/2}-\frac{\Delta}{\sigma}\sqrt{\frac{n}{2}}\right)$$

Under $\mu_X-\mu_Y=\Delta>0$

$$K(\Delta)= \mathrm{P}(Z_0 > Z_{\alpha};\Delta) = \Phi\left(-Z_{\alpha}+\frac{|\Delta|}{\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}}\right) \xrightarrow{n=m,\sigma_X=\sigma_Y=\sigma} \Phi\left(-Z_{\alpha}+\frac{|\Delta|}{\sigma}\sqrt{\frac{n}{2}}\right)$$

Under $\mu_X-\mu_Y=\Delta<0$

$$K(\Delta)= \mathrm{P}(Z_0 < -Z_{\alpha};\Delta) = \Phi\left(-Z_{\alpha}+\frac{|\Delta|}{\sqrt{\frac{\sigma_X^2}{n}+\frac{\sigma_Y^2}{m}}}\right) \xrightarrow{n=m,\sigma_X=\sigma_Y=\sigma} \Phi\left(-Z_{\alpha}+\frac{|\Delta|}{\sigma}\sqrt{\frac{n}{2}}\right)$$

Testing Equality of Two Normal Means

$n = m$,
$\sigma_X = \sigma_Y = \sigma$,

Null Hypothesis

$$H_0: \mu_X = \mu_Y$$

Test Statistic

$$Z_0 = \frac{\bar{X} - \bar{Y}}{\sigma\sqrt{2/n}}$$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

Alternative Hypothesis	Critical Region (H_0 is rejected)	Power under $\mu_X - \mu_Y = \Delta$	Sample size needed for $K(\Delta) = 1 - \beta; K(0) = \alpha$
$H_1: \mu_X \neq \mu_Y$	$\lvert Z_0 \rvert > Z_{\alpha/2}$	$\Phi\left(-Z_{\alpha/2} + \frac{\Delta}{\sigma}\sqrt{\frac{n}{2}}\right) + \Phi\left(-Z_{\alpha/2} - \frac{\Delta}{\sigma}\sqrt{\frac{n}{2}}\right)$	$n = 2\frac{\left(Z_\beta + Z_{\alpha/2}\right)^2 \sigma^2}{\Delta^2}$
$H_1: \mu_X > \mu_Y$	$Z_0 > Z_\alpha$	$\Phi\left(-Z_\alpha + \frac{\lvert\Delta\rvert}{\sigma}\sqrt{\frac{n}{2}}\right)$	$n = 2\frac{\left(Z_\beta + Z_\alpha\right)^2 \sigma^2}{\Delta^2}$
$H_1: \mu_X < \mu_Y$	$Z_0 < -Z_\alpha$		

Power for proportions

Testing Proportion (large sample size)

Data samples

$$X_1, X_2, \cdots, X_n \sim Bernoulli(p)$$

Null Hypothesis

$$H_0 : p = p_0$$

Alternative Hypothesis

$$H_1 : p \neq p_0$$
$$H_1 : p > p_0$$
$$H_1 : p < p_0$$

Test Statistic

$$Z_0 = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Critical region

C: $|Z_0| > Z_{\alpha/2}$

$Z_0 > Z_\alpha$

$Z_0 < -Z_\alpha$

Distribution under H_0

$$Z_0 \sim N(0,1)$$

Power function $K(p_1) = 1 - \beta = P(Z_0 \in C; p_1)$

Under $p = p_1$

Under $p = p_1$ $\quad \frac{\hat{p} - p_1}{\sqrt{\frac{p_1(1-p_1)}{n}}} \sim N(0,1)$

$$Z_0 = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \sim N\left(\frac{p_1 - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}, \frac{p_1(1-p_1)}{p_0(1-p_0)}\right)$$

Testing Proportion (large sample size)

Null Hypothesis	Test Statistic	Distribution under H_0
$H_0 : p = p_0$	$Z_0 = \dfrac{\hat{p} - p_0}{\sqrt{\dfrac{p_0(1-p_0)}{n}}}$	$Z_0 \sim N(0,1)$

Alternative Hypothesis	Critical Region (H_0 is rejected)	Power under $p = p_1$	Sample size needed for $K(\Delta) = 1-\beta; K(0) = \alpha$
$H_1 : p \neq p_0$	$\lvert Z_0 \rvert > Z_{\alpha/2}$	$\Phi\left(\dfrac{-Z_{\alpha/2}\sqrt{p_0(1-p_0)} + \sqrt{n}(p_1-p_0)}{\sqrt{p_1(1-p_1)}}\right) + \Phi\left(\dfrac{-Z_{\alpha/2}\sqrt{p_0(1-p_0)} - \sqrt{n}(p_1-p_0)}{\sqrt{p_1(1-p_1)}}\right)$	$n = \dfrac{\left(Z_\beta\sqrt{p_1(1-p_1)} + Z_{\alpha/2}\sqrt{p_0(1-p_0)}\right)^2}{(p_1-p_0)^2}$
$H_1 : p > p_0$	$Z_0 > Z_\alpha$	$\Phi\left(\dfrac{-Z_\alpha\sqrt{p_0(1-p_0)} + \sqrt{n}\lvert p_1-p_0\rvert}{\sqrt{p_1(1-p_1)}}\right)$	$n = \dfrac{\left(Z_\beta\sqrt{p_1(1-p_1)} + Z_\alpha\sqrt{p_0(1-p_0)}\right)^2}{(p_1-p_0)^2}$
$H_1 : p < p_0$	$Z_0 < -Z_\alpha$		

Power for test of proportion

small sample size – exact calculation of power

n=12.

$$H_0: p=0.5$$
$$H_1: p<0.5$$

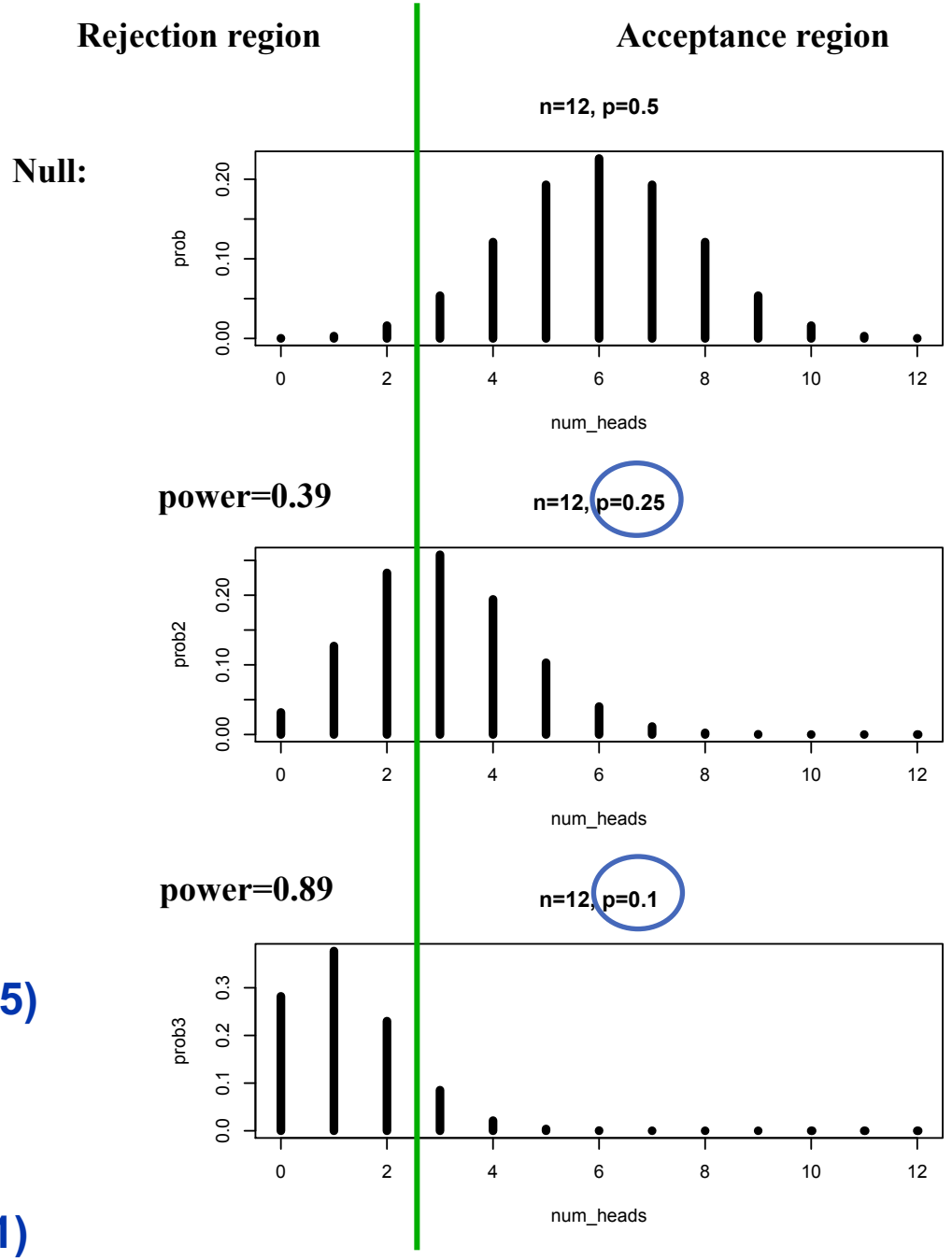
Rejection region is $x \leq 2$

$$\sum_{x=0}^{2} \binom{12}{x} .5^x .5^{12-x} = 0.019$$ **Type-I**

$$Prob(reject\ H_0 | p = 0.25) = Prob(x \leq 2 | p = 0.25)$$
$$= \sum_{x=0}^{2} \binom{12}{x} .25^x .75^{12-x} = 0.39$$ **Power (p=0.25)**

$$Prob(reject\ H_0 | p = 0.1) = \sum_{x=0}^{2} \binom{12}{x} .1^x .9^{12-x} = 0.89$$

Power (p=0.1)



Exercises

Example

6.4-8. Let $f(x;\theta) = (1/\theta)x^{(1-\theta)/\theta}$, $0 < x < 1$, $0 < \theta < \infty$.

(a) Show that the maximum likelihood estimator of θ is $\widehat{\theta} = -(1/n)\sum_{i=1}^{n} \ln X_i$.

(b) Show that $E(\widehat{\theta}) = \theta$ and thus that $\widehat{\theta}$ is an unbiased estimator of θ.

Example

Likelihood $\mathcal{L}(\theta) = \frac{1}{\theta^n}\left[\prod_{i=1}^{n} x_i\right]^{\frac{1-\theta}{\theta}}$

Log-likelihood $\ell(\theta) = -n\ln\theta + \frac{1-\theta}{\theta}\cdot\sum_{i=1}^{n}\ln(x_i)$

$$\ell'(\theta) = \frac{-n}{\theta} + \sum_{i=1}^{n}\ln(x_i)\left(-\frac{1}{\theta^2}\right) = 0$$

$$\Rightarrow \quad -n = \frac{1}{\theta}\cdot\sum_{i=1}^{n}\ln x_i$$

$$\Rightarrow \quad \hat{\theta}_{MLE} = -\frac{1}{n}\sum_{i=1}^{n}\ln x_i$$

Example

(b) $\mathbb{E}(\hat{\theta}_{MLE}) = -\mathbb{E}(\ln X)$

$$\mathbb{E}(\ln X) = \int_0^1 \ln x \cdot \frac{1}{\theta} x^{\frac{1}{\theta}-1} dx$$

$$= \int_0^1 \ln x \, d(x^{\frac{1}{\theta}})$$

$$= (\ln x) \, x^{\frac{1}{\theta}} \Big|_0^1 - \int_0^1 x^{\frac{1}{\theta}} \cdot \frac{1}{x} dx$$

$$= -\int_0^1 x^{\frac{1}{\theta}-1} dx = -\theta x^{\frac{1}{\theta}} \Big|_0^1 = -\theta$$

$\Rightarrow \mathbb{E}(\hat{\theta}_{MLE}) = \theta$ unbiased.

Example

(C) Method of Moments

$$\mathbb{E}(X) = \int_0^1 x \cdot \frac{1}{\theta} x^{\frac{1}{\theta}-1} dx = \frac{1}{\theta} \int_0^1 x^{\frac{1}{\theta}} dx$$

$$= \frac{1}{\theta} \times \frac{1}{\frac{1}{\theta}+1} x^{\frac{1}{\theta}+1} \Big|_0^1$$

$$= \frac{1}{\theta+1}$$

$$\frac{1}{\theta+1} = \bar{x} \Rightarrow \hat{\theta}_{mom} = \frac{1}{\bar{x}} - 1$$

Example

7.2-8. Let $\overline{X}$, $\overline{Y}$, S_X^2, and S_Y^2 be the respective sample means and unbiased estimates of the variances obtained from independent samples of sizes n and m from the normal distributions $N(\mu_X, \sigma_X^2)$ and $N(\mu_Y, \sigma_Y^2)$, where μ_X, μ_Y, σ_X^2, and σ_Y^2 are unknown. If $\sigma_X^2/\sigma_Y^2 = d$, a known constant,

(a) Argue that $\dfrac{(\overline{X} - \overline{Y}) - (\mu_X - \mu_Y)}{\sqrt{d\sigma_Y^2/n + \sigma_Y^2/m}}$ is $N(0,1)$.

(b) Argue that $\dfrac{(n-1)S_X^2}{d\sigma_Y^2} + \dfrac{(m-1)S_Y^2}{\sigma_Y^2}$ is $\chi^2(n+m-2)$.

(c) Argue that the two random variables in (a) and (b) are independent.

(d) With these results, construct a random variable (not depending upon σ_Y^2) that has a t distribution and that can be used to construct a confidence interval for $\mu_X - \mu_Y$.

Example

$$\frac{\sigma_X^2}{\sigma_Y^2} = d \quad \Rightarrow \quad \sigma_X^2 = d\sigma_Y^2$$

$$\bar{X} - \bar{Y} \sim N\left(\mu_X - \mu_Y, \quad \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}\right)$$

$$= N\left(\mu_X - \mu_Y, \quad \frac{d\sigma_Y^2}{n} + \frac{\sigma_Y^2}{m}\right)$$

$$\Rightarrow \quad \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{d\sigma_Y^2/n + \sigma_Y^2/m}} \sim N(0,1)$$

Example

(b) $\dfrac{(n-1)S_X^2}{\sigma_X^2} = \dfrac{(n-1)S_X^2}{d\sigma_Y^2} \sim \chi^2(n-1)$

$$\frac{(m-1)S_Y^2}{\sigma_Y^2} \sim \chi^2(m-1)$$

$$\Rightarrow \frac{(n-1)S_X^2}{d\sigma_Y^2} + \frac{(m-1)S_Y^2}{\sigma_Y^2} \sim \chi^2(n-1+m-1)$$

Example

$$\frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{\sqrt{\frac{d\sigma_Y^2}{n}+\frac{\sigma_Y^2}{m}}} \sim N(0,1)$$

$$\frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{\sqrt{\frac{\frac{(n-1)S_X^2}{d\sigma_Y^2}+\frac{(m-1)S_Y^2}{\sigma_Y^2}}{n+m-2}}} \sim t(n+m-2)$$

$\frac{(n-1)S_X^2}{d\sigma_Y^2}+\frac{(m-1)S_Y^2}{\sigma_Y^2} \sim \chi^2(n+m-2)$

$$= \frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{\sqrt{\frac{d}{n}+\frac{1}{m}}} \Big/ \sqrt{\frac{\frac{(n-1)S_X^2}{d(n+m-2)}+\frac{(m-1)S_Y^2}{n+m-2}}{}} \sim t(n+m-2)$$

$$= \frac{\bar{X}-\bar{Y}-(\mu_X-\mu_Y)}{\sqrt{\frac{(n-1)S_X^2}{d(n+m-2)}+\frac{(m-1)S_Y^2}{n+m-2}}\cdot\sqrt{\frac{d}{n}+\frac{1}{m}}} \sim t(n+m-2)$$

Example

7.3-10. A candy manufacturer selects mints at random from the production line and weighs them. For one week, the day shift weighed $n_1 = 194$ mints and the night shift weighed $n_2 = 162$ mints. The numbers of these mints that weighed at most 21 grams was $y_1 = 28$ for the day shift and $y_2 = 11$ for the night shift. Let p_1 and p_2 denote the proportions of mints that weigh at most 21 grams for the day and night shifts, respectively.

(a) Give a point estimate of p_1.

(b) Give the endpoints for a 95% confidence interval for p_1.

(c) Give a point estimate of $p_1 - p_2$.

(d) Find a one-sided 95% confidence interval that gives a lower bound for $p_1 - p_2$.

Example

$n_1 = 194 \qquad n_2 = 162$

(a) $\hat{p}_1 = \frac{28}{194} \qquad \hat{p}_2 = \frac{11}{162}$

(b) C.I for p_1. $\qquad \alpha = 0.05 \qquad z_{\alpha/2} = 1.96$

$$\hat{p}_1 \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}}$$

$$0.144 \pm 1.96 \times \sqrt{\frac{0.144 \times (1-0.144)}{194}}$$

$$[0.0949,\ 0.1938]$$

Example

(c) $\hat{p}_1 - \hat{p}_2 = 0.144 - 0.068 = 0.076$

(d) $\alpha = 0.05 \quad z_\alpha = 1.645$

$$\hat{p}_1 - \hat{p}_2 - z_\alpha \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$$0.076 - 1.645 \times \sqrt{\frac{0.144 \times (1-0.144)}{194} + \frac{0.068 \times (1-0.068)}{162}}$$

$= 0.023$

$\Rightarrow [0.023, 1]$

Example

7.4-6. A manufacturer sells a light bulb that has a mean life of 1450 hours with a standard deviation of 33.7 hours. A new manufacturing process is being tested, and there is interest in knowing the mean life μ of the new bulbs. How large a sample is required so that $\bar{x} \pm 5$ is a 95% confidence interval for μ? You may assume that the change in the standard deviation is minimal.

$z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq 5 \qquad \alpha = 0.05 \qquad z_{\alpha/2} = 1.96$

$\Rightarrow 1.96 \times \frac{33.7}{\sqrt{n}} \leq 5$

$n \geq \left(\frac{1.96 \times 33.7}{5}\right)^2 = 174.5$

$n = 175$

Example

7.4-8. Some college professors and students examined 137 Canadian geese for patent schistosome in the year they hatched. Of these 137 birds, 54 were infected. The professors and students were interested in estimating p, the proportion of infected birds of this type. For future studies, determine the sample size n so that the estimate of p is within $\varepsilon = 0.04$ of the unknown p with 90% confidence.

Example

$n = 137$. $\hat{p} = \frac{54}{137}$.

$z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}} \le \varepsilon = 0.04$ $\quad \alpha = 0.1$. $z_{\alpha/2} = 1.645$

Method 1: $p(1-p) \le \frac{1}{4}$

$$\Rightarrow n \ge \left(\frac{z_{\alpha/2}}{\varepsilon}\right)^2 \cdot \frac{1}{4} = \left(\frac{1.645}{0.04}\right)^2 \times \frac{1}{4} = 422.8$$

$n = 423$

using $n = 846$, $\hat{p}$ is within $\varepsilon = 0.04$ of the unknown p with at least 90% confidence. (for any p)

Method 2. $n \ge \left(\frac{z_{\alpha/2}}{\varepsilon}\right)^2 \cdot \hat{p}(1-\hat{p}) = \left(\frac{1.645}{0.04}\right)^2 \times \frac{54}{137} \times \left(1 - \frac{54}{137}\right)$

$= 403.9$

$n = 404$.

Example

8.1-15. Let $X_1, X_2, \ldots, X_{19}$ be a random sample of size $n = 19$ from the normal distribution $N(\mu, \sigma^2)$.

(a) Find a critical region, C, of size $\alpha = 0.05$ for testing $H_0: \sigma^2 = 30$ against $H_1: \sigma^2 = 80$.

(b) Find the approximate value of β, the probability of a Type II error, for the critical region C of part (a).

(a) $\frac{(n-1)S^2}{\sigma_0^2} \sim \chi^2(n-1)$ under $H_0: \sigma = \sigma_0^2 = 30$

view it as a one-sided test ($80 > 30$)

Critical region. $\frac{(n-1)S^2}{\sigma_0^2} > \chi^2_\alpha(n-1)$ $\alpha = 0.05$

Example

(b) $\beta = P\left(\frac{(n-1)S^2}{\sigma_0^2} < \chi^2_{0.05}(18),\ \sigma^2 = 80\right)$

$= P\left(\frac{18S^2}{80} \times \frac{80}{30} < \chi^2_{0.05}(18);\ \sigma^2 = 80\right)$

$\frac{18S^2}{80} \sim \chi^2(18)$

$= P\left(\chi^2(18) < \frac{30}{80} \times \chi^2_{0.05}(18)\right)$

$= P\left(\chi^2(18) < \frac{3}{8} \times 28.87 = 10.83\right)$

≈ 0.1

Example

8.2-11. Let X equal the fill weight in April and Y the fill weight in June for an 8-pound box of bleach. We shall test the null hypothesis $H_0: \mu_X - \mu_Y = 0$ against the alternative hypothesis $H_1: \mu_X - \mu_Y > 0$ given that $n = 90$ observations of X yielded $\bar{x} = 8.10$ and $s_x = 0.117$ and $m = 110$ observations of Y yielded $\bar{y} = 8.07$ and $s_y = 0.054$.

(a) What is your conclusion if $\alpha = 0.05$?
HINT: Do the variances seem to be equal?

(b) What is the approximate p-value of this test?

Example

$n = 90, \quad \bar{x} = 8.1 \quad S_X = 0.117$

$m = 110 \quad \bar{Y} = 8.07 \quad S_Y = 0.054$

(a) F- test $\quad \frac{S_X^2}{S_Y^2} = \frac{0.117^2}{0.054^2} = 4.7 > 1.47$

$F_{\alpha/2}(89, 109) \approx 1.47$

$F_{1-\alpha/2}(89, 109) \approx 0.67$

$\Rightarrow$ Reject H_0

$H_0: \sigma_X^2 = \sigma_Y^2$

Example

use Welch's t-test

$$r = \left\lfloor \frac{\left(\frac{S_X^2}{n} + \frac{S_Y^2}{m}\right)^2}{\frac{1}{n-1}\left(\frac{S_X^2}{n}\right)^2 + \frac{1}{m-1}\left(\frac{S_Y^2}{m}\right)^2} \right\rfloor = \lfloor 119.76 \rfloor = 119$$

Test statistic $$\frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}} = \frac{8.1 - 8.07}{\sqrt{\frac{0.117^2}{90} + \frac{0.054^2}{110}}} = 2.24$$

$t_{0.05}(119) = 1.658$ (or $\approx z_{0.05} = 1.645$)

since $2.24 > 1.658$ Reject H_0

p value $= P(t(119) > 2.24) \approx 0.013 < 0.05$

Example

8.5-9. Let p denote the probability that, for a particular tennis player, the first serve is good. Since $p = 0.40$, this player decided to take lessons in order to increase p. When the lessons are completed, the hypothesis $H_0: p = 0.40$ will be tested against $H_1: p > 0.40$ on the basis of $n = 25$ trials. Set the significance level as 0.05.

(a) Find the power function $K(p)$ for this test.

(b) Suppose that after the lessons, the true probability that the first serve is good has increased to 0.5, what is the sample size needed to achieve a power of 90%?

Example

H_0: $p = 0.4$ $\quad p_0 = 0.4$, $n = 25$

H_1: $p > 0.4$ $\quad \alpha = 0.05$

(a) under H_1: $p = p_1$.

$$\frac{\hat{P} - p_1}{\sqrt{\frac{p_1(1-p_1)}{n}}} \sim N(0,1)$$

Test statistic $\quad \dfrac{\hat{P} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$

critical region: $\quad \dfrac{\hat{P} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} > z_\alpha$

Example

$$k(p_1) = P\left(\frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} > z_\alpha \text{, under. } \frac{\hat{p} - p_1}{\sqrt{\frac{p_1(1-p_1)}{n}}} \sim N(0,1) \right)$$

$$= P\left(\hat{p} > p_0 + z_\alpha \sqrt{\frac{p_0(1-p_0)}{n}} \text{, when } \hat{p} \sim N\left(p_1, \sqrt{\frac{p_1(1-p_1)}{n}}\right) \right)$$

$$= P\left(\frac{\hat{p} - p_1}{\sqrt{\frac{p_1(1-p_1)}{n}}} > \frac{p_0 - p_1}{\sqrt{\frac{p_1(1-p_1)}{n}}} + z_\alpha \sqrt{\frac{p_0(1-p_0)}{p_1(1-p_1)}} \right)$$

$$= \Phi\left(\frac{p_1 - p_0}{\sqrt{\frac{p_1(1-p_1)}{n}}} - z_\alpha \sqrt{\frac{p_0(1-p_0)}{p_1(1-p_1)}} \right)$$

$$= \Phi\left(\frac{\sqrt{n}(p_1 - p_0) - z_\alpha \sqrt{p_0(1-p_0)}}{\sqrt{p_1(1-p_1)}} \right)$$

Example

(b) $p_1 = 0.5 \quad p_0 = 0.4 \quad 1-\beta = 0.9$

$$\Phi\left(\frac{\sqrt{n}(p_1-p_0) - z_\alpha \sqrt{p_0(1-p_0)}}{\sqrt{p_1(1-p_1)}} \right) = 1-\beta$$

$$\Rightarrow \frac{\sqrt{n}(p_1-p_0) - z_\alpha \sqrt{p_0(1-p_0)}}{\sqrt{p_1(1-p_1)}} = z_\beta$$

$$\Rightarrow n = \left[\frac{z_\beta \sqrt{p_1(1-p_1)} + z_\alpha \sqrt{p_0(1-p_0)}}{p_1-p_0} \right]^2$$

$$= \left[\frac{1.28 \times 0.5 + 1.645 \times \sqrt{0.24}}{0.1} \right]^2$$

$$= 209.1$$

$$n = 210$$

Contents after midterm

Definition

CDF and PDF of order statistics

Expectation

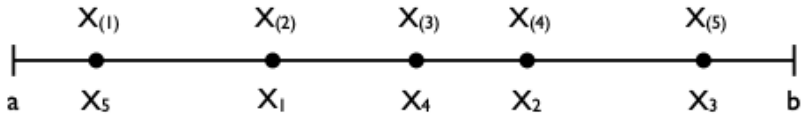
Q-Q plot

ORDER STATISTICS AND Q-Q PLOT

Order Statistics

Let $X_1, X_2, \ldots, X_n$ be i.i.d. random samples from a common distribution f.

$$X_1, X_2, \ldots, X_n \sim f$$



Definition

Denote the ***ordered*** sample values $X_{(1)} < X_{(2)} < \cdots < X_{(n)}$ as the order statistic

For each k, the k-th order statistic is

$$X_{(k)} = k\text{-th smallest of } X_1, X_2, \ldots, X_n$$

Example

$$X_{(1)} = \min\{X_1, X_2, \ldots, X_n\},$$
$$X_{(2)} = \textit{second smallest of } X_1, \ldots, X_n,$$
$$X_{(n)} = \max\{X_1, X_2, \ldots, X_n\}.$$

Density of Order Statistics

Let $X_1, X_2, \ldots, X_n$ be i.i.d. random samples from a common distribution with CDF $F(x)$ and PDF $f(x)$. What is the CDF and PDF of any order statistic $X_{(k)}$?

Theorem 1 *Suppose that $X_1, \ldots, X_n$ are i.i.d. continuous random variables with common pdf $f(x)$ and cdf $F(x)$. For $k = 1, \ldots, n$, denote the cdf and pdf of the kth order statistic $X_{(k)}$ to be respectively $F_{X_{(k)}}$ and $f_{X_{(k)}}$. They can be written as*

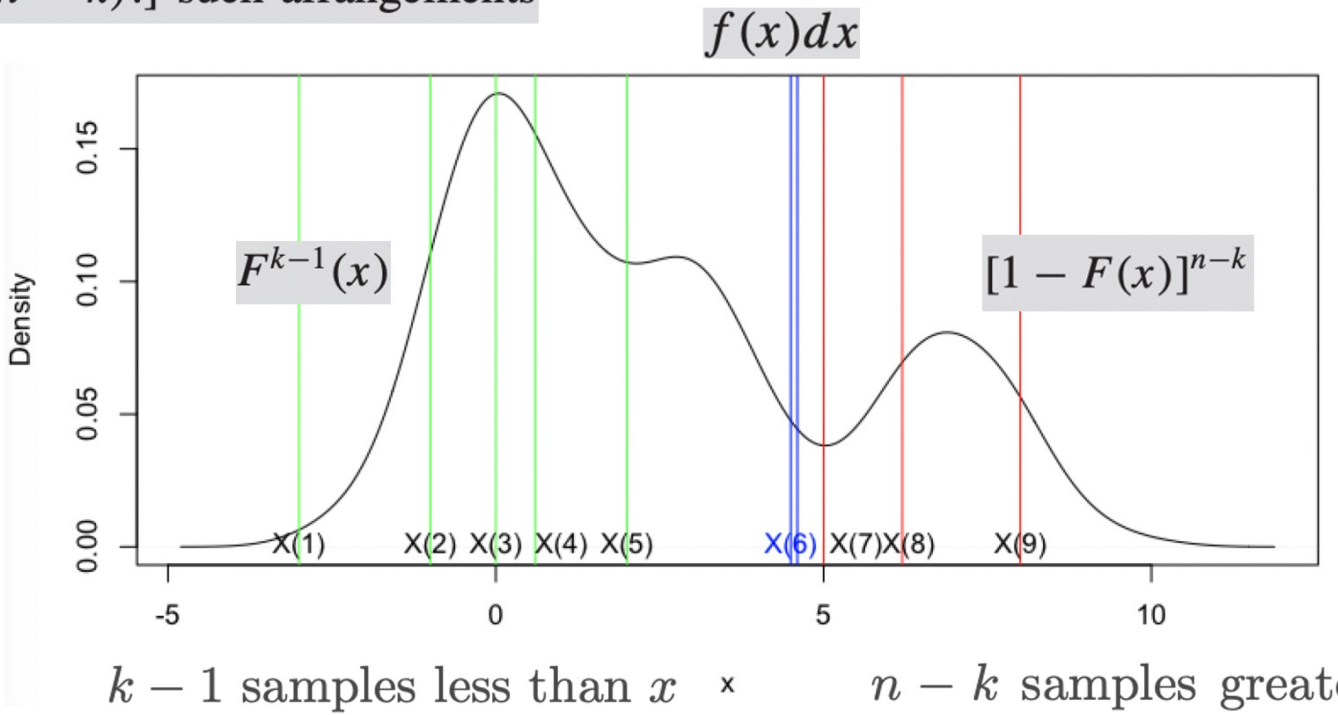
$$F_{X_{(k)}}(x) = \sum_{l=k}^{n} \binom{n}{l} F(x)^l (1 - F(x))^{n-l},$$

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!1!(n-k)!} [F(x)]^{k-1} f(x) [1 - F(x)]^{n-k}.$$

Density of Order Statistics

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!1!(n-k)!}[F(x)]^{k-1} f(x)[1-F(x)]^{n-k}.$$

$n!/[(k-1)!1!(n-k)!]$ such arrangements



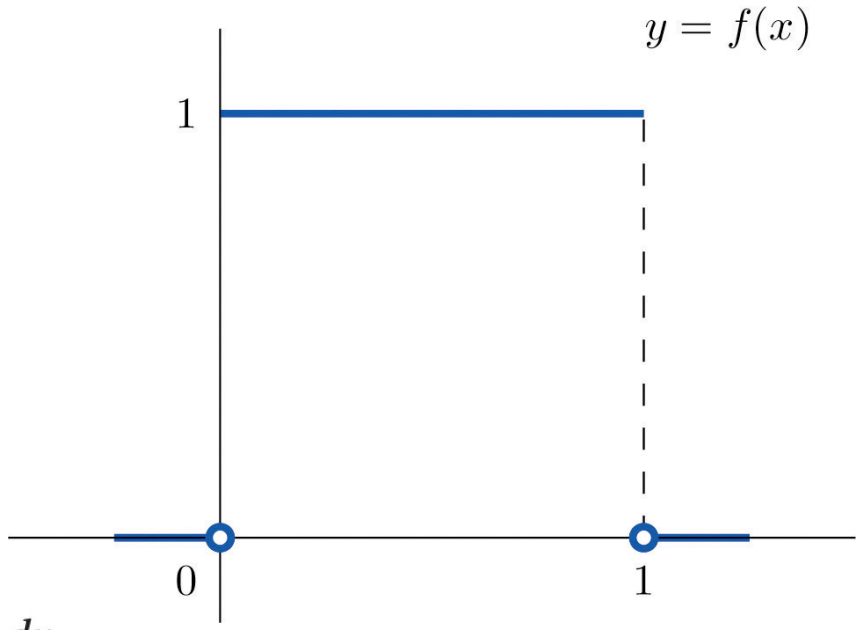
Example – Uniform distribution

Let $X_1, X_2, \ldots, X_n$ be i.i.d. samples from the uniform distribution on [0,1].

$$X_1, X_2, \ldots, X_n \sim \text{Uniform}[0,1]$$

$$F(x) = x, \quad f(x) = 1, \forall 0 \le x \le 1$$

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!\,1!\,(n-k)!} x^{k-1}(1-x)^{n-k}$$



Beta distribution

$$X \sim Beta(\alpha, \beta) \qquad f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}, 0 \le x \le 1$$

$$B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)} \qquad \Gamma(t) = \int_0^\infty y^{t-1}e^{-y}\,dy,$$

$$\Gamma(n) = (n-1)!$$

$$X_{(k)} \sim Beta(k, n-k+1)$$

Example – Uniform distribution

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!\,1!\,(n-k)!} x^{k-1}(1-x)^{n-k} \qquad X_{(k)} \sim Beta(k, n-k+1)$$

The mean and variance of $X_{(k)}$

Beta distribution

$$X \sim Beta(\alpha, \beta) \Rightarrow E(X) = \frac{\alpha}{\alpha+\beta} \quad Var(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

$$\alpha = k, \beta = n-k+1$$

$$\Rightarrow E(X_{(k)}) = \frac{\alpha}{\alpha+\beta} = \frac{k}{n+1}$$

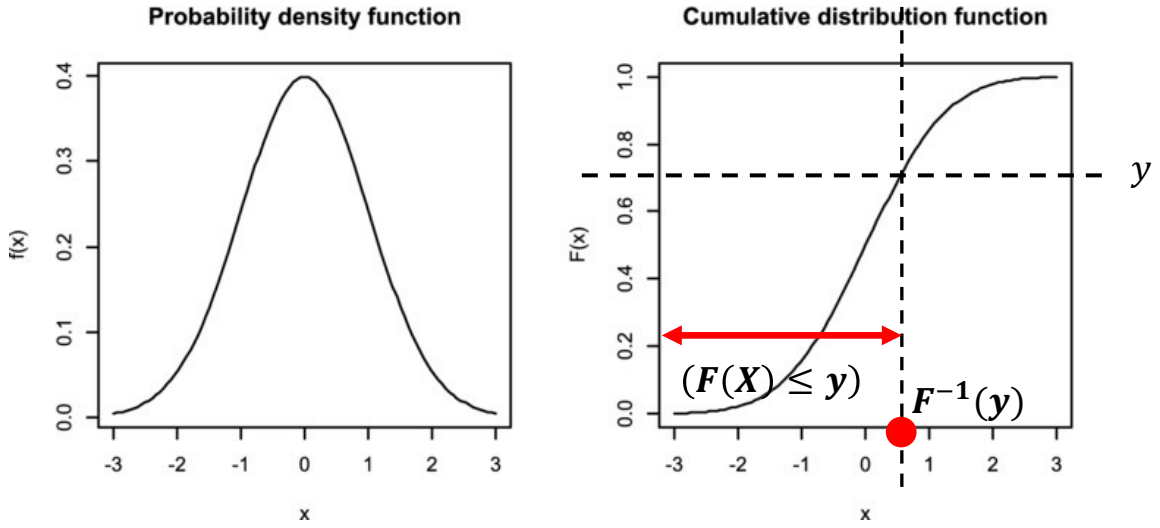
$$Var(X_{(k)}) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} = \frac{k(n+1-k)}{(n+1)^2(n+2)}$$

Expectation of $F(X_{(k)})$

- Let X be a random variable with PDF $f(x)$ and *strictly increasing* CDF $F(x)$.
- Let $Y = F(X)$ be a new random variable. $0 \leq Y \leq 1$.
- The CDF of Y:

$$P(Y \leq y) = P(F(X) \leq y) = P\left(X \leq F^{-1}(y)\right) = F\left(F^{-1}(y)\right) = y$$

- Therefore, $Y \sim \text{Uniform}[0,1]$



Expectation of $F(X_{(k)})$

Let $X_1, X_2, \ldots, X_n$ be i.i.d. random samples from a common distribution with PDF $f(x)$ and *strictly increasing* CDF $F(x)$.

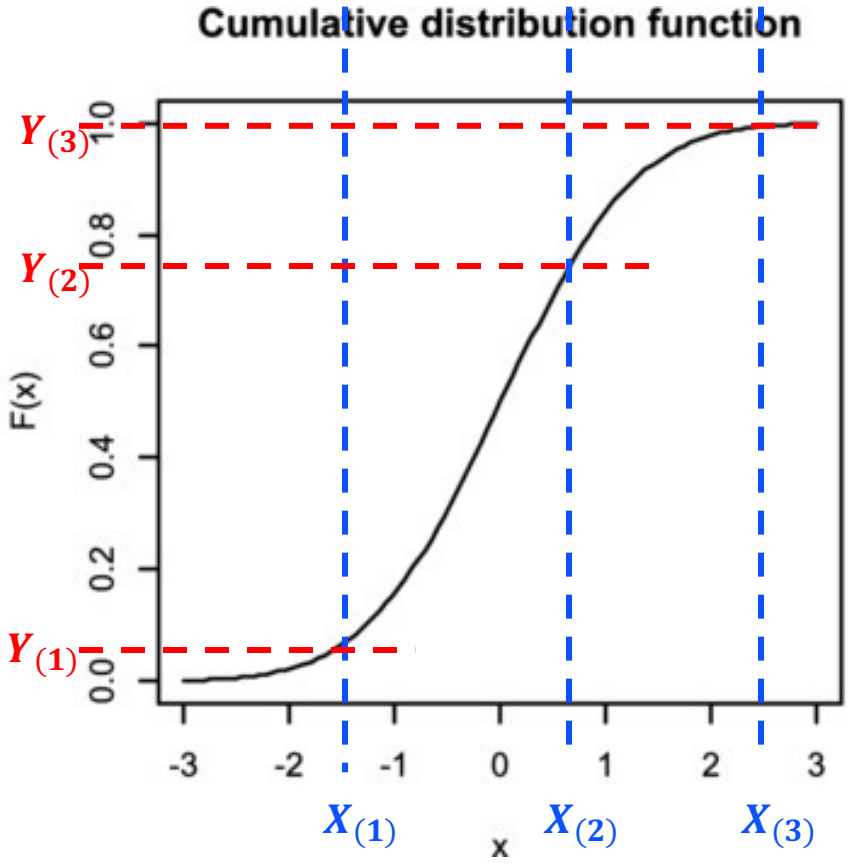
$$X_1, X_2, \ldots, X_n \sim f$$

Let $Y_1 = F(X_1), Y_2 = F(X_2), \ldots, Y_n = f(X_n)$

$$Y_1, Y_2, \ldots, Y_n \sim \text{Uniform}[0,1]$$

$Y_{(k)} = F(X_{(k)})$ is the k-th order statistic for uniform distributions:

$$E(Y_{(k)}) = E\left(F(X_{(k)})\right) = \frac{k}{n+1}$$



Example: Exponential Distribution

Let $X_1, X_2, \ldots, X_n$ be i.i.d. random samples from an exponential distribution

$$X_1, X_2, \ldots, X_n \sim Exp(\frac{1}{\theta}) \qquad f(x) = \frac{1}{\theta} e^{-x/\theta}, \qquad 0 \le x < \infty$$

What is the distribution of $X_{(1)}$? Its expectation and variance?

$f(x) = \frac{1}{\theta} e^{-\frac{x}{\theta}}$

$F(x) = \int_0^x \frac{1}{\theta} e^{-\frac{w}{\theta}} dw = -e^{-\frac{w}{\theta}} \Big|_0^x = 1 - e^{-\frac{x}{\theta}}$

$1 - F(x) = e^{-\frac{x}{\theta}}$

$f_{X_{(1)}}(x) = n(1-F(x))^{n-1} f(x) = n e^{-\frac{(n-1)x}{\theta}} \cdot \frac{1}{\theta} e^{-\frac{x}{\theta}}$

$= \frac{n}{\theta} e^{-\frac{nx}{\theta}}$

$\Rightarrow X_{(1)} \sim Exp(\frac{\theta}{n})$

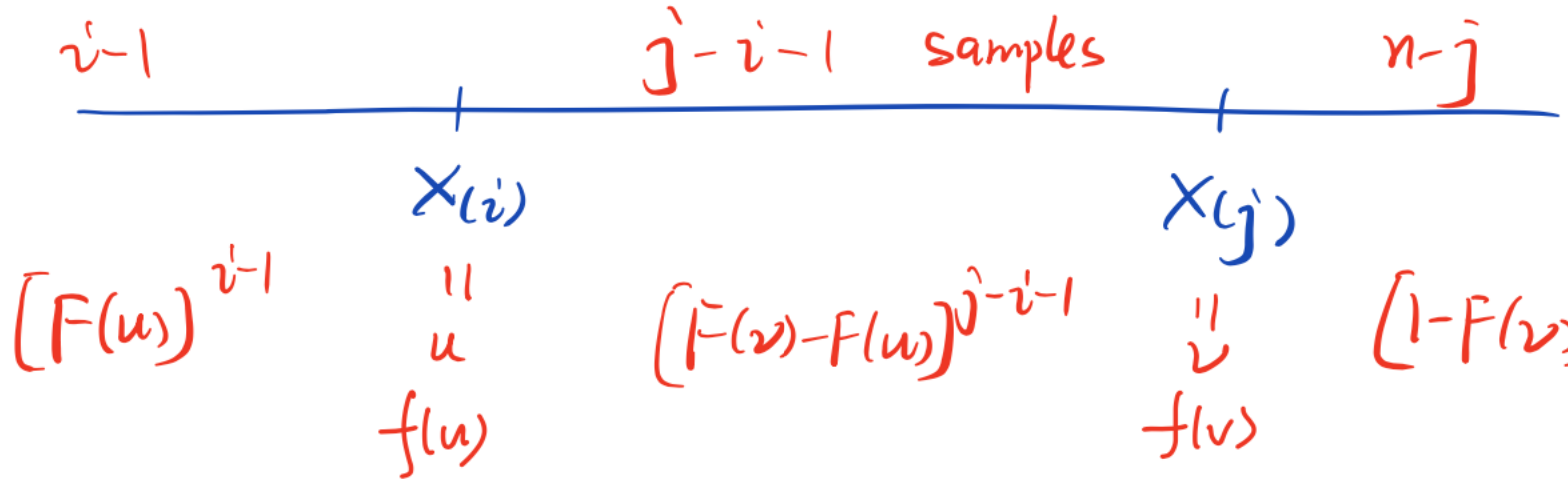
$E(X_{(1)}) = \frac{\theta}{n}$ $\quad Var(X_{(1)}) = (\frac{\theta}{n})^2$

Example

6.3-10. Use the heuristic argument to show that the joint pdf of the two order statistics $Y_i < Y_j$ is

$$g(y_i, y_j) = \frac{n!}{(i-1)!(j-i-1)!(n-j)!} \times [F(y_i)]^{i-1}[F(y_j) - F(y_i)]^{j-i-1} \times [1 - F(y_j)]^{n-j} f(y_i) f(y_j), \quad -\infty < y_i < y_j < \infty.$$

Example



$$\Rightarrow g(u,v) =$$

$$\frac{n!}{(i-1)!\,(j-i-1)!\,(n-j)!}[F(u)]^{i-1}[F(v)-F(u)]^{j-i-1}[1-F(v)]^{n-j}f(u)f(v)$$

(joint PDF of $(X_{(i)}, X_{(j)})$).

Theoretical vs Sample Quantiles

- The p-th **theoretical quantile** of the distribution F is $\boldsymbol{\pi_p}$:

$$F(\pi_p) = p \qquad \pi_p = F^{-1}(p)$$

$$E\left(F\left(X_{(k)}\right)\right) = \frac{k}{n+1}$$

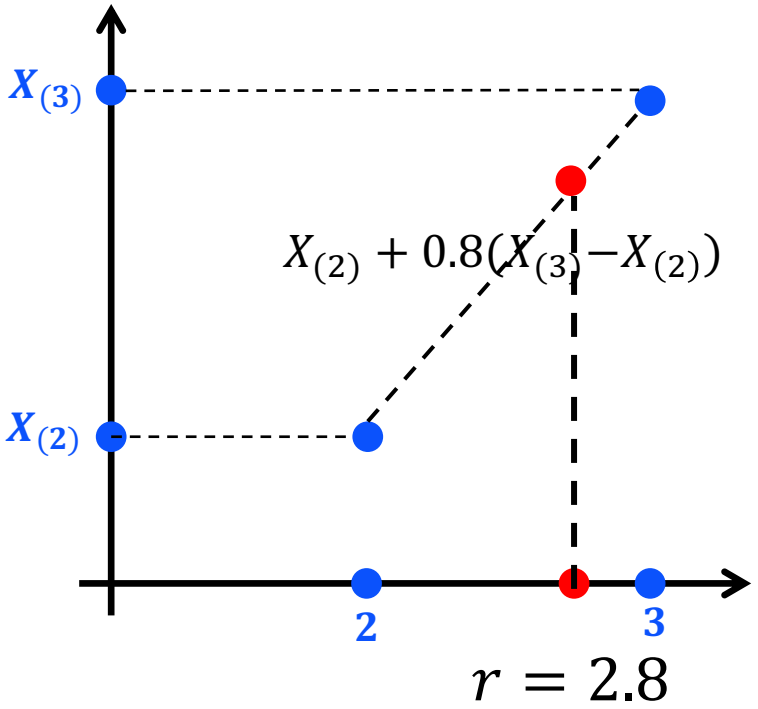
- $X_{(k)}, k = (n+1)p$ is the p-th **sample quantile** $\boldsymbol{\hat{\pi}_p}$.

Theoretical vs Sample Quantiles

Calculation of the p-th sample quantile

- Order the samples $X_{(1)} < X_{(2)} < \cdots < X_{(n)}$
- $r = (n+1)p$
- If r is an integer, the sample quantile is $\hat{\pi}_p = X_{(r)}$
- If r is not an integer, $r = \lfloor r \rfloor + (r - \lfloor r \rfloor)$, the sample quantile is

$$\hat{\pi}_p = X_{(\lfloor r \rfloor)} + (r - \lfloor r \rfloor)(X_{(\lfloor r \rfloor + 1)} - X_{(\lfloor r \rfloor)})$$



Quantile-Quantile (Q-Q) plot

Given data $x_1, x_2, \ldots, x_n$, we suspect that they are coming from a common distribution F, say, normal distribution or exponential distribution or gamma distribution. We can then compute the theoretical quantiles $\pi_{i/(n+1)} = F^{-1}\left(\frac{i}{n+1}\right)$ for $i = 1,2, \ldots, \mathrm{n}$. If the data is indeed statistically similar to the distribution, then we expect that

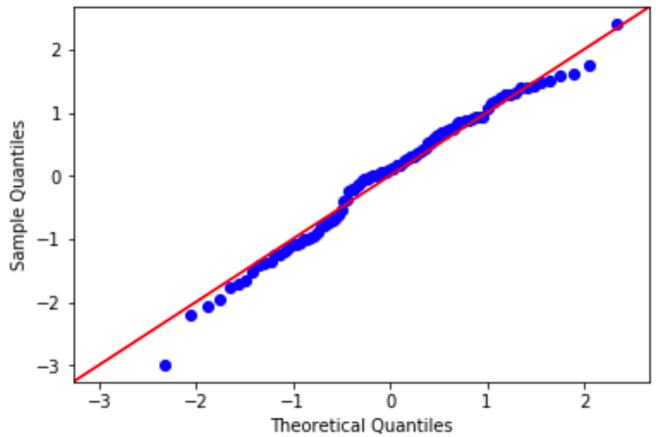
$$\underbrace{\widehat{\pi}_{i/(n+1)} = x_{(i)}}_{\text{sample quantiles}} \approx \underbrace{\pi_{i/(n+1)}}_{\text{theoretical quantiles}}$$

If we plot a scatterplot of the pairs $(\pi_{i/(n+1)}, x_{(i)})$, it should be close to the line y = x.

This is what we call **quantiles-quantiles plot or QQ plots**, as we plot the theoretical quantiles against the sample quantiles.

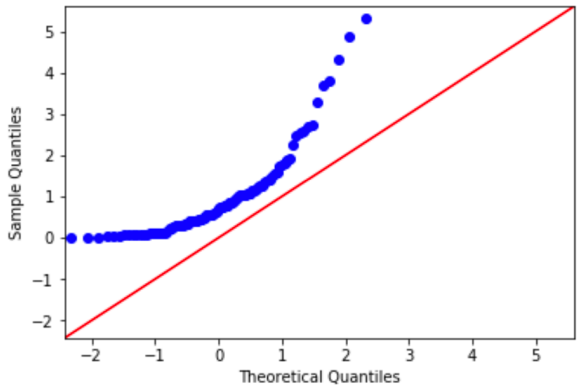
Examples

Hypothesis: the data is coming from standard normal.

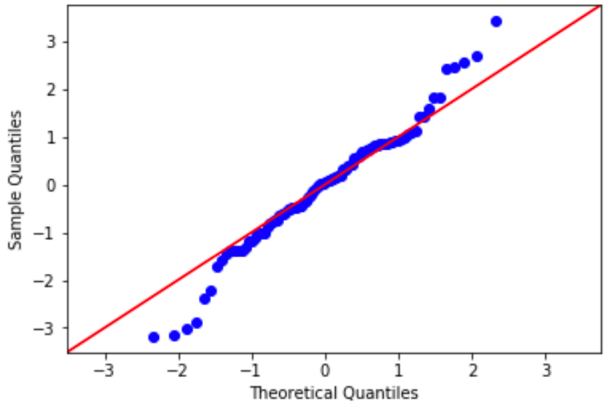


This dataset looks like a good fit.

Hypothesis: the data is coming from standard normal.



This is a poor fit, since the data is clearly non-negative, and is more likely to take on larger values than normal distribution (larger "right tail").



This is a poor fit, since the data is more likely to take on extremely large or extremely small values than standard normal (a heavy-tailed distribution)

Use order statistics to construct distribution-free CI for percentiles

CLT approximation when n is large

DISTRIBUTION-FREE CI

CI for Median

Let $X_1, X_2, \ldots, X_n$ be i.i.d. random samples from a common distribution F.

$$X_1, X_2, \ldots, X_n \sim F$$

The **median** $\boldsymbol{m = \pi_{1/2}}$: $\boxed{F(m) = 1/2}$ $\quad P(X_i \le m) = 1/2$

Ordered statistics $X_{(1)} < X_{(2)} < \cdots < X_{(n)}$ ⟹ ***CI***: $(X_{(i)}, X_{(j)})$

Confidence level of $(X_{(i)}, X_{(j)})$

$$W = \sum 1\{X_i \le m\}$$
$$W \sim Bin\left(n, \frac{1}{2}\right)$$

$$P(X_{(i)} < m < X_{(j)}) = P(i \le W \le j-1)$$

$$= \sum_{l=i}^{j-1} \binom{n}{l} \left(\frac{1}{2}\right)^l \left(1-\frac{1}{2}\right)^{n-l} = \sum_{l=i}^{j-1} \binom{n}{l} \left(\frac{1}{2}\right)^n$$

CLT Approximation for Large n

$$W = \sum 1\{X_i \le m\}$$

$$W \sim Bin\left(n, \frac{1}{2}\right)$$

$$\xrightarrow{\text{CLT}} \quad W \sim N\left(\frac{n}{2}, \frac{n}{4}\right)$$

Confidence level of $(X_{(i)}, X_{(j)})$

Required

With Half-unit continuity correction

$$P(X_{(i)} < m < X_{(j)}) = P(i \le W \le j-1)$$

$$= P(i - 1/2 \le W \le j - 1/2)$$

$$= P\left(\frac{i-1/2-n/2}{\sqrt{n/4}} \le \underbrace{\frac{W-n/2}{\sqrt{n/4}}}_{N(0,1)} \le \frac{j-1/2-n/2}{\sqrt{n/4}}\right)$$

$$\approx \Phi\left(\frac{j-1/2-n/2}{\sqrt{n/4}}\right) - \Phi\left(\frac{i-1/2-n/2}{\sqrt{n/4}}\right)$$

CI for any Percentiles π_p

- The p-th **theoretical percentile** of the distribution F is $\boldsymbol{\pi_p}$:

$$F(\pi_p) = p$$

$$W = \sum 1\{X_i \leq \pi_p\}$$
$$W \sim Bin(n, p)$$

Confidence level of $(X_{(i)}, X_{(j)})$

$$P(X_{(i)} < \pi_p < X_{(j)}) = P(i \leq W \leq j-1)$$
$$= \sum_{l=i}^{j-1} \binom{n}{l} p^l (1-p)^{n-l}$$

CLT approximation

- The p-th **theoretical percentile** of the distribution F is $\boldsymbol{\pi_p}$:

$$F(\pi_p) = p$$

$$W = \sum 1\{X_i \le \pi_p\}$$
$$W \sim Bin(n, p)$$

CLT $\Rightarrow$ $W \sim N(np, np(1-p))$

Confidence level of $(X_{(i)}, X_{(j)})$

With Half-unit continuity correction

$$P(X_{(i)} < \pi_p < X_{(j)}) = P(i \le W \le j-1) = P(i - 1/2 \le W \le j - 1/2)$$

$$= P\left(\frac{i - 1/2 - np}{\sqrt{np(1-p)}} \le \frac{W - np}{\sqrt{np(1-p)}} \le \frac{j - 1/2 - np}{\sqrt{np(1-p)}}\right)$$

($\frac{W - np}{\sqrt{np(1-p)}} \sim N(0,1)$)

$$\approx \Phi\left(\frac{j - 1/2 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{i - 1/2 - np}{\sqrt{np(1-p)}}\right)$$

Sign test

Wilcoxon (signed rank) test

DISTRIBUTION-FREE HYPOTHESIS TEST

Sign test for median

$X_1, \ldots, X_n \sim F$ $\quad$ $F(m) = 1/2$ $\quad\Rightarrow\quad$ $P(X_i \le m) = 1/2$

Null Hypothesis $\quad$ **Test Statistic** $\quad$ **Distribution under H_0**

$H_0: m = m_0$ $\quad$ $W = \sum 1\{X_i \le m_0\}$ $\quad$ $W \sim Bin\left(n, \frac{1}{2}\right)$

Alternative Hypothesis	**Rejection/Critical Region form**	**p-value**	p-value under CLT (half-unit correction)
$H_1: m \neq m_0$	$W \ge c_1$ or $W \le c_2$	$2\min\{\mathrm{P}(W \ge W_0), \mathrm{P}(\mathrm{W} \le \mathrm{W}_0)\}$	$2\min\{\mathrm{P}(W \ge W_0 - 0.5), \mathrm{P}(\mathrm{W} \le \mathrm{W}_0 + 0.5)\}$
$H_1: m > m_0$	$W \le c_3$	$\mathrm{P}(\mathrm{W} \le \mathrm{W}_0)$	$\mathrm{P}(\mathrm{W} \le \mathrm{W}_0 + 0.5)$
$H_1: m < m_0$	$W \ge c_4$	$\mathrm{P}(\mathrm{W} \ge \mathrm{W}_0)$	$\mathrm{P}(\mathrm{W} \ge \mathrm{W}_0 - 0.5)$

Here c needs to be determined by significance level (α)

H_0 is rejected when p-value<α

$W \sim N\left(\frac{n}{2}, \frac{n}{4}\right)$

CLT approximation

W_0: the observed test statistic

Example

For the same dataset, say we still want to test

$$H_0: m = 6.2 \text{ against the alternative hypothesis } H_1: m < 6.2.$$

What is the p-value of the test?

6.8	5.7	6.9	5.3	4.1	9.8	1.7	7.0
2.1	19.0	18.9	16.9	10.4	44.1	2.9	2.4
4.8	18.9	4.8	7.9				

$y = 9$

Large value of *y* favors H_1

p-value $= P(y \geq 9 | y \sim Bin(20,0.5)) = \sum_{l=9}^{20} \binom{n}{l} (0.5)^l (0.5)^{n-l} = 0.7483 \gg 0.05$

p-value > significance level, we fail to reject H_0.

Example

Again for the same dataset, say we want to test

$$H_0 : m = 7.2, \quad H_1 : m \neq 7.2.$$

For such two-sided test, we construct a two-sided rejection region C:

Reject H_0 when $y \geq 15$ or $y \leq 5$

significance level

$$\alpha = P(y \geq 15 \text{ or } y \leq 5 | y \sim Bin(20,0.5)) = 1 - \sum_{l=6}^{14} \binom{n}{l} (0.5)^l (0.5)^{n-l} = 0.0414$$

6.8	5.7	6.9	5.3	4.1	9.8	1.7	7.0
2.1	19.0	18.9	16.9	10.4	44.1	2.9	2.4
4.8	18.9	4.8	7.9				

$y = 12$

fail to reject H_0.

p-value $= 2P(y \geq 12 | y \sim Bin(20,0.5)) = 0.5034.$ ⇨ fail to reject H_0.

Wilcoxon Test

$X_1, \ldots, X_n \sim F$ $\quad$ $F(m) = 1/2$

Null Hypothesis

$H_0: m = m_0$

Wilcoxon test procedure

Magnitudes (not used in sign test)

1) Rank the absolute values $|X_1 - m_0|, |X_2 - m_0|, \ldots, |X_n - m_0|$, and denote the rank of $|X_k - m_0|$ as R_k.
2) With each R_k, we calculate the **signed ranks**
   - If $X_k - m_0 > 0$, we use R_k
   - If $X_k - m_0 < 0$, we use $-R_k$
3) The Wilcoxon signed rank statistic

$$W = \sum_{k=1}^{n} \text{signed ranks}$$

Example

Example 8.4-2

Suppose the lengths of $n = 10$ sunfish are

$$x_i: \ 5.0 \ 3.9 \ 5.2 \ 5.5 \ 2.8 \ 6.1 \ 6.4 \ 2.6 \ 1.7 \ 4.3$$

We shall test H_0: $m = 3.7$ against the alternative hypothesis H_1: $m > 3.7$. Thus, we have

$x_k - m_0$:	1.3,	0.2,	1.5,	1.8,	−0.9,	2.4,	2.7,	−1.1,	−2.0,	0.6
$\lvert x_k - m_0 \rvert$:	1.3,	0.2,	1.5,	1.8,	0.9,	2.4,	2.7,	1.1,	2.0,	0.6
Ranks:	5,	1,	6,	7,	3,	9,	10,	4,	8,	2
Signed Ranks:	5,	1,	6,	7,	−3,	9,	10,	−4,	−8,	2

Therefore, the Wilcoxon statistic is equal to

$$W = 5 + 1 + 6 + 7 - 3 + 9 + 10 - 4 - 8 + 2 = 25.$$

Distribution of W Under H_0

- Assume the random variable X is of the continuous type
- Assume that the pdf of X is **symmetric** about the median

Null Hypothesis

$$X_1, \ldots, X_n \sim F \qquad H_0: m = m_0 \qquad W = \sum_{k=1}^{n} \text{signed ranks}$$

Under H_0:

➡ ***W* has the same distribution as *V*.**

$$V = \sum_{k=1}^{n} V_k \qquad P(V_k = k) = P(V_k = -k) = \frac{1}{2}, \qquad k = 1, 2, \ldots, n.$$

$$E(V_k) = -k\left(\frac{1}{2}\right) + k\left(\frac{1}{2}\right) = 0$$

$$\text{Var}(V_k) = E(V_k^2) = (-k)^2\left(\frac{1}{2}\right) + (k)^2\left(\frac{1}{2}\right) = k^2$$

Distribution of W Under H_0

- Assume the random variable X is of the continuous type
- Assume that the pdf of X is **symmetric** about the median

Null Hypothesis

$$X_1, \ldots, X_n \sim F \qquad H_0: m = m_0 \qquad W = \sum_{k=1}^{n} \text{signed ranks}$$

Under H_0

$$E(W) = E(V) = \sum_{k=1}^{n} E(V_k) = 0.$$

$$\text{Var}(W) = \text{Var}(V) = \sum_{k=1}^{n} \text{Var}(V_k) = \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

CLT Approximation of W Under H_0

- Assume the random variable X is of the continuous type
- Assume that the pdf of X is **symmetric** about the median

$$X_1, \ldots, X_n \sim F$$

Null Hypothesis

$$H_0: m = m_0$$

$$W = \sum_{k=1}^{n} \text{signed ranks}$$

Under H_0

CLT (Lyapunov) approximation:

$$Z = \frac{W - 0}{\sqrt{n(n+1)(2n+1)/6}} \sim N(0,1)$$

Note: *W* always differ by 2

(Whole-unit) continuity correction:

$$P(W \geq k) = P(W \geq k-1) = P\left(\underbrace{\frac{W - 0}{\sqrt{n(n+1)(2n+1)/6}}}_{N(0,1)} \geq \frac{k-1}{\sqrt{n(n+1)(2n+1)/6}}\right) = 1 - \Phi\left(\frac{k-1}{\sqrt{n(n+1)(2n+1)/6}}\right)$$

$$P(W \leq k) = P(W \leq k+1) = \Phi\left(\frac{k+1}{\sqrt{n(n+1)(2n+1)/6}}\right)$$

Example

Example 8.4-2

Suppose the lengths of $n = 10$ sunfish are

x_i: 5.0 3.9 5.2 5.5 2.8 6.1 6.4 2.6 1.7 4.3

We shall test H_0: $m = 3.7$ against the alternative hypothesis H_1: $m > 3.7$. Thus, we have

$x_k - m_0$:	1.3,	0.2,	1.5,	1.8,	−0.9,	2.4,	2.7,	−1.1,	−2.0,	0.6
$\lvert x_k - m_0 \rvert$:	1.3,	0.2,	1.5,	1.8,	0.9,	2.4,	2.7,	1.1,	2.0,	0.6
Ranks:	5,	1,	6,	7,	3,	9,	10,	4,	8,	2
Signed Ranks:	5,	1,	6,	7,	−3,	9,	10,	−4,	−8,	2

Therefore, the Wilcoxon statistic is equal to

$$W = 5 + 1 + 6 + 7 - 3 + 9 + 10 - 4 - 8 + 2 = 25.$$

$n = 10$, and so $n(n+1)(2n+1)/6 = 385$. The test is

$$H_0 : m = 3.7, \quad H_1 : m > 3.7.$$

The Wilcoxon signed rank statistic is

$$w = 25.$$

The p-value of the test is

(Whole-unit) continuity correction:

$$P(W \geq 25; H_0) = 1 - P(W \leq 23; H_0) \approx 1 - \Phi\left(\frac{24}{\sqrt{385}}\right) = 0.11,$$

we fail to reject H_0 at 5% significance.

Extension: data with ties

- If we have $|X_i - m_0| = 0$, simply delete this data and perform the test on remaining samples
- If we have $|X_i - m_0| = |X_j - m_0|$ for some $i \neq j$, then each observation is assigned the average of the corresponding ranks

Example: $H_0: m = 0$ $H_1: m > 0$

1.8 −3.1 0.1 1.1 0.6 −5.1 9.2 0.2 0.4
0.0 1.9 −0.4 −1.5 1.4 −1.0 2.2 0.8 −0.4
2.0 −5.8 −3.4 −2.3 3.0 2.7 0.2 3.2

Raw data

Table 8.4-2 Ordered absolute values, changes in percentage of body fat

0.1	0.2	0.2	0.4	0.4	0.4	0.6	0.8	1.0	1.1	1.4	1.5	1.8
1	2.5	2.5	5	-5	-5	7	8	-9	10	11	-12	13
1	2.5	2.5	5	5	5	7	8	9	10	11	12	13

1.9	2.0	2.2	2.3	2.7	3.0	3.1	3.2	3.4	5.1	5.8	9.2
14	15	16	-17	18	19	-20	21	-22	-23	-24	25
14	15	16	17	18	19	20	21	22	23	24	25

Signed ranks

those that were originally negative underlined

$$w = 1 + 2.5 + 2.5 + 5 - 5 - 5 + \cdots + 25 = 51.$$

$$p\text{-value} = P(W \geq 51) = P(W \geq 50) \quad \text{(Whole-unit) continuity correction}$$

$$\approx P\left(Z \geq \frac{50 - 0}{\sqrt{(25)(26)(51)/6}}\right) = P(Z \geq 0.673) = 0.2505.$$

Not Required

Wilcoxon rank-sum Test (Mann-Whitney test)

Two samples: Assume F and G have the same shape and spread but possibly different locations, i.e. $G(x) = F(x + c)$

$$X_1, \dots, X_n \sim F$$
$$Y_1, \dots, Y_m \sim G$$

Null Hypothesis

$$H_0: F = G \quad (m_X = m_Y)$$

Alternative Hypothesis

$$H_1: m_X < m_Y$$
$$H_1: m_X > m_Y$$
$$H_1: m_X \neq m_Y$$

Intuition:

1) Rank the two samples together $\{X_1, \dots, X_n, Y_1, \dots, Y_m\}$
2) If $F = G$, the rank of Y are random samples from $\{1,2 \dots, n + m\}$, and each rank is uniform in $\{1,2 \dots, n + m\}$
3) We calculate the sum of ranks for Y, denoted as T_Y, and reject the null hypothesis if the sum is too large or too small.

Let T_Y denote the sum of the ranks of $Y_1, Y_2, \dots, Y_m$.

Not Required

Wilcoxon rank-sum Test (Mann-Whitney test)

Under $H_0: m = m_0$

$$E(T_Y) = \frac{m(m+n+1)}{2}$$

$$\mathrm{Var}(T_Y) = \frac{mn(m+n+1)}{12}$$

When m and n are sufficiently large (>7)

$$\frac{T_Y - E(T_Y)}{\sqrt{Var(T_Y)}} \sim N(0,1)$$

When m and n are small, tables of critical values computed from combinatorial are available.

Chi-square statistics

Chi-square test with known parameter

Chi-square test with unknown parameter

CHI-SQUARE GOODNESS-OF-FIT TEST

Overview

- Given *n* samples
- Discretize the sample space into *k* intervals
- Let O_i be the observed frequency in the *i*-th interval
- From hypothesized probability distribution, we compute the expected frequency in the *i*-th interval, denoted as E_i
- The chi-square test statistic

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

- Under the null, $\chi^2 \sim \chi^2(k-1-d)$ (approximately)
- *d* is the number of parameters that have been estimated

(d=0 if all parameters are known)

Example

- Let *X* denote the number of heads that occur when four coins are tossed at random. Under the assumption that the four coins are independent and the probability of heads on each coin is 1/2, $X \sim Bin\left(4, \frac{1}{2}\right)$.
- 100 repetitions of this experiment
- Do these results support the assumption?

	observed	expected
$A_1 = \{0\}$	7	6.26
$A_2 = \{1\}$	18	25
$A_3 = \{2\}$	40	37.5
$A_4 = \{3\}$	31	25
$A_5 = \{4\}$	4	6.25

$$H_0: p_i = p_{i0}, \qquad i = 1, 2, \ldots, 5,$$

k=5

d=0

(no unknown parameter)

$$p_{10} = p_{50} = \binom{4}{0}\left(\frac{1}{2}\right)^4 = \frac{1}{16} = 0.0625,$$

$$p_{20} = p_{40} = \binom{4}{1}\left(\frac{1}{2}\right)^4 = \frac{4}{16} = 0.25,$$

$$p_{30} = \binom{4}{2}\left(\frac{1}{2}\right)^4 = \frac{6}{16} = 0.375.$$

Example

	observed	expected
$A_1 = \{0\}$	7	6.26
$A_2 = \{1\}$	18	25
$A_3 = \{2\}$	40	37.5
$A_4 = \{3\}$	31	25
$A_5 = \{4\}$	4	6.25

- Chi-square test statistic:

k=5

$$\frac{(7-6.25)^2}{6.25} + \frac{(18-25)^2}{25} + \frac{(40-37.5)^2}{37.5} + \frac{(31-25)^2}{25} + \frac{(4-6.25)^2}{6.25} = 4.47 < \chi^2_{0.05}(4) = 9.488$$

- Fail to reject the null hypothesis.

Goodness-of-fit test (with unspecified parameter)

Example

Data

$$X_1, \dots, X_n \sim \text{Bin}(4, p)$$

***p* unknown**

Null Hypothesis

$$H_0 : p_i = p_{i0}, \quad i = 1, 2, \dots, k,$$

$$p_{i0} = P(A_i) = \frac{4!}{(i-1)!\,(5-i)!} p^{i-1}(1-p)^{5-i}, i = 1,2, \dots, 5$$

p_{0i} also unknown (depend on *p*)

	observed	expected
$A_1 = \{0\}$	7	$100(1-p)^4$
$A_2 = \{1\}$	18	$100 * 4p(1-p)^3$
$A_3 = \{2\}$	40	$100 * 6p^2(1-p)^2$
$A_4 = \{3\}$	31	$100 * 4p^3(1-p)^1$
$A_5 = \{4\}$	4	$100p^4$

Estimate *p* from data and plug-in

minimum chi-square estimator $\tilde{p}$

k=5

d=1

(1 unknown parameter *p*)

$$\chi^2(\tilde{p}) \sim \chi^2(5-1-1)$$

the degrees of freedom is further reduced by one for each parameter estimated by the minimum chi-square technique.

Goodness-of-fit test (with unspecified parameter)

Data

$$Y_1, \ldots, Y_n \sim \text{Multinomial}(n, p_1, \ldots, p_k)$$

k categories

Null Hypothesis

$$H_0 : p_i = p_{i0}, \quad i = 1, 2, \ldots, k,$$

p_{0i} unknown (depend on parameters)

d unknown parameters

⇩

Estimate *d* parameters using maximum likelihood estimate
(or MOM, whichever easier to compute)

⇩

(Chi-square) Test Statistic

$$\chi^2 = \sum_{i=1}^{k} \frac{(y_i - n\tilde{p}_{0i})^2}{n\tilde{p}_{0i}}$$

Distribution under H_0

$$\chi^2 \sim \chi^2(k - 1 - d)$$

Critical (Rejection) region

$$\chi^2 > \chi^2_\alpha(k - 1 - d)$$

at significance level α

p-value $P(\chi^2_\alpha(k - 1 - d) > \chi^2)$

$\tilde{p}_{0i}$ is calculated using the maximum likelihood estimate in practice

Testing Poisson

Poisson Distribution

The number of defects in printed circuit boards is hypothesized to follow a Poisson distribution. A random sample of $n = 60$ printed boards has been collected, and the following number of defects observed.

Number of Defects	Observed Frequency
0	32
1	15
2	9
3	4

Estimate (MLE) $\hat{\lambda} = 0.75$

$$p_1 = P(X = 0) = \frac{e^{-0.75}(0.75)^0}{0!} = 0.472$$

$$p_2 = P(X = 1) = \frac{e^{-0.75}(0.75)^1}{1!} = 0.354$$

$$p_3 = P(X = 2) = \frac{e^{-0.75}(0.75)^2}{2!} = 0.133$$

$$p_4 = P(X \geq 3) = 1 - (p_1 + p_2 + p_3) = 0.041$$

Number of Defects	Observed Frequency	Expected Frequency
0	32	28.32
1	15	21.24
2 (or more)	13	10.44

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} = \frac{(32 - 28.32)^2}{28.32} + \frac{(15 - 21.24)^2}{21.24} + \frac{(13 - 10.44)^2}{10.44} = 2.94$$

k=3

d=1

Conclusion: Since $\chi^2 < \chi^2_{0.05}(1) = 3.84$, fail to reject H_0

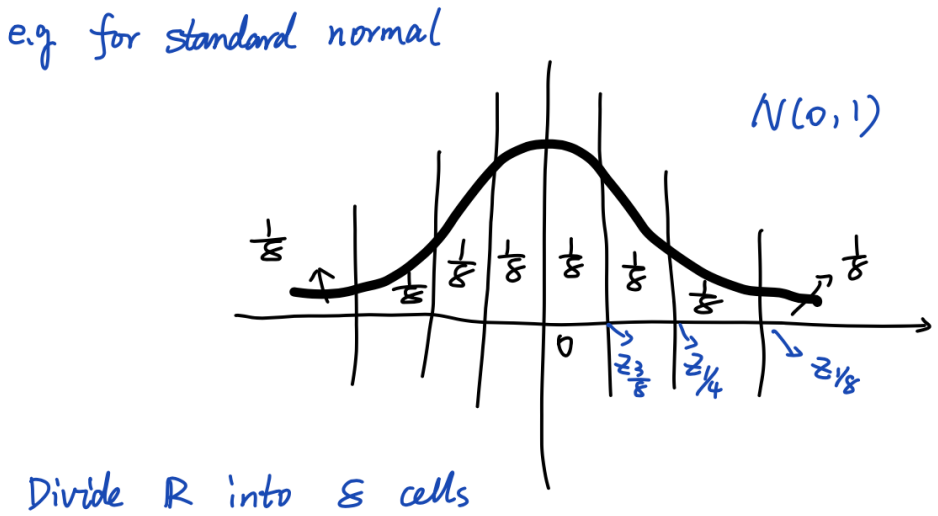
p-value: $P(\chi^2(1) > 2.94) = 0.086$

Example: Testing normal

> A manufacturing engineer is testing a power supply used in a notebook computer and, using $\alpha = 0.05$, wishes to determine whether output voltage is adequately described by a normal distribution. Sample estimates of the mean and standard deviation of $\bar{x} = 5.04\,\text{V}$ and $s = 0.08\,\text{V}$ are obtained from a random sample of $n = 100$ units.

Estimate *d*=2 parameters

- A common practice in constructing the time intervals used in the chi-square test is to choose the intervals so that the expected frequencies p_i are equal for all cells.
- Suppose we want to choose $k = 8$ cells, then the 8 intervals will be chosen such that p_i=1/8



Example: Testing normal

If $X \sim N(\mu, \sigma^2)$

probability Poi

A_1: $(-\infty, \quad -z_{1/8}\sigma+\mu)$ 0.125

A_2: $[-z_{1/8}\sigma+\mu, \; -z_{1/4}\sigma+\mu)$ 0.125

A_3: $[-z_{1/4}\sigma+\mu, \; -z_{3/8}\sigma+\mu)$ 0.125

A_4: $[-z_{3/8}\sigma+\mu, \; \mu)$ 0.125

A_5: $[\mu, \quad z_{3/8}\sigma+\mu)$ 0.125

A_6 $[z_{3/8}\sigma+\mu, \; z_{1/4}\sigma+\mu)$ 0.125

A_7: $[z_{1/4}\sigma+\mu, \; z_{1/8}\sigma+\mu)$ 0.125

A_8 $[z_{1/8}\sigma+\mu, \; +\infty)$ 0.125

Class Interval	Observed Frequency o_i	Expected Frequency E_i
$x < 4.948$	12	12.5
$4.948 \le x < 4.986$	14	12.5
$4.986 \le x < 5.014$	12	12.5
$5.014 \le x < 5.040$	13	12.5
$5.040 \le x < 5.066$	12	12.5
$5.066 \le x < 5.094$	11	12.5
$5.094 \le x < 5.132$	12	12.5
$5.132 \le x$	14	12.5
Totals	100	100

$\hat{\mu} = 5.04$. $\hat{\sigma} = 0.08$

$z_{1/8} = 1.15$ $z_{1/4} = 0.67$ $z_{3/8} = 0.32$

$\Rightarrow$ A_1: $(-\infty, \; -1.15\times 0.08 + 5.04) = (-\infty, 4.948)$

A_2: $[4.948, \; -0.67\times 0.08 + 5.04) = [4.948, 4.986)$

$\vdots$

Example: Testing normal

Null hypothesis: H_0: The form of the distribution is normal.

Alternative hypothesis: H_1: The form of the distribution is nonnormal.

Test statistic: The test statistic is

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} = \frac{(12 - 12.5)^2}{12.5} + \frac{(14 - 12.5)^2}{12.5} + \cdots + \frac{(14 - 12.5)^2}{12.5} = 0.64$$

Class Interval	Observed Frequency o_i	Expected Frequency E_i
$x < 4.948$	12	12.5
$4.948 \le x < 4.986$	14	12.5
$4.986 \le x < 5.014$	12	12.5
$5.014 \le x < 5.040$	13	12.5
$5.040 \le x < 5.066$	12	12.5
$5.066 \le x < 5.094$	11	12.5
$5.094 \le x < 5.132$	12	12.5
$5.132 \le x$	14	12.5
Totals	100	100

Reject H_0 if: Since two parameters in the normal distribution have been estimated, the chi-square statistic above will have $k - d - 1 = 8 - 2 - 1 = 5$ degrees of freedom. We will use a fixed significance level test with $\alpha = 0.05$. Therefore, we will reject H_0 If $\chi^2 > \chi^2_{0.05}(5)$.

Conclusion: Since $\chi^2 < \chi^2_{0.05}(5) = 11.07$, fail to reject H_0

p-value: $P(\chi^2(5) > 0.64) = 0.9861$

Test for homogeneity

Test for independence

CONTINGENCY TABLES

Test for Homogeneity

- Two independent experiments, each experiment has *k* mutually exclusive and exhaustive outcomes, $A_1, A_2, \ldots, A_k$

	A_1	A_2	A_3	$\cdots$	A_k
First experiment	$p_{1,1}$	$p_{1,2}$	$p_{1,3}$	$\cdots$	$p_{1,k}$
Second experiment	$p_{2,1}$	$p_{2,2}$	$p_{2,3}$	$\cdots$	$p_{2,k}$

Two multinomial distributions

$$p_{1,1} + p_{1,2} + \cdots p_{1,k} = 1$$

$$p_{2,1} + p_{2,2} + \cdots p_{2,k} = 1$$

- Test homogeneity (the corresponding probabilities associated with the two independent experiments are equal)

$$H_0: \ p_{1,1} = p_{2,1}, \ p_{1,2} = p_{2,2}, \ p_{1,3} = p_{2,3}, \ldots, \ p_{1,k} = p_{2,k}$$

The Contingency Table (summarize)

	A_1	A_2	A_3	$\cdots$	A_k	Total
First experiment	$Y_{1,1}$ (Observed) $n_1\hat{p}_1$ (expected)	$Y_{1,2}$ $n_1\hat{p}_2$	$Y_{1,3}$ $n_1\hat{p}_3$	$\cdots$	$Y_{1,k}$ $n_1\hat{p}_k$	n_1
Second experiment	$Y_{2,1}$ $n_2\hat{p}_1$	$Y_{2,2}$ $n_2\hat{p}_2$	$Y_{2,3}$ $n_3\hat{p}_3$	$\cdots$	$Y_{2,k}$ $n_2\hat{p}_k$	n_2
Total	$Y_{1,1}+Y_{2,1}$ $\left(\hat{p}_1=\frac{Y_{1,1}+Y_{2,1}}{n_1+n_2}\right)$	$Y_{1,2}+Y_{2,2}$ $\left(\hat{p}_2=\frac{Y_{1,2}+Y_{2,2}}{n_1+n_2}\right)$	$Y_{1,3}+Y_{2,3}$ $\left(\hat{p}_3=\frac{Y_{1,3}+Y_{2,3}}{n_1+n_2}\right)$		$Y_{1,k}+Y_{2,k}$ $\left(\hat{p}_k=\frac{Y_{1,k}+Y_{2,k}}{n_1+n_2}\right)$	n_1+n_2

$$\chi^2=\sum_{i=1}^{2}\sum_{j=1}^{k}\frac{\left(y_{i,j}-n_i\hat{p}_j\right)^2}{n_i\hat{p}_j}\sim \boldsymbol{\chi^2(k-1)}$$

Critical (Rejection) region

$$\chi^2>\chi^2_\alpha(k-1)$$

at significance level α

p-value

$$P(\chi^2(k-1)>\chi^2)$$

h groups and k categories

- h (e.g., $h = 2$ in previous examples) independent experiments, each experiment has *k* mutually exclusive and exhaustive outcomes, $A_1, A_2, \dots, A_k$

	A_1	A_2	A_3	$\cdots$	A_k
First experiment	$p_{1,1}$	$p_{1,2}$	$p_{1,3}$	$\cdots$	$p_{1,k}$
Second experiment	$p_{2,1}$	$p_{2,2}$	$p_{2,3}$	$\cdots$	$p_{2,k}$
……	…	…	…	…	…
h-th experiment	$p_{h,1}$	$p_{h,2}$	$p_{h,3}$	$\cdots$	$p_{h,k}$

multinomial distributions

$$p_{1,1} + p_{1,2} + \cdots p_{1,k} = 1$$
$$\vdots$$
$$p_{h,1} + p_{h,2} + \cdots p_{h,k} = 1$$

- Test homogeneity (the corresponding probabilities associated with the *h* independent experiments are equal)

$$H_0: \ p_{i,j} = p_j, \forall i = 1,2, \dots, h, \qquad j = 1,2, \dots, k$$

$h \times k$ Contingency Table

	A_1	A_2	A_3	$\cdots$	A_k	Total
First	$Y_{1,1}$ (Observed) $n_1\hat{p}_1$ (expected)	$Y_{1,2}$ $n_1\hat{p}_2$	$Y_{1,3}$ $n_1\hat{p}_3$	$\cdots$	$Y_{1,k}$ $n_1\hat{p}_k$	n_1
Second	$Y_{2,1}$ $n_2\hat{p}_1$	$Y_{2,2}$ $n_2\hat{p}_2$	$Y_{2,3}$ $n_3\hat{p}_3$	$\cdots$	$Y_{2,k}$ $n_2\hat{p}_k$	n_2
…	…	…	…		…	…
h-th	$Y_{h,1}$ $n_h\hat{p}_1$	$Y_{h,2}$ $n_h\hat{p}_2$	$Y_{h,3}$ $n_h\hat{p}_3$		$Y_{h,k}$ $n_h\hat{p}_k$	n_k
Total	$Y_{1,1} + \cdots + Y_{h,1}$ $\hat{p}_1 = \frac{Y_{1,1} + \cdots + Y_{h,1}}{n_1 + \cdots + n_h}$	$Y_{1,2} + \cdots + Y_{h,2}$ $\hat{p}_2 = \frac{Y_{1,2} + \cdots + Y_{h,2}}{n_1 + \cdots + n_h}$	$Y_{1,3} + \cdots + Y_{h,3}$ $\hat{p}_3 = \frac{Y_{1,3} + \cdots + Y_{h,3}}{n_1 + \cdots + n_h}$		$Y_{1,k} + \cdots + Y_{h,k}$ $\hat{p}_k = \frac{Y_{1,k} + \cdots + Y_{h,k}}{n_1 + \cdots + n_h}$	$n_1 + \cdots + n_h$

$$\chi^2 = \sum_{i=1}^{h}\sum_{j=1}^{k} \frac{\left(y_{i,j} - n_i\hat{p}_j\right)^2}{n_i\hat{p}_j} \sim \boldsymbol{\chi^2\left((h-1)(k-1)\right)}$$

Critical (Rejection) region

$$\chi^2 > \chi^2_\alpha((h-1)(k-1))$$

p-value

$$P(\chi^2_\alpha((h-1)(k-1)) > \chi^2)$$

at significance level α

Example

- Suppose we have the following 3×2 Contingency Table, test the homogeneity

Group	L	H	Totals
1	9	1	10
2	5	5	10
3	1	9	10
Totals	15	15	30

Example.

Group	Category L	H	Total
1	9 (5)	1 (5)	10
2	5 (5)	5 (5)	10
3	1 (5)	9 (5)	10
Total	15	15	30
	$\hat{P}_L = \frac{1}{2}$	$\hat{P}_H = \frac{1}{2}$	

expected. $10 \times \hat{P}_L = 10 \times \hat{P}_H = 5$

Example

chi-square statistics

$$\chi^2 = \frac{(9-5)^2}{5} + \frac{(1-5)^2}{5} + \frac{(5-5)^2}{5} + \frac{(5-5)^2}{5} + \frac{(1-5)^2}{5} + \frac{(9-5)^2}{5} = 12.5$$

degree of freedom: $(h-1)(k-1) = (3-1)\times(2-1) = 2$

critical region: $\{\chi^2 > \chi^2_{0.05}(2) = 5.99\}$.

Reject H_0.

p-value $= P(\chi^2(2) > 12.5) = 0.002 < 0.05$

Testing Independence

- two different attributes
- The first attribute is assigned to one and only one of *h* mutually exclusive and exhaustive event—say $A_1, A_2, \dots, A_h$ —and the second attribute falls into one and only one of *k* mutually exclusive and exhaustive events—say $B_1, B_2, \dots, B_k$

$$p_{i,j} = P(A_i \cap B_j), \qquad i = 1,2, \dots, h, j = 1,2, \dots, k$$

- Denote $p_{i,\cdot} = \sum_{j=1}^{k} p_{i,j}$ as the marginal distribution for *A*
- Denote $p_{\cdot,j} = \sum_{i=1}^{h} p_{i,j}$ as the marginal distribution for *B*

$$\Rightarrow \quad H_0: \ p_{i,j} = p_{i,\cdot} \times p_{\cdot,j}, \qquad \forall\, i = 1,2, \dots, h, \forall\, j = 1,2, \dots, k$$

A and B are independent

$h \times k$ Contingency Table

	B_1	B_2	$\cdots$	B_k	Total
A_1	$Y_{1,1}$ (Observed) $n\hat{p}_{1,\cdot}\hat{p}_{\cdot,1}$ (expected)	$Y_{1,2}$ $n\hat{p}_{1,\cdot}\hat{p}_{\cdot,2}$	$\cdots$	$Y_{1,k}$ $n\hat{p}_{1,\cdot}\hat{p}_{\cdot,k}$	$Y_{1,\cdot}$ $\hat{p}_{1,\cdot} = Y_{1,\cdot}/n$
A_2	$Y_{2,1}$ $n\hat{p}_{2,\cdot}\hat{p}_{\cdot,1}$	$Y_{2,2}$ $n\hat{p}_{2,\cdot}\hat{p}_{\cdot,2}$	$\cdots$	$Y_{2,k}$ $n\hat{p}_{2,\cdot}\hat{p}_{\cdot,k}$	$Y_{2,\cdot}$ $\hat{p}_{1,\cdot} = Y_{2,\cdot}/n$
$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
A_h	$Y_{h,1}$ $n\hat{p}_{h,\cdot}\hat{p}_{\cdot,1}$	$Y_{h,2}$ $n\hat{p}_{h,\cdot}\hat{p}_{\cdot,2}$	$\cdots$	$Y_{h,k}$ $n\hat{p}_{h,\cdot}\hat{p}_{\cdot,k}$	$Y_{h,\cdot}$ $\hat{p}_{h,\cdot} = Y_{h,\cdot}/n$
Total	$Y_{\cdot,1}$ $\hat{p}_{\cdot,1} = Y_{\cdot,1}/n$	$Y_{\cdot,2}$ $\hat{p}_{\cdot,2} = Y_{\cdot,2}/n$		$Y_{\cdot,k}$ $\hat{p}_{\cdot,k} = Y_{\cdot,k}/n$	n

$$\chi^2 = \sum_{i=1}^{h}\sum_{j=1}^{k} \frac{\left(y_{i,j} - n\hat{p}_{i,\cdot}\hat{p}_{\cdot,j}\right)^2}{n\hat{p}_{i,\cdot}\hat{p}_{\cdot,j}} \sim \boldsymbol{\chi^2\left((h-1)(k-1)\right)}$$

Critical (Rejection) region

$$\chi^2 > \chi^2_\alpha((h-1)(k-1))$$

p-value

$$P(\chi^2_\alpha((h-1)(k-1)) > \chi^2)$$

Summary

- **Homogeneity Tests**
  - h groups of experiments, and k categories
  - Test whether the multinomial distribution for each group is the same

$$\chi^2 = \sum_{i=1}^{h}\sum_{j=1}^{k}\frac{\left(y_{i,j} - n_i\hat{p}_j\right)^2}{n_i\hat{p}_j} \sim \boldsymbol{\chi^2\left((h-1)(k-1)\right)}$$

- **Independence Tests**
  - Experiments has two attributes each with h and k possible outcomes, test whether they are independent

$$\chi^2 = \sum_{i=1}^{h}\sum_{j=1}^{k}\frac{\left(y_{i,j} - n\hat{p}_{i,\cdot}\hat{p}_{\cdot,j}\right)^2}{n\hat{p}_{i,\cdot}\hat{p}_{\cdot,j}} \sim \boldsymbol{\chi^2\left((h-1)(k-1)\right)}$$

One-way ANOVA

Two-way ANOVA

Two-way ANOVA with one observation per cell

ANOVA

One-factor ANOVA

- *m* treatments, *i*-th treatment result in observations sampled from a distribution $N(\mu_i, \sigma^2)$
- n_i samples from the *i*-th treatment

$$X_{ij} \overset{i.i.d.}{\sim} N(\mu_i, \sigma^2), \quad j = 1, 2, \ldots, n_i. \qquad i = 1, \ldots, m$$

grand total $n = \sum_{i=1}^{m} n_i$

- Test the equality of the *m* means

$$H_0 : \mu_1 = \mu_2 = \ldots = \mu_m = \mu, \quad H_1 : H_0 \text{ is not true}$$

Table 9.3-1 One-factor random samples

					Means
X_1:	X_{11}	X_{12}	$\cdots$	X_{1n_1}	$\overline{X}_{1\cdot}$
X_2:	X_{21}	X_{22}	$\cdots$	X_{2n_2}	$\overline{X}_{2\cdot}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_m:	X_{m1}	X_{m2}	$\cdots$	X_{mn_m}	$\overline{X}_{m\cdot}$
Grand Mean:					$\overline{X}_{\cdot\cdot}$

Notations

- Sample mean of the ith distribution:(treatments)

$$\boxed{\overline{X}_{i\cdot} = \frac{1}{n_i}\sum_{j=1}^{n_i} X_{ij}} \quad i = 1, 2, \ldots, m$$

- Overall sample mean (aka **grand mean**) of the entire data table:

$$\boxed{\overline{X}_{\cdot\cdot} = \frac{1}{n}\sum_{i=1}^{m}\sum_{j=1}^{n_i} X_{ij}} = \frac{1}{n}\sum_{i=1}^{m} n_i \overline{X}_{i\cdot}$$

- Overall sum-of-squares or **total sum-of-squares** $SS(TO)$:

$$\boxed{SS(TO) = \sum_{i=1}^{m}\sum_{j=1}^{n_i} \left(X_{ij} - \overline{X}_{\cdot\cdot}\right)^2}$$

Notations

- **Between-treatment sum-of-squares** $SS(T)$:

$$\boxed{SS(T) = \sum_{i=1}^{m} n_i \left(\overline{X}_{i\cdot} - \overline{X}_{\cdot\cdot}\right)^2}$$

- **Error sum-of-squares** or within-treatment sum-of-squares $SS(E)$:

$$\boxed{SS(E) = \sum_{i=1}^{m} \sum_{j=1}^{n_i} \left(X_{ij} - \overline{X}_{i\cdot}\right)^2}$$

- Within-treatment sum-of-squares of treatment i for $i = 1, 2, \ldots, m$, denoted by $SS(E_i)$:

$$SS(E_i) = \sum_{j=1}^{n_i} \left(X_{ij} - \overline{X}_{i\cdot}\right)^2$$

$$SS(E) = \sum_{i=1}^{m} SS(E_i)$$

Relations and distributions

$$H_0 : \mu_1 = \mu_2 = \ldots = \mu_m = \mu, \quad H_1 : H_0 \text{ is not true}$$

Under H_0: $X_{i,j} \sim N(\mu, \sigma^2), \quad i = 1, \ldots, m, j = 1, \ldots, n_j$ grand total $n = \sum_{i=1}^{m} n_i$

$$\mathrm{SS(TO)} = \mathrm{SS(E)} + \mathrm{SS(T)}.$$

$$\frac{\mathrm{SS(TO)}}{\sigma^2} \sim \chi^2(n-1)$$

$$\frac{\mathrm{SS(E)}}{\sigma^2} \sim \chi^2(n-m)$$

$$\frac{\mathrm{SS(T)}}{\sigma^2} \sim \chi^2(m-1)$$

SS(E) and SS(T) are independent

Hypothesis Testing

Hypothesis

$$H_0 : \mu_1 = \mu_2 = \ldots = \mu_m = \mu, \quad H_1 : H_0 \text{ is not true}$$

Test statistics

$$F = \frac{\mathrm{SS(T)}/(m-1)}{\mathrm{SS(E)}/(n-m)} = \frac{[\mathrm{SS(T)}/\sigma^2]/(m-1)}{[\mathrm{SS(E)}/\sigma^2]/(n-m)}$$

Under H_0:

$$F \sim F(m-1, n-m)$$

$$\mathrm{SS(TO)} = \mathrm{SS(E)} + \mathrm{SS(T)}.$$

$\chi^2(n-1)$ $\chi^2(n-m)$ $\chi^2(m-1)$

SS(E) and SS(T) are independent

Critical (rejection) region:

$$F \geq F_\alpha(m-1, n-m).$$

***p*-value:**

$$P\{F(m-1, n-m) > F\}$$

One-factor ANOVA Table

Table 9.3-2 Analysis-of-variance table

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F Ratio
Treatment	SS(T)	$m-1$	$\text{MS(T)} = \dfrac{\text{SS(T)}}{m-1}$	$\dfrac{\text{MS(T)}}{\text{MS(E)}}$
Error	SS(E)	$n-m$	$\text{MS(E)} = \dfrac{\text{SS(E)}}{n-m}$	
Total	SS(TO)	$n-1$		

Two-factor ANOVA with c>1 observations per cell

- Two factors (attributes), one of which has a levels and the other b levels; There are thus n = ab possible interactions
- For the (i,j)-th combination, we have c observations that are sampled from a distribution

$$X_{i,j,k} \sim N(\mu_{i,j}, \sigma^2), k = 1, \dots, c$$

- Assume

$$\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$$

α_i: i-th row effect; β_j: j-th column effect; γ_{ij}: (i,j)-th interaction effect

Remark: without loss of generality, we assume $\sum_{i=1}^{a} \alpha_i = 0$ and $\sum_{j=1}^{b} \beta_j = 0$

$\sum_{i=1}^{a} \gamma_{ij} = 0$ and $\sum_{j=1}^{b} \gamma_{ij} = 0$.

Notations

Sample mean of the (i,j)-th interaction effect $\overline{X}_{ij\cdot} = \frac{1}{c}\sum_{k=1}^{c} X_{ijk},$

Sample mean of the i-th row factor $\overline{X}_{i\cdot\cdot} = \frac{1}{bc}\sum_{j=1}^{b}\sum_{k=1}^{c} X_{ijk},$

Sample mean of the j-th column factor $\overline{X}_{\cdot j\cdot} = \frac{1}{ac}\sum_{i=1}^{a}\sum_{k=1}^{c} X_{ijk},$

Overall sample mean $\overline{X}_{\cdots} = \frac{1}{abc}\sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{c} X_{ijk},$

Notations

- Overall sum-of-squares or total sum-of-squares

$$SS(TO) = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{c}\left(X_{ijk} - \overline{X}...\right)^2$$

- Sum-of-squares of Factor A (row factor)

$$SS(A) = bc\sum_{i=1}^{a}\left(\overline{X}_{i..} - \overline{X}...\right)^2$$

- Sum-of-squares of Factor B (column factor)

$$SS(B) = ac\sum_{j=1}^{b}\left(\overline{X}_{.j.} - \overline{X}...\right)^2$$

Notations

- Interaction sum-of-squares

$$SS(AB) = c\sum_{i=1}^{a}\sum_{j=1}^{b}(\overline{X}_{ij\cdot} - \overline{X}_{i\cdot\cdot} - \overline{X}_{\cdot j\cdot} + \overline{X}_{\cdot\cdot\cdot})^2$$

- Error sum-of-squares

$$SS(E) = \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{c}\left(X_{ijk} - \overline{X}_{ij\cdot}\right)^2$$

- Relations

$$SS(TO) = SS(A) + SS(B) + SS(E) + SS(AB)$$

Two-factor Hypothesis Testing

We want to test:

$$\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$$

i-th row effect; j-th column effect; (i,j)-th interaction effect

1. (Is there an effect on μ due to row?)

$$H_{0A} : \alpha_1 = \alpha_2 = \ldots = \alpha_a = 0, \quad H_{1A} : H_{0A} \text{ is not true}$$

2. (Is there an effect on μ due to column?)

$$H_{0B} : \beta_1 = \beta_2 = \ldots = \beta_b = 0, \quad H_{1B} : H_{0B} \text{ is not true}$$

3. (Is there an effect on μ due to interactions between the row and the column?)

$$H_{0AB} : \gamma_{ij} = 0, \text{ for all } i, j, \quad H_{1AB} : H_{0AB} \text{ is not true}$$

Hypothesis Testing

$$H_{0A} : \alpha_1 = \alpha_2 = \ldots = \alpha_a = 0, \quad H_{1A} : H_{0A} \text{ is not true}$$

$$H_{0B} : \beta_1 = \beta_2 = \ldots = \beta_b = 0, \quad H_{1B} : H_{0B} \text{ is not true}$$

$$H_{0AB} : \gamma_{ij} = 0, \text{ for all } i, j, \quad H_{1AB} : H_{0AB} \text{ is not true}$$

If H_{0A}, H_{0B} and H_{0AB} are true, then

$$\underbrace{\frac{SS(TO)}{\sigma^2}}_{\sim\chi^2(abc-1)} = \underbrace{\frac{SS(A)}{\sigma^2}}_{\sim\chi^2(a-1)} + \underbrace{\frac{SS(B)}{\sigma^2}}_{\sim\chi^2(b-1)} + \underbrace{\frac{SS(E)}{\sigma^2}}_{\sim\chi^2(ab(c-1))} + \underbrace{\frac{SS(AB)}{\sigma^2}}_{\sim\chi^2((a-1)(b-1))},$$

Hypothesis Testing

Hypothesis

$H_{0A}: \alpha_1 = \alpha_2 = \ldots = \alpha_a = 0, \quad H_{1A}: H_{0A}$ is not true

Test statistics

$$F_A = \frac{\mathrm{SS(A)}/(a-1)}{\mathrm{SS(E)}/[ab(c-1)]}$$

Under H_{0A}:

$$F_A \sim F(a-1, ab(c-1))$$

Critical (rejection) region:

$$F_A \geq F_\alpha(a-1, ab(c-1))$$

***p*-value:**

$$P\{F(a-1, ab(c-1)) \geq F_A\}$$

Hypothesis

$H_{0B}: \beta_1 = \beta_2 = \ldots = \beta_b = 0, \quad H_{1B}: H_{0B}$ is not true

Test statistics

$$F_B = \frac{\mathrm{SS(B)}/(b-1)}{\mathrm{SS(E)}/[ab(c-1)]}$$

Under H_{0B}:

$$F_B \sim F(b-1, ab(c-1))$$

Critical (rejection) region:

$$F_B \geq F_\alpha(b-1, ab(c-1))$$

***p*-value:**

$$P\{F(b-1, ab(c-1)) \geq F_B\}$$

Hypothesis Testing

Hypothesis

$$H_{0AB}: \gamma_{ij} = 0, \text{ for all } i, j, \quad H_{1AB}: H_{0AB} \text{ is not true}$$

Test statistics

$$F_{AB} = \frac{\text{SS(AB)}/[(a-1)(b-1)]}{\text{SS(E)}/[ab(c-1)]}$$

Under H_{0AB}:

$$F_{AB} \sim F\big((a-1)(b-1), ab(c-1)\big)$$

Critical (rejection) region:

$$F_{AB} \geq F_{\alpha}\big((a-1)(b-1), ab(c-1)\big)$$

***p*-value:**

$$P\big\{F\big((a-1)(b-1), ab(c-1)\big) \geq F_{AB}\big\}$$

Two-factor ANOVA Table

Table 9.4-4 Two-way ANOVA table, c observations per cell

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F
Factor A (row)	SS(A)	$a-1$	$\text{MS(A)} = \dfrac{\text{SS(A)}}{a-1}$	$\dfrac{\text{MS(A)}}{\text{MS(E)}}$
Factor B (column)	SS(B)	$b-1$	$\text{MS(B)} = \dfrac{\text{SS(B)}}{b-1}$	$\dfrac{\text{MS(B)}}{\text{MS(E)}}$
Factor AB (interaction)	SS(AB)	$(a-1)(b-1)$	$\text{MS(AB)} = \dfrac{\text{SS(AB)}}{(a-1)(b-1)}$	$\dfrac{\text{MS(AB)}}{\text{MS(E)}}$
Error	SS(E)	$ab(c-1)$	$\text{MS(E)} = \dfrac{\text{SS(E)}}{ab(c-1)}$	
Total	SS(TO)	$abc-1$		

Two-factor ANOVA with one observation per cell

- Two factors (attributes), one of which has a levels and the other b levels; There are thus n = ab possible combinations
- For the (i,j)-th combination, data are sampled from a distribution

$$X_{i,j} \sim N(\mu_{i,j}, \sigma^2)$$

- **One** samples from the (i,j)-th combination of treatment
- Assume

$$\mu_{ij} = \mu + \alpha_i + \beta_j$$

i-th row effect j-th column effect

Remark: without loss of generality, we assume $\sum_{i=1}^{a} \alpha_i = 0$ and $\sum_{j=1}^{b} \beta_j = 0$

Notations

- Sample mean of the i-th row factor

$$\overline{X}_{i\cdot} = \frac{1}{b}\sum_{j=1}^{b} X_{ij}$$

- Sample mean of the j-th column factor

$$\overline{X}_{\cdot j} = \frac{1}{a}\sum_{i=1}^{a} X_{ij}$$

- Overall sample mean of the entire data table

$$\overline{X}_{\cdot\cdot} = \frac{1}{ab}\sum_{i=1}^{a}\sum_{j=1}^{b} X_{ij}$$

Notations

- Overall sum-of-squares or total sum-of-squares

$$SS(TO) = \sum_{i=1}^{a}\sum_{j=1}^{b}\left(X_{ij} - \overline{X}_{..}\right)^2$$

- Sum-of-squares of Factor A (row factor)

$$SS(A) = b\sum_{i=1}^{a}\left(\overline{X}_{i.} - \overline{X}_{..}\right)^2$$

- Sum-of-squares of Factor B (column factor)

$$SS(B) = a\sum_{j=1}^{b}\left(\overline{X}_{.j} - \overline{X}_{..}\right)^2$$

Notations

- Error sum-of-squares

$$SS(E) = \sum_{i=1}^{a} \sum_{j=1}^{b} \left[X_{ij} - \underbrace{(\overline{X}_{i\cdot} - \overline{X}_{\cdot\cdot})}_{\widehat{\alpha}_i} - \underbrace{(\overline{X}_{\cdot j} - \overline{X}_{\cdot\cdot})}_{\widehat{\beta}_j} - \underbrace{\overline{X}_{\cdot\cdot}}_{\widehat{\mu}} \right]^2$$

- Relations

$$SS(TO) = SS(A) + SS(B) + SS(E)$$

Hypothesis Testing

$H_{0A}: \alpha_1 = \alpha_2 = \ldots = \alpha_a = 0, \quad H_{1A}: H_{0A}$ is not true

Under H_{0A} :

$$SS(A) = b\sum_{i=1}^{a}(\overline{X}_{i.} - \overline{X}_{..})^2$$

$$\frac{\mathrm{SS(A)}}{\sigma^2} \sim \chi^2(a-1)$$

$H_{0B}: \beta_1 = \beta_2 = \ldots = \beta_b = 0, \quad H_{1B}: H_{0B}$ is not true

Under H_{0B}:

$$SS(B) = a\sum_{j=1}^{b}(\overline{X}_{.j} - \overline{X}_{..})^2$$

$$\frac{\mathrm{SS(B)}}{\sigma^2} \sim \chi^2(b-1)$$

When both H_{0A} and H_{0B} are true:

$$SS(TO) = SS(A) + SS(B) + SS(E)$$

$$\frac{\mathrm{SS(TO)}}{\sigma^2} \sim \chi^2(ab-1)$$

$$\frac{\mathrm{SS(E)}}{\sigma^2} \sim \chi^2\big((a-1)(b-1)\big)$$

Hypothesis Testing

Hypothesis

$$H_{0A}: \alpha_1 = \alpha_2 = \ldots = \alpha_a = 0, \quad H_{1A}: H_{0A} \text{ is not true}$$

Test statistics

$$F_A = \frac{\mathrm{SS(A)}/[\sigma^2(a-1)]}{\mathrm{SS(E)}/[\sigma^2(a-1)(b-1)]} = \frac{\mathrm{SS(A)}/(a-1)}{\mathrm{SS(E)}/[(a-1)(b-1)]}$$

Under H_{0A}:

$$F_A \sim F(a-1, (a-1)(b-1))$$

Critical (rejection) region:

$$F_A \geq F_\alpha(a-1, (a-1)(b-1))$$

***p*-value:**

$$P\{F\ \ (a-1, (a-1)(b-1)) \geq F_A\}$$

Hypothesis

$$H_{0B}: \beta_1 = \beta_2 = \ldots = \beta_b = 0, \quad H_{1B}: H_{0B} \text{ is not true}$$

Test statistics

$$F_B = \frac{\mathrm{SS(B)}/[\sigma^2(b-1)]}{\mathrm{SS(E)}/[\sigma^2(a-1)(b-1)]} = \frac{\mathrm{SS(B)}/(b-1)}{\mathrm{SS(E)}/[(a-1)(b-1)]}$$

Under H_{0B}:

$$F_B \sim F(b-1, (a-1)(b-1))$$

Critical (rejection) region:

$$F_B \geq F_\alpha(b-1, (a-1)(b-1))$$

***p*-value:**

$$P\{F\ \ (b-1, (a-1)(b-1)) \geq F_B\}$$

Two-factor ANOVA Table

Table 9.4-1 Two-way ANOVA table, one observation per cell

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F
Factor A (row)	SS(A)	$a-1$	$\text{MS(A)} = \dfrac{\text{SS(A)}}{a-1}$	$\dfrac{\text{MS(A)}}{\text{MS(E)}}$
Factor B (column)	SS(B)	$b-1$	$\text{MS(B)} = \dfrac{\text{SS(B)}}{b-1}$	$\dfrac{\text{MS(B)}}{\text{MS(E)}}$
Error	SS(E)	$(a-1)(b-1)$	$\text{MS(E)} = \dfrac{\text{SS(E)}}{(a-1)(b-1)}$	
Total	SS(TO)	$ab-1$		

Summary

- **One-factor ANOVA**

Table 9.3-2 Analysis-of-variance table

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F Ratio
Treatment	SS(T)	$m-1$	$\text{MS(T)} = \frac{\text{SS(T)}}{m-1}$	$\frac{\text{MS(T)}}{\text{MS(E)}}$
Error	SS(E)	$n-m$	$\text{MS(E)} = \frac{\text{SS(E)}}{n-m}$	
Total	SS(TO)	$n-1$		

- **Two-factor ANOVA**

Table 9.4-1 Two-way ANOVA table, one observation per cell

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F
Factor A (row)	SS(A)	$a-1$	$\text{MS(A)} = \frac{\text{SS(A)}}{a-1}$	$\frac{\text{MS(A)}}{\text{MS(E)}}$
Factor B (column)	SS(B)	$b-1$	$\text{MS(B)} = \frac{\text{SS(B)}}{b-1}$	$\frac{\text{MS(B)}}{\text{MS(E)}}$
Error	SS(E)	$(a-1)(b-1)$	$\text{MS(E)} = \frac{\text{SS(E)}}{(a-1)(b-1)}$	
Total	SS(TO)	$ab-1$		

Table 9.4-4 Two-way ANOVA table, c observations per cell

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F
Factor A (row)	SS(A)	$a-1$	$\text{MS(A)} = \frac{\text{SS(A)}}{a-1}$	$\frac{\text{MS(A)}}{\text{MS(E)}}$
Factor B (column)	SS(B)	$b-1$	$\text{MS(B)} = \frac{\text{SS(B)}}{b-1}$	$\frac{\text{MS(B)}}{\text{MS(E)}}$
Factor AB (interaction)	SS(AB)	$(a-1)(b-1)$	$\text{MS(AB)} = \frac{\text{SS(AB)}}{(a-1)(b-1)}$	$\frac{\text{MS(AB)}}{\text{MS(E)}}$
Error	SS(E)	$ab(c-1)$	$\text{MS(E)} = \frac{\text{SS(E)}}{ab(c-1)}$	
Total	SS(TO)	$abc-1$		

Least square / maximum likelihood estimates

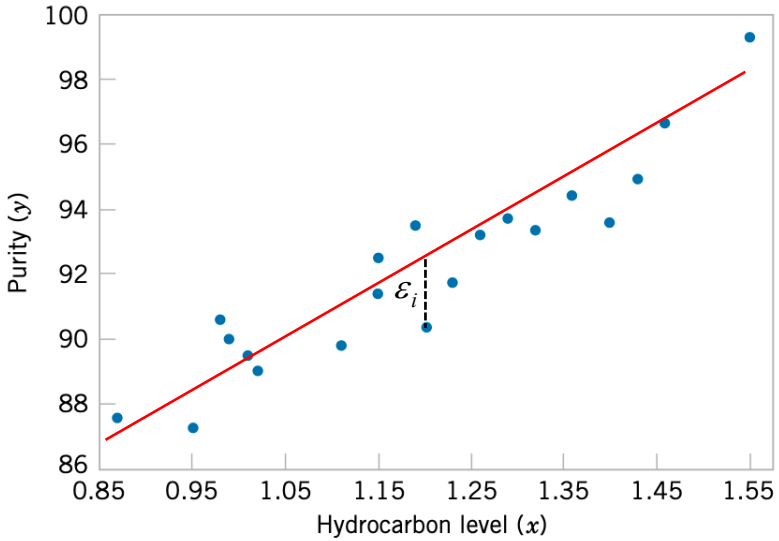
Confidence interval

Hypothesis testing

SIMPLE LINEAR REGRESSION

Simple Linear Regression

Simple linear regression model:



Response **Regressor or Predictor**

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \qquad i = 1,2,\cdots,n$$

Intercept Slope Random error

$$\varepsilon_i \sim N(0,\sigma^2)$$

Mean response $E(Y|x) = \beta_0 + \beta_1 x$

$$V(Y|x) = \sigma^2$$

the slope β_1 and intercept β_0 of the line are called regression coefficients.

Least Square/Maximum Likelihood Estimates

The least squares estimates (also MLE) of the intercept and slope in the simple linear regression model are

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i x_i - \frac{\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}$$

where $\bar{y} = (1/n)\sum_{i=1}^{n} y_i$ and $\bar{x} = (1/n)\sum_{i=1}^{n} x_i$.

$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ Fitted (estimated) regression model

$e_i = y_i - \hat{y}_i$ is called the **residual.**

Distribution of the estimates

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i x_i - \frac{\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}$$

where $\bar{y} = (1/n)\sum_{i=1}^{n} y_i$ and $\bar{x} = (1/n)\sum_{i=1}^{n} x_i$.

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}$$

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

$$Y_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2) \qquad i = 1, 2, \cdots, n$$

$$\hat{\beta}_1 = \frac{1}{S_{xx}} \sum_{i=1}^{n} y_i (x_i - \bar{x})$$

$$\hat{\beta}_0 = \sum_{i=1}^{n} y_i \left(\frac{1}{n} - \frac{\bar{x}}{S_{xx}} (x_i - \bar{x})\right)$$

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)\right)$$

Estimation of Variance (σ^2)

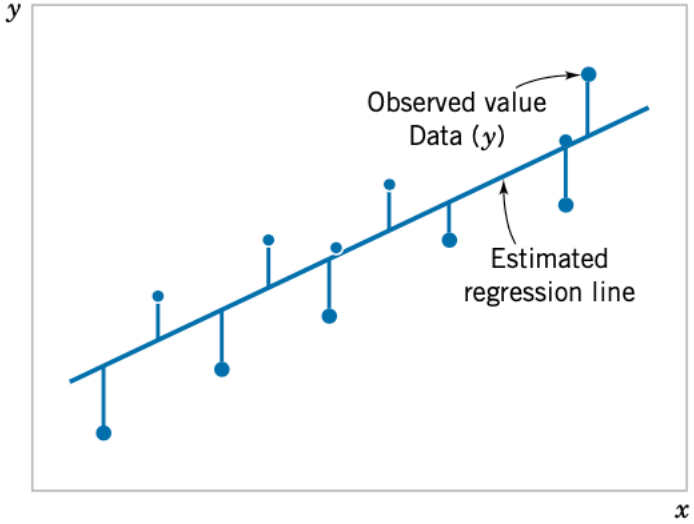
The **error sum of squares** is

$$SS_E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$SS_E = \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2$$

$$\frac{\sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2}{\sigma^2} \sim \chi^2(n-2)$$

$$\Rightarrow \quad E(SS_E) = (n-2)\sigma^2.$$



An unbiased estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{SS_E}{n-2}$$

Distribution of the estimate

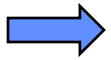
$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)\right)$$

$\hat{\sigma}^2, \hat{\beta}_0, \hat{\beta}_1$ are mutually independent

$$\hat{\sigma}^2 := \frac{1}{n-2}\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2$$

$$\frac{\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2}{\sigma^2} \sim \chi^2(n-2)$$



$$\frac{\hat{\beta}_1 - \beta_1}{\hat{\sigma}/\sqrt{S_{xx}}} \sim t(n-2)$$

$$\frac{\hat{\beta}_0 - \beta_0}{\hat{\sigma}/\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}} \sim t(n-2)$$

Confidence Interval for Regression Coefficients

Mean and variance of the slope estimator $E(\hat{\beta}_1) = \beta_1$ $\quad V(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}$

Use this to find standard error

Under the assumption that the observations are normally and independently distributed, a $100(1-\alpha)\%$ **confidence interval on the slope** β_1 in simple linear regression is

$$\hat{\beta}_1 - t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} \le \beta_1 \le \hat{\beta}_1 + t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}$$

Mean and variance of the intercept estimator $E(\hat{\beta}_0) = \beta_0$ and $V(\hat{\beta}_0) = \sigma^2\left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right]$

Similarly, a $100(1-\alpha)\%$ **confidence interval on the intercept** β_0 is

$$\hat{\beta}_0 - t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n}+\frac{\bar{x}^2}{S_{xx}}\right)} \le \beta_0 \le \hat{\beta}_0 + t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n}+\frac{\bar{x}^2}{S_{xx}}\right)}$$

$$S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}$$

$\text{cov}(\hat{\beta}_0, \hat{\beta}_1) = -\sigma^2\bar{x}/S_{xx}.$

The width of confidence interval indicates the overall quality of regression line.

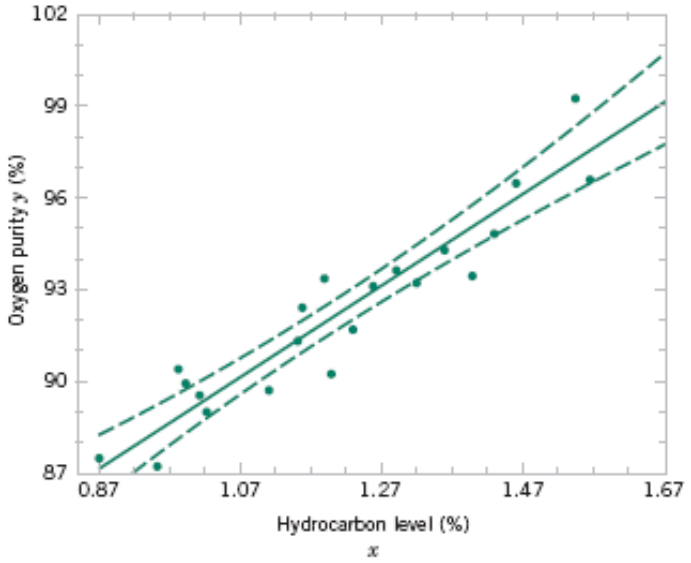
Confidence Interval on Mean Response

A $100(1-\alpha)\%$ **confidence interval about the mean response** at the value of $x = x_0$, say $\mu_{Y|x_0}$, is given by

$$\hat{\mu}_{Y|x_0} - t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)}$$

$$\leq \mu_{Y|x_0} \leq \hat{\mu}_{Y|x_0} + t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)}$$

where $\hat{\mu}_{Y|x_0} = \hat{\beta}_0 + \hat{\beta}_1 x_0$ is computed from the fitted regression model.



$$E(Y|x_0) = \mu_{Y|x_0} = \beta_0 + \beta_1 x_0$$

Prediction interval

- Predicting response for **new** observation
- The **new** observation is **independent** of data used to build linear regression model

A $100(1-\alpha)$ % **prediction interval on a future observation** Y_0 at the value x_0 is given by

$$\hat{y}_0 - t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(1+\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)} \leq Y_0 \leq \hat{y}_0 + t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(1+\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)}$$

The value $\hat{y}_0$ is computed from the regression model $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

An Alternative Perspective

- Sometimes, in order to simplify the model and computation, we may write

New Regressor or Predictor X_i'

$$Y_i = (\beta_0) + \beta_1(X_i - \bar{X}) + \epsilon_i$$

New Intercept — Slope (the same) — Random error

$$\varepsilon_i \sim N(0,\sigma^2)$$

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

Original model

- The least-square (also the maximum likelihood) estimate:

$$\hat{\beta}_0 = \bar{y}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i x_i - \frac{\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}$$

- Distributions:

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

$$\hat{\beta}_0 \sim N\left(\beta_0, \frac{\sigma^2}{n}\right)$$

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}$$

An Alternative Perspective

- Confidence Interval

$$Y_i = \beta_0 + \beta_1(X_i - \bar{X}) + \epsilon_i$$

Intercept $$[\hat{\beta}_0 - t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{n}}, \hat{\beta}_0 + t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{n}}]$$

Slope $$[\hat{\beta}_1 - t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}, \hat{\beta}_1 + t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}]$$

$$\hat{\sigma}^2 := \frac{1}{n-2}\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1(x_i - \bar{x})\right)^2$$

$$S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n}x_i^2 - \frac{\left(\sum_{i=1}^{n}x_i\right)^2}{n}$$

- Confidence Interval on mean response

$$[\hat{\mu}_{Y|x_0} - t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}, \hat{\mu}_{Y|x_0} + t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}]$$

$$\hat{y}_0 = \hat{\mu}_{Y|x_0} = \hat{\beta}_0 + \hat{\beta}_1(x_0 - \bar{x})$$

- Prediction Interval

$$[\hat{y}_0 - t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}, \hat{y}_0 + t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}]$$

Summary: Hypothesis testing on the Slope

Null Hypothesis $H_0: \beta_1 = \beta_{1,0}$

Test Statistic $t_0 = \dfrac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{\hat{\sigma}^2/S_{xx}}}$

Distribution under H_0 $t_0 \sim t(n-2)$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1: \beta_1 \neq \beta_{1,0}$	$\lvert t_0 \rvert > t_{\alpha/2}(n-2)$	$2\mathrm{P}(T > \lvert t_0 \rvert)$	$[\hat{\beta}_1 - t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}, \hat{\beta}_1 + t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}]$
$H_1: \beta_1 > \beta_{1,0}$	$t_0 > t_\alpha(n-2)$	$\mathrm{P}(T > t_0)$	$[\hat{\beta}_1 - t_\alpha(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}, +\infty)$
$H_1: \beta_1 < \beta_{1,0}$	$t_0 < -t_\alpha(n-2)$	$\mathrm{P}(T < t_0)$	$(-\infty, \hat{\beta}_1 + t_\alpha(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include $\beta_{1,0}$

Summary: Hypothesis testing on Intercept

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

Original model

Null Hypothesis

$$H_0: \beta_0 = \beta_{0,0}$$

Test Statistic

$$t_0 = \frac{\hat{\beta}_0 - \beta_{0,0}}{\sqrt{\hat{\sigma}^2\left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right]}}$$

Distribution under H_0

$$t_0 \sim t(n-2)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
Distribution under H_0	$\lvert t_0 \rvert > t_{\alpha/2}(n-2)$	$2\mathrm{P}(T > \lvert t_0 \rvert)$	$[\hat{\beta}_0 - t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)}, \hat{\beta}_0 + t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)}]$
$H_1: \beta_0 > \beta_{0,0}$	$t_0 > t_\alpha(n-2)$	$\mathrm{P}(T > t_0)$	$[\hat{\beta}_0 - t_\alpha(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)}, +\infty)$
$H_1: \beta_0 < \beta_{0,0}$	$t_0 < -t_\alpha(n-2)$	$\mathrm{P}(T < t_0)$	$(-\infty, \hat{\beta}_0 + t_\alpha(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)}]$

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include $\beta_{0,0}$

Summary: Inference on the Intercept

$$Y_i = \beta_0 + \beta_1(X_i - \bar{X}) + \epsilon_i$$

Null Hypothesis

$$H_0: \beta_0 = \beta_{0,0}$$

Test Statistic

$$t_0 = \frac{\hat{\beta}_0 - \beta_{0,0}}{\hat{\sigma}/\sqrt{n}}$$

Distribution under H_0

$$t_0 \sim t(n-2)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1: \beta_0 \neq \beta_{0,0}$	$\lvert t_0 \rvert > t_{\alpha/2}(n-2)$	$2\mathrm{P}(T > \lvert t_0 \rvert)$	$[\hat{\beta}_0 - t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{n}}, \hat{\beta}_0 + t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{n}}]$
$H_1: \beta_0 > \beta_{0,0}$	$t_0 > t_\alpha(n-2)$	$\mathrm{P}(T > t_0)$	$[\hat{\beta}_0 - t_\alpha(n-2)\sqrt{\frac{\hat{\sigma}^2}{n}}, +\infty)$
$H_1: \beta_0 < \beta_{0,0}$	$t_0 < -t_\alpha(n-2)$	$\mathrm{P}(T < t_0)$	$(-\infty, \hat{\beta}_0 + t_\alpha(n-2)\sqrt{\frac{\hat{\sigma}^2}{n}}]$

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include $\beta_{0,0}$

Testing Correlation

Testing correlation

- Let X, Y have a bivariate normal distribution

- Observe *n* pairs of data (each pair is sampled from the bivariate normal distribution and independent from other pairs):

$$(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$$

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = \sigma_{XY} \qquad \rho = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

- Test:

$$H_0: \rho = 0 \quad \text{v.s.} \quad H_1: \rho \neq 0 \quad H_1: \rho > 0 \quad H_1: \rho < 0$$

Null hypothesis

Sample Correlation Coefficient

Define sample correlation coefficient:

$$R = \frac{[1/(n-1)]\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{[1/(n-1)]\sum_{i=1}^{n}(X_i - \overline{X})^2}\sqrt{[1/(n-1)]\sum_{i=1}^{n}(Y_i - \overline{Y})^2}} = \frac{S_{XY}}{S_X S_Y}.$$

$$\frac{R\sqrt{n-2}}{\sqrt{1-R^2}} \sim t(n-2)$$

Sample covariance: $s_{XY} = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})$

Sample variances:

$$s_X^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 \qquad s_Y^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2$$

Testing correlation

To test $H_0 : \rho = 0$, we have two options:

1. Compute r, the sample correlation coefficient,

$$\boxed{r = \frac{[1/(n-1)] \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{[1/(n-1)] \sum_{i=1}^{n} (x_i - \bar{x})^2} \sqrt{[1/(n-1)] \sum_{i=1}^{n} (y_i - \bar{y})^2}} = \frac{s_{XY}}{s_X s_Y}},$$

   and then compute

$$\boxed{\frac{r\sqrt{n-2}}{\sqrt{1-r^2}}}.$$

   Compare this value with a $t(n-2)$ quantile using the t-table.

   $t_{\alpha/2}(n-2)$

2. Compute r and use Table IX in Appendix B of the book with $n-2$ degree of freedom.

   $r_{\alpha/2}(n-2)$

Table IX Distribution Function of the Correlation Coefficient R, $\rho = 0$

Summary: testing correlation

Null Hypothesis

$$H_0: \rho = 0$$

Test Statistic

$$t = \frac{R\sqrt{n-2}}{\sqrt{1-R^2}}$$

Distribution under H_0

$$t \sim t(n-2)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value
$H_1: \rho \neq 0$	$\lvert t \rvert > t_{\alpha/2}(n-2)$	$2\mathrm{P}(T > \lvert t \rvert)$
$H_1: \rho > 0$	$t > t_{\alpha}(n-2)$	$\mathrm{P}(T > t)$
$H_1: \rho < 0$	$t < -t_{\alpha}(n-2)$	$\mathrm{P}(T < t)$

H_0 is rejected when p-value<α

Test Statistic

$$R$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)
$H_1: \rho \neq 0$	$\lvert R \rvert > r_{\alpha/2}(n-2)$
$H_1: \rho > 0$	$R > r_{\alpha}(n-2)$
$H_1: \rho < 0$	$R < -r_{\alpha}(n-2)$

Testing correlation

What if we want to test $H_0 : \rho = \rho_0$ for some $\rho_0 \neq 0$? It turns out that we can prove

$$\frac{1}{2}\ln\left(\frac{1+R}{1-R}\right) \overset{approx}{\sim} N\left(\frac{1}{2}\ln\left(\frac{1+\rho}{1-\rho}\right), \frac{1}{n-3}\right).$$

As a result, we can use

$$\boxed{\frac{\frac{1}{2}\ln\left(\frac{1+R}{1-R}\right) - \frac{1}{2}\ln\left(\frac{1+\rho_0}{1-\rho_0}\right)}{\sqrt{1/(n-3)}}}$$

as our test statistic, and compare this with z_α or $-z_\alpha$ or $z_{\alpha/2}$ depending on H_1.

(approximate) Confidence interval for ρ

$$\frac{1}{2}\ln\left(\frac{1+R}{1-R}\right) \overset{approx}{\sim} N\left(\frac{1}{2}\ln\left(\frac{1+\rho}{1-\rho}\right), \frac{1}{n-3}\right)$$

$$P\left(-c \leq \frac{(1/2)\ln[(1+R)/(1-R)] - (1/2)\ln[(1+\rho)/(1-\rho)]}{\sqrt{1/(n-3)}} \leq c\right) \approx 1-\alpha.$$

$$c = z_{\alpha/2}$$

$$P\left(\frac{1+R-(1-R)\exp(2c/\sqrt{n-3})}{1+R+(1-R)\exp(2c/\sqrt{n-3})} \leq \rho \leq \frac{1+R-(1-R)\exp(-2c/\sqrt{n-3})}{1+R+(1-R)\exp(-2c/\sqrt{n-3})}\right) \approx 1-\alpha.$$

Best Critical Region (Neyman-Pearson Lemma)

Likelihood ratio tests

LIKELIHOOD RATIO TESTS

Composite Hypothesis Testing

- Assume that the pdf of the random variable X is f(x;θ), where θ represents one or more unknown parameters

- We let Ω denote the total parameter space — that is, the set of all possible values of the parameter θ given by either H_0 or H_1

- Test the hypotheses

Null Hypothesis **Alternative Hypothesis**

$$H_0 : \theta \in \omega, \qquad H_1 : \theta \in \omega',$$

$$\Omega = \omega \cup \omega'$$

- We use the likelihood ratio as our test statistic

Likelihood Ratio

- We use the likelihood ratio as our test statistic

Null Hypothesis **Alternative Hypothesis**

$$H_0 : \theta \in \omega, \qquad H_1 : \theta \in \omega',$$

Definition 8.7-1

The **likelihood ratio** is the quotient

$$\lambda = \frac{L(\hat{\omega})}{L(\hat{\Omega})},$$

where $L(\hat{\omega})$ is the maximum of the likelihood function with respect to θ when $\theta \in \omega$ and $L(\hat{\Omega})$ is the maximum of the likelihood function with respect to θ when $\theta \in \Omega$.

- $0 \leq \lambda \leq 1$
- A small value of λ will lead to the rejection of H_0

Critical (rejection) Region

$$H_0: \theta \in \omega, \qquad H_1: \theta \in \omega',$$

- $0 \leq \lambda \leq 1$
- A small value of λ will lead to the rejection of H_0

Definition 8.7-2

To test $H_0: \theta \in \omega$ against $H_1: \theta \in \omega'$, the **critical region for the likelihood ratio test** is the set of points in the sample space for which

$$\lambda = \frac{L(\hat{\omega})}{L(\hat{\Omega})} \leq k,$$

where $0 < k < 1$ and k is selected so that the test has a desired significance level α.

Testing normal mean (variance known)

- How about one-sided test? For example:

$$H_0: \mu = \mu_0 \qquad H_1: \mu > \mu_0$$

$$w = \{ \mu = \mu_0, \quad \sigma^2 = \sigma_0^2 \}$$

$$w' = \{ \mu_0 < \mu < +\infty, \quad \sigma^2 = \sigma_0^2 \}$$

$$\text{let } \Omega = w \cup w' = \{ \mu \geq \mu_0, \quad \sigma^2 = \sigma_0^2 \}$$

Testing normal mean (variance known)

$$H_0: \mu = \mu_0 \qquad H_1: \mu > \mu_0$$

under ω. $L(\mu_0) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x_i-\mu_0)^2}$

under Ω $L(\mu) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x_i-\mu)^2}$

$$\ell(\mu) = \log L(\mu) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

Therefore, $\max_{\mu \geq \mu_0} L(\mu)$ is obtained at

$$\underset{\mu \geq \mu_0}{\arg\max}\, L(\mu) = \begin{cases} \bar{x} & \text{if } \bar{x} \geq \mu_0 \\ \mu_0 & \text{if } \bar{x} < \mu_0 \end{cases}$$

denotes the value of $\mu \geq \mu_0$ such that $L(\mu)$ is maximized.

Testing normal mean (variance known)

$$H_0: \mu = \mu_0 \qquad H_1: \mu > \mu_0$$

$$\Rightarrow \lambda = \begin{cases} \dfrac{\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma_0^2}} e^{-\frac{1}{2\sigma_0^2}(x_i-\mu_0)^2}}{\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma_0^2}} e^{-\frac{1}{2\sigma_0^2}(x_i-\bar{x})^2}} & \text{if } \bar{x} \geq \mu_0 \\ \dfrac{\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma_0^2}} e^{-\frac{1}{2\sigma_0^2}(x_i-\mu_0)^2}}{\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma_0^2}} e^{-\frac{1}{2\sigma_0^2}(x_i-\mu_0)^2}} = 1 & \text{if } \bar{x} < \mu_0 \end{cases}$$

$$= \begin{cases} e^{-\frac{n}{2\sigma_0^2}(\bar{x}-\mu_0)^2} & \text{if } \bar{x} \geq \mu_0 \\ 1 & \text{if } \bar{x} < \mu_0 \end{cases}$$

$\Rightarrow$

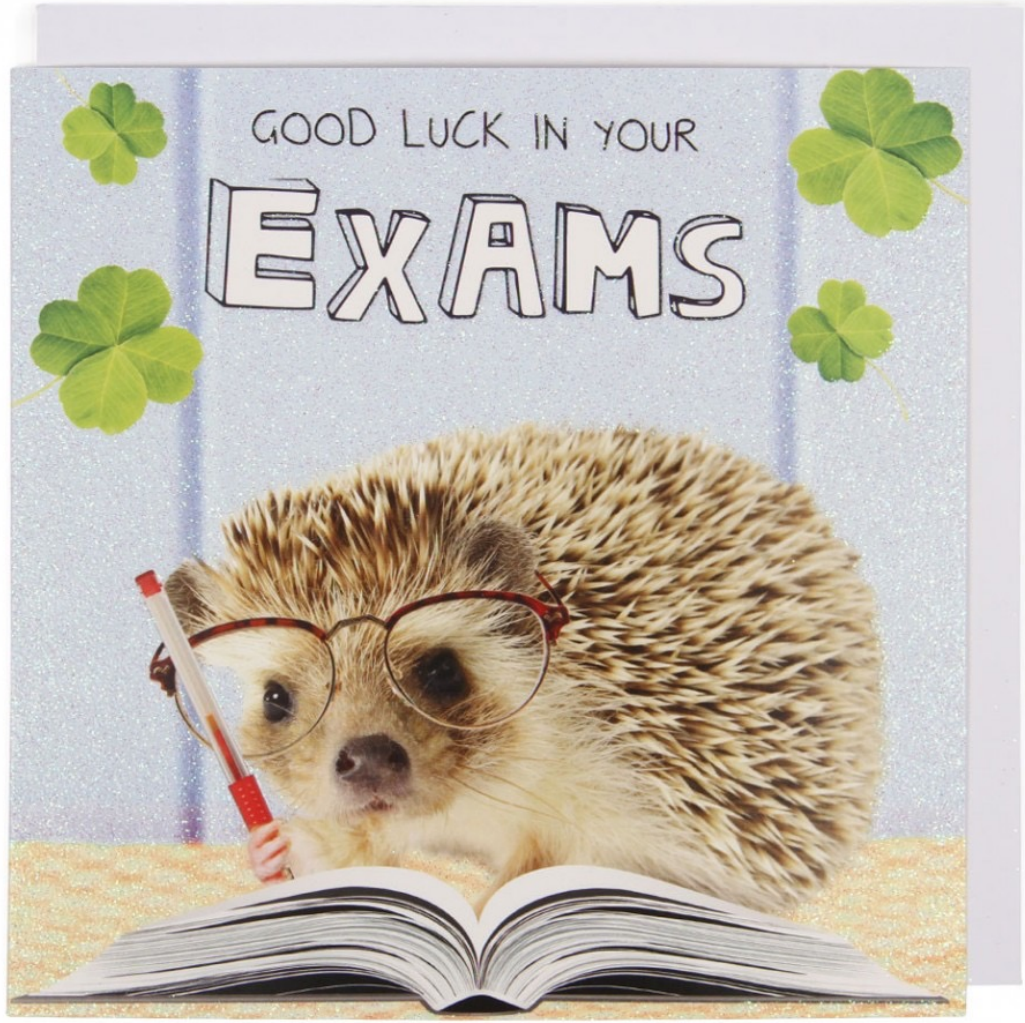
$\lambda \leq k \Rightarrow e^{-\frac{n}{2\sigma_0^2}(\bar{x}-\mu_0)^2} \leq k \qquad (\bar{x} \geq \mu_0)$

$\bar{x} - \mu_0 \geq \sqrt{\frac{-2\sigma_0^2 \ln k}{n}}$ (because $\bar{x} \geq \mu_0$)

$P(\lambda \leq k \mid \text{under } H_0) = \alpha$

$\Rightarrow P\left(\bar{x} - \mu_0 \geq \sqrt{\frac{-2\sigma_0^2 \ln k}{n}}\right) = \alpha.$

$\frac{\sqrt{(-2\sigma_0^2 \ln k)/n}}{\sigma_0/\sqrt{n}} = z_\alpha$


GOOD LUCK IN YOUR
EXAMS